

Model Testing Report

Experiment: model_testing

Generated: 2026-04-08 17:37:05

Segmentation Column: value_date

Segments Defined: 4

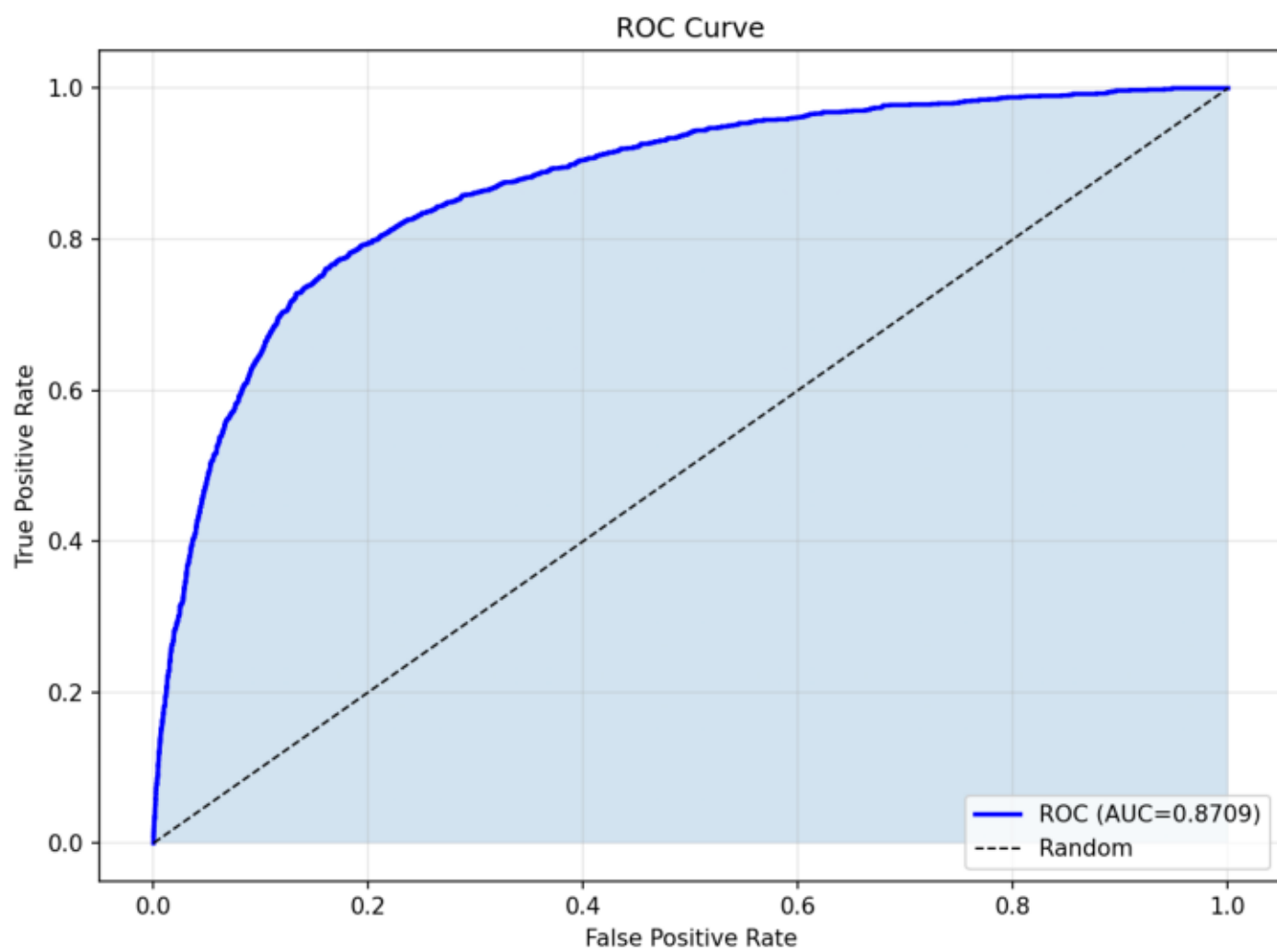
- value_date_quarter=2024Q1: 2122 rows (completed)
- value_date_quarter=2024Q2: 2448 rows (completed)
- value_date_quarter=2024Q3: 2652 rows (completed)
- value_date_quarter=2024Q4: 2778 rows (completed)

Overall - 1) Model Effectiveness

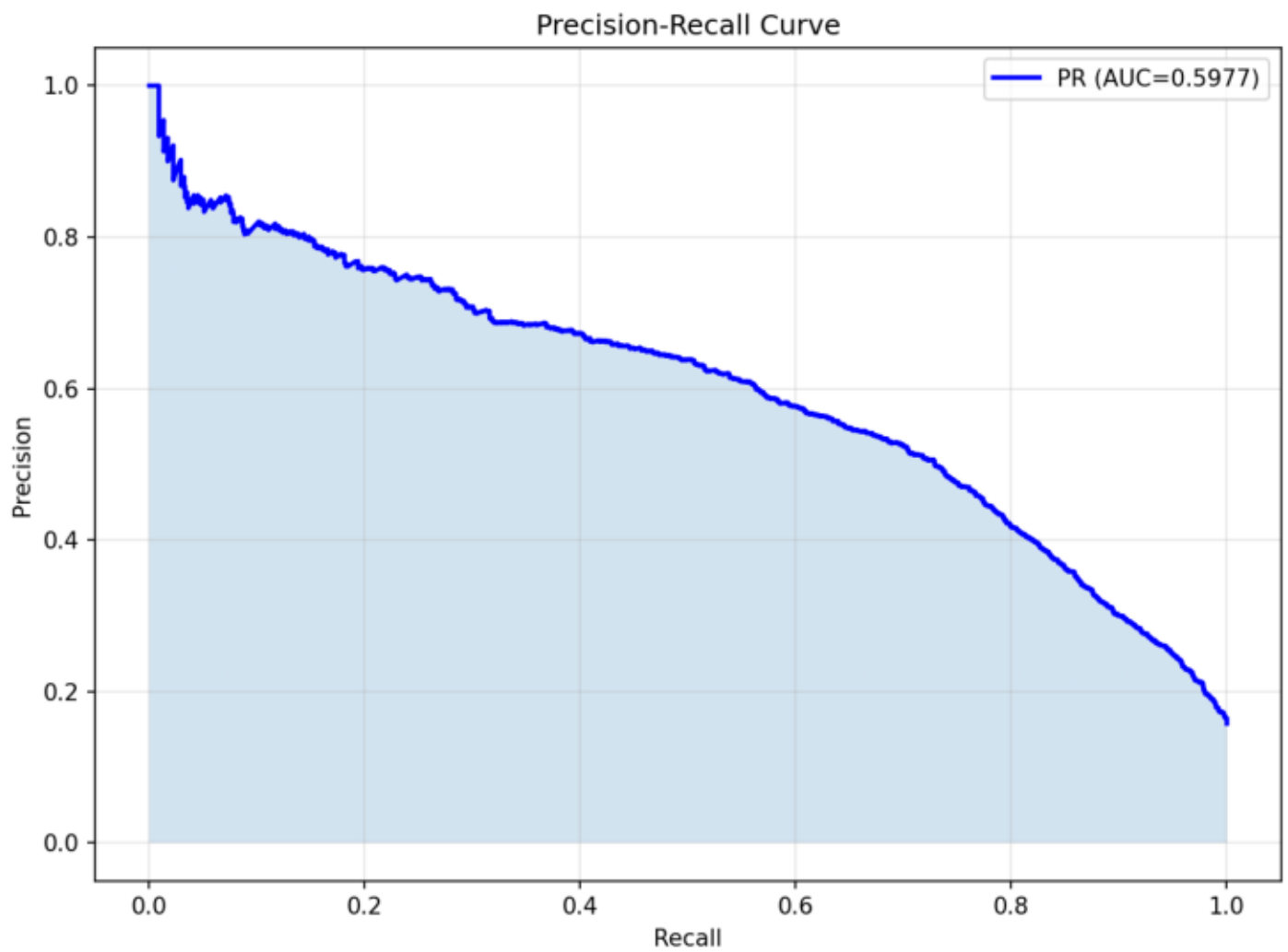
- auc_roc: 0.8709
- auc_pr: 0.5977
- precision: 0.6630
- recall: 0.4225
- f1: 0.5161
- ks_statistic: 0.5994
- ks_threshold: 0.1800
- confusion_matrix.TN: 8088
- confusion_matrix.FP: 338
- confusion_matrix.FN: 909
- confusion_matrix.TP: 665
- precision_at_k.10: 1.0000
- precision_at_k.50: 0.9000
- precision_at_k.100: 0.8400
- precision_at_k.200: 0.8150
- precision_at_k.500: 0.7500
- recall_at_k.10: 0.0064
- recall_at_k.50: 0.0286
- recall_at_k.100: 0.0534
- recall_at_k.200: 0.1036
- recall_at_k.500: 0.2382

Overall - 1) Model Effectiveness - Explanations

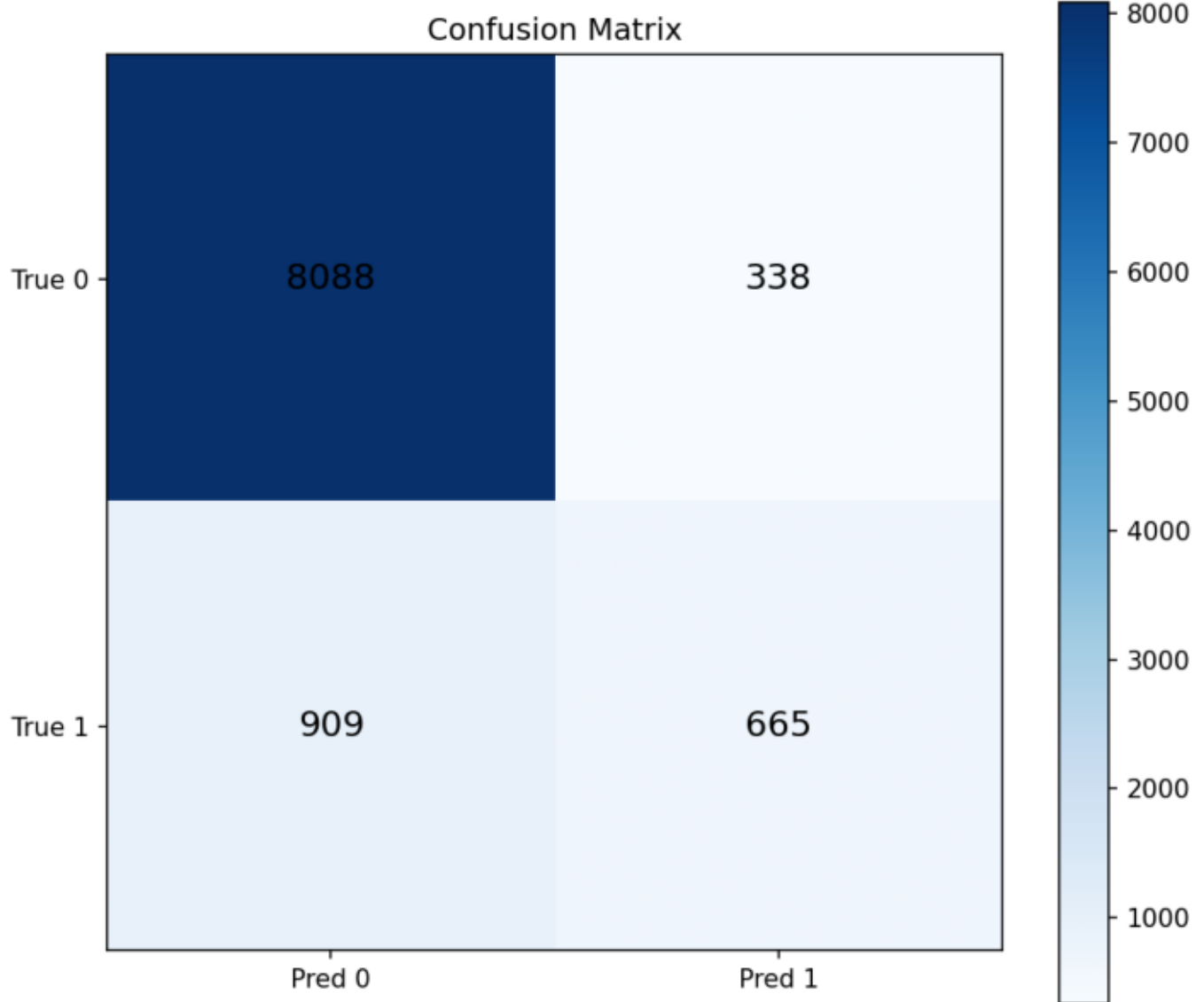
- Positive rate is 15.74%; AUC-ROC is 0.8709 (strong ranking).
- AUC-PR is 0.5977, above the base rate 15.74%, indicating meaningful lift.
- At threshold 0.50, precision=0.6630, recall=0.4225, F1=0.5161 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=665, FP=338, FN=909, TN=8088 (FPR=4.01%, FNR=57.75%).
- KS is 0.5994 at threshold 0.18, indicating very strong separation.
- Top-10: precision=1.0000, recall=0.0064 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.7500, recall=0.2382 shows how much coverage you get with a larger queue.



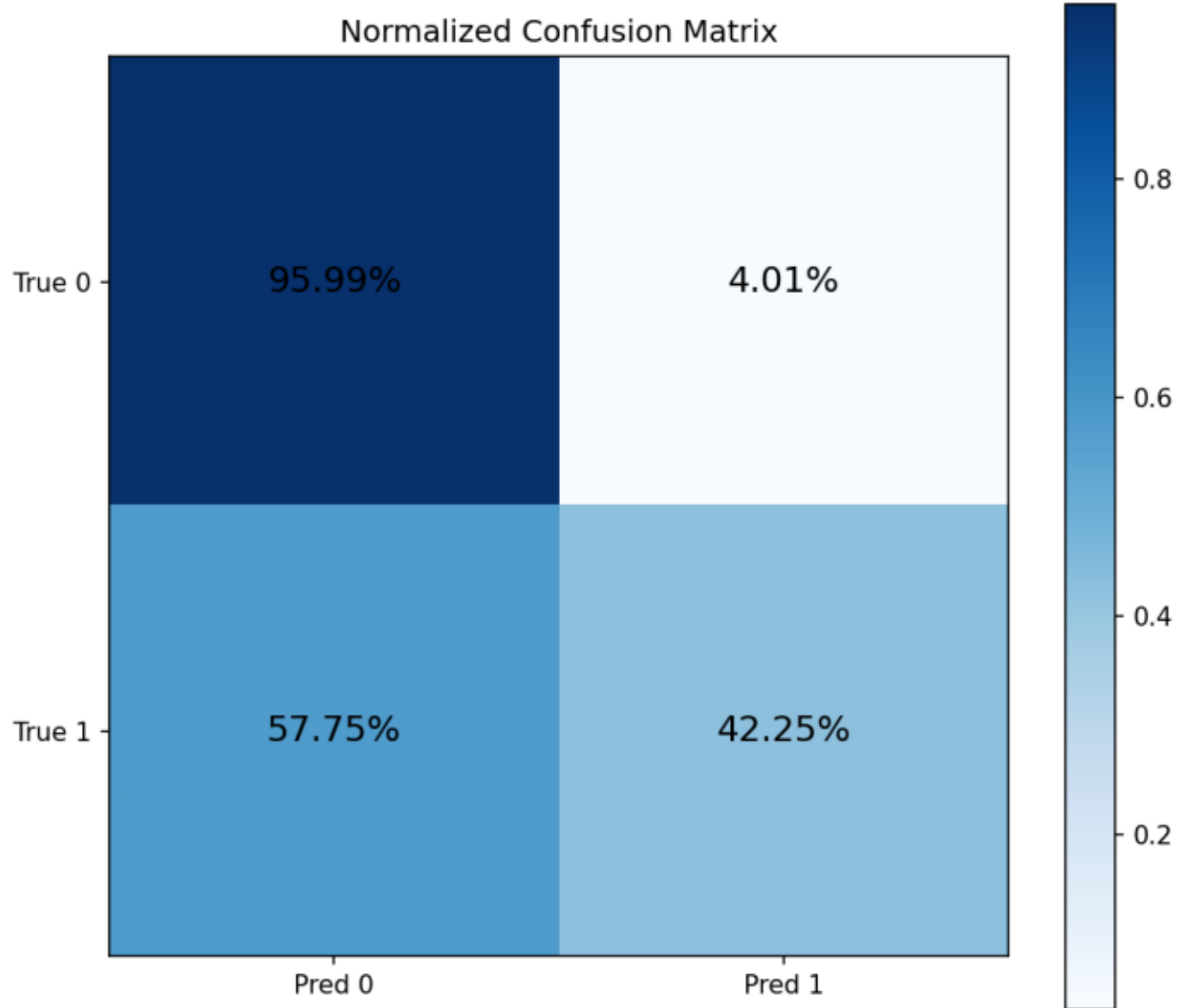
roc_curve: AUC-ROC=0.8709; curve above diagonal indicates strong separation.



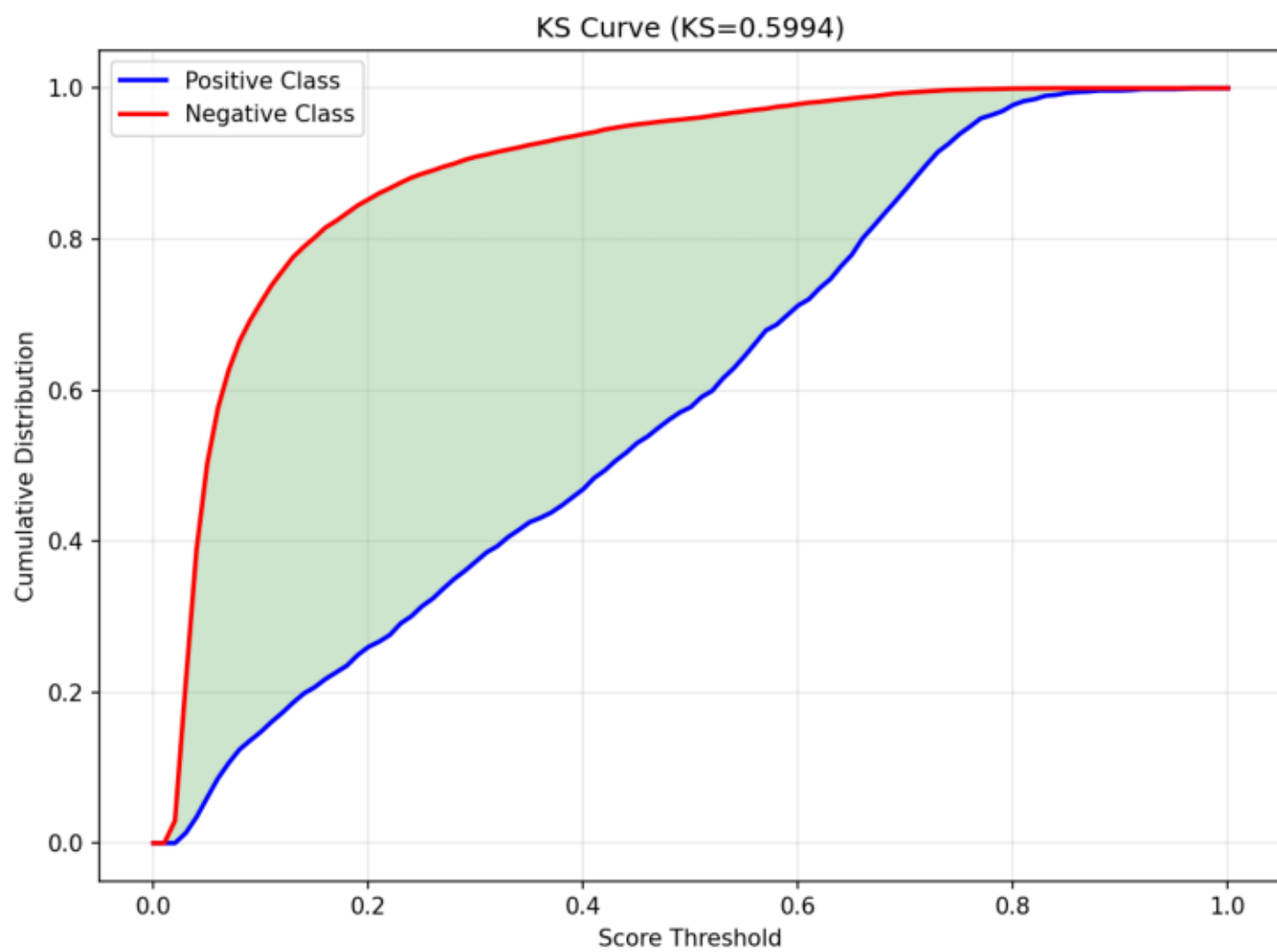
pr_curve: AUC-PR=0.5977 vs baseline 15.74%; higher curve implies better precision at recall.



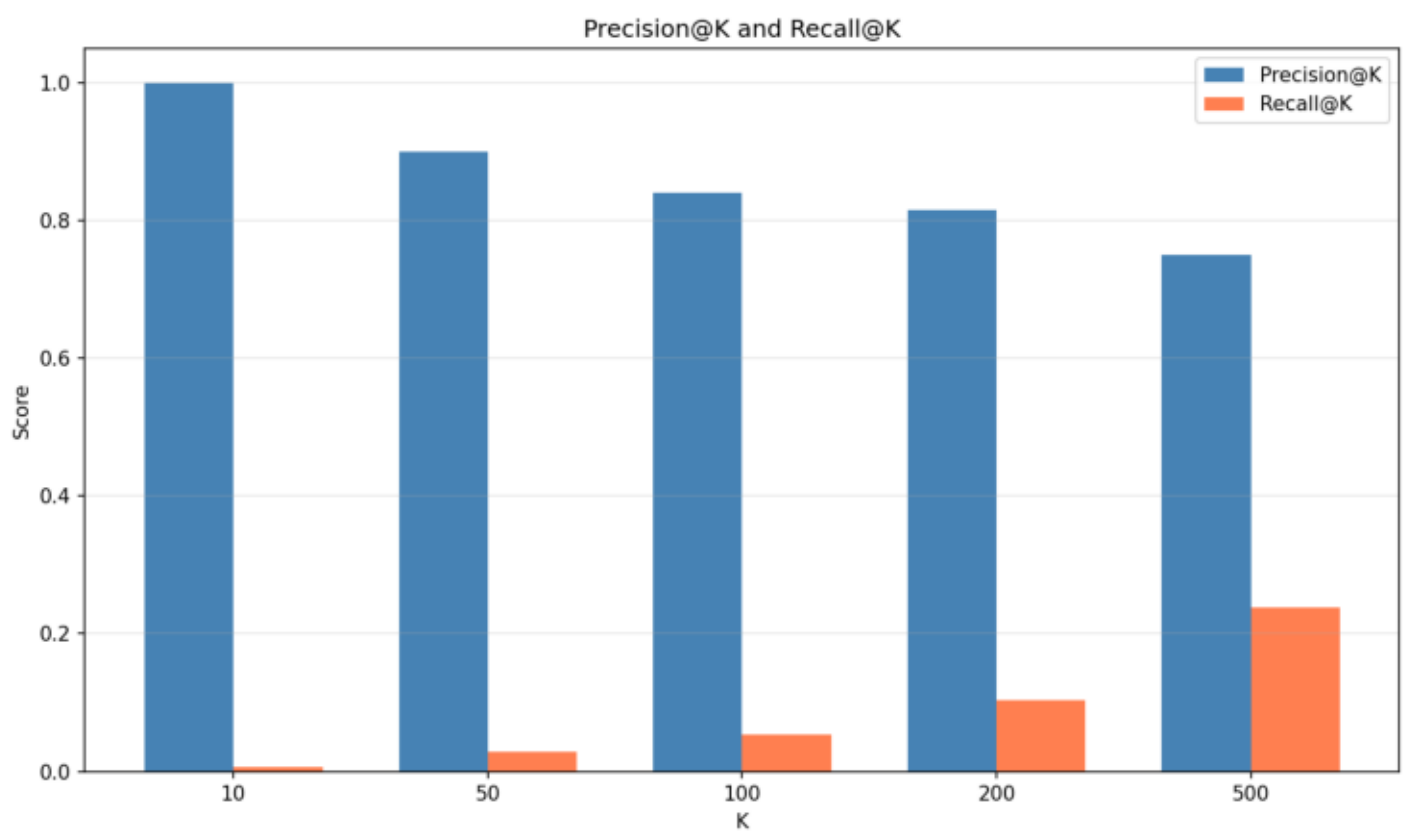
confusion_matrix: At threshold 0.50: TP=665, FP=338, FN=909, TN=8088.



confusion_matrix_norm: Normalized rates: TPR=42.25%, TNR=95.99%.



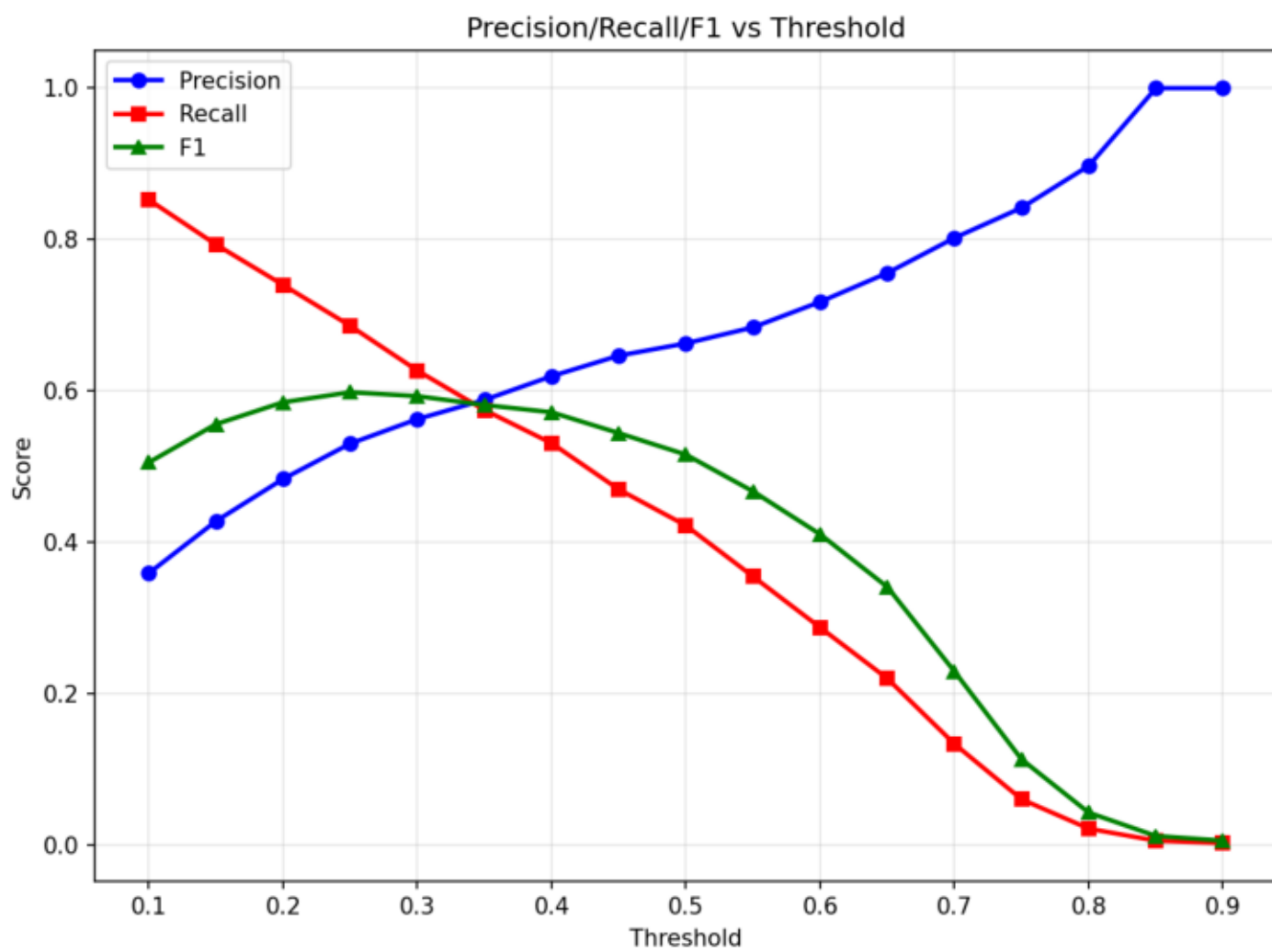
ks_curve: Maximum separation KS=0.5994 at threshold 0.18.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.412, negative=0.108 (larger gap is better).



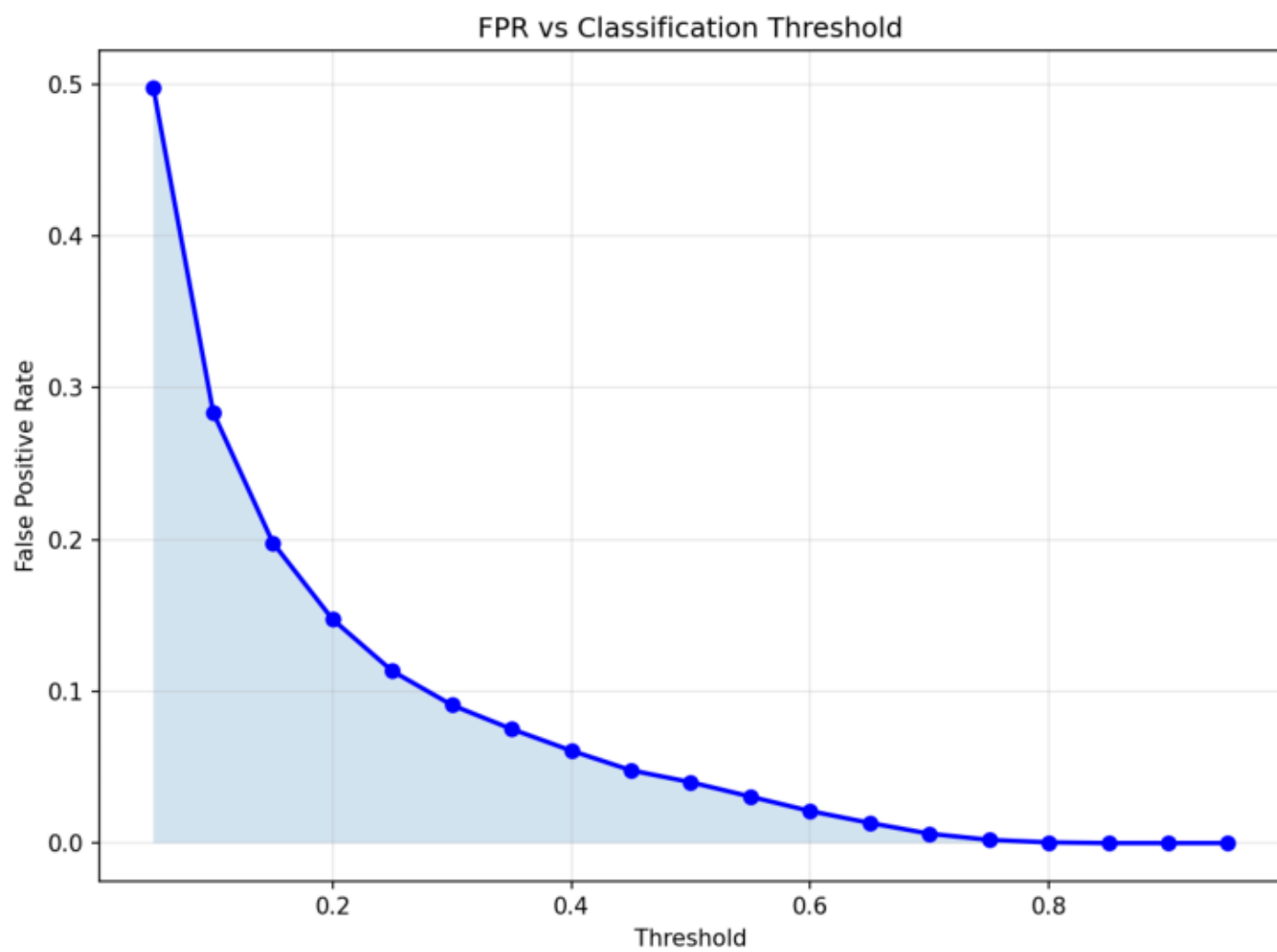
threshold_analysis: Best F1 in tested grid is 0.5985 at threshold 0.25.

Overall - 2) Model Efficiency

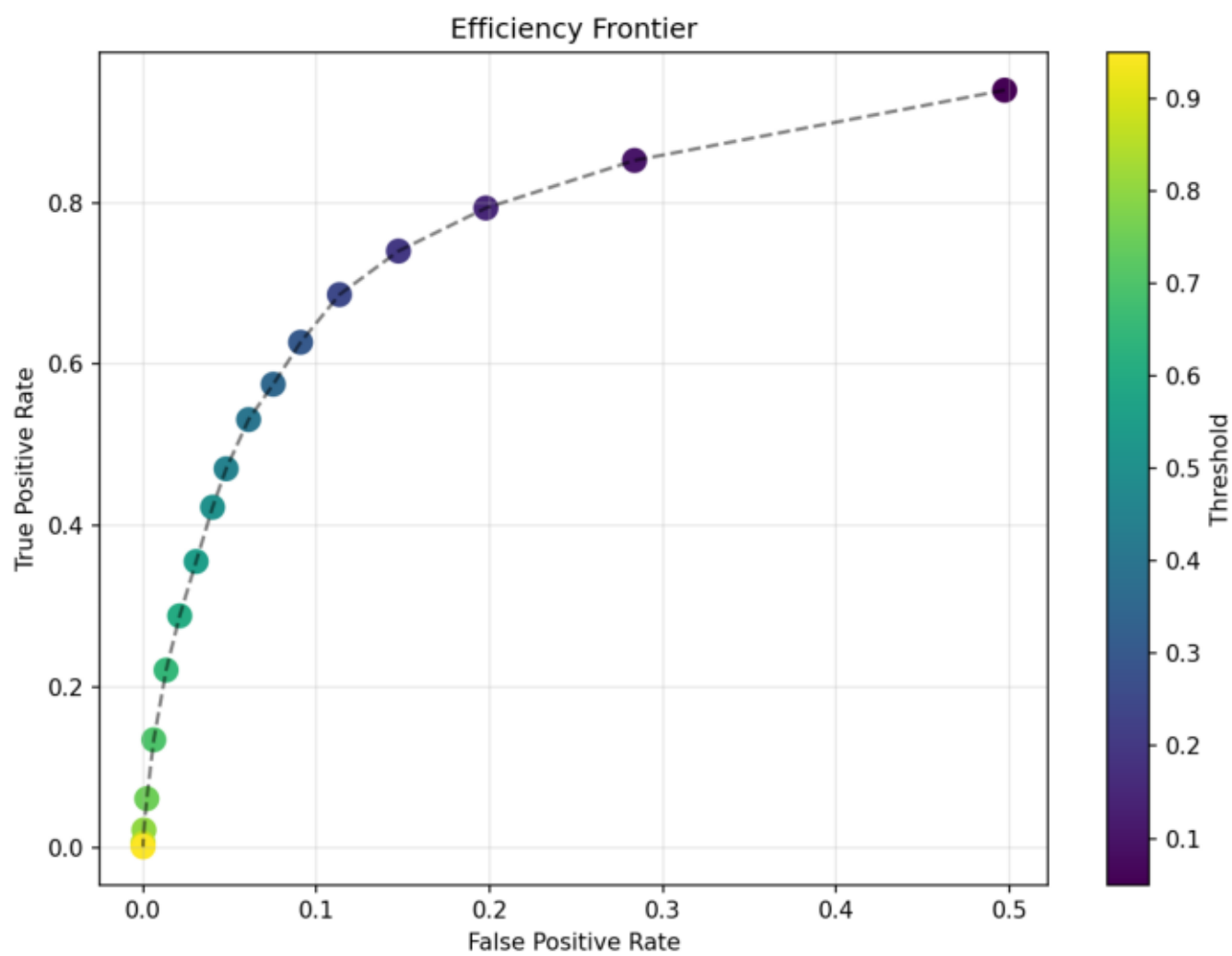
- fpr: 0.0401
- tn: 8088
- fp: 338
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.4973
- fpr_at_thresholds.t_0.10: 0.2836
- fpr_at_thresholds.t_0.15: 0.1978
- fpr_at_thresholds.t_0.20: 0.1474
- fpr_at_thresholds.t_0.25: 0.1133
- fpr_at_thresholds.t_0.30: 0.0909
- fpr_at_thresholds.t_0.35: 0.0751
- fpr_at_thresholds.t_0.40: 0.0609
- fpr_at_thresholds.t_0.45: 0.0479
- fpr_at_thresholds.t_0.50: 0.0401
- fpr_at_thresholds.t_0.55: 0.0306
- fpr_at_thresholds.t_0.60: 0.0211
- fpr_at_thresholds.t_0.65: 0.0133
- fpr_at_thresholds.t_0.70: 0.0062
- fpr_at_thresholds.t_0.75: 0.0021
- fpr_at_thresholds.t_0.80: 0.0005
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

Overall - 2) Model Efficiency - Explanations

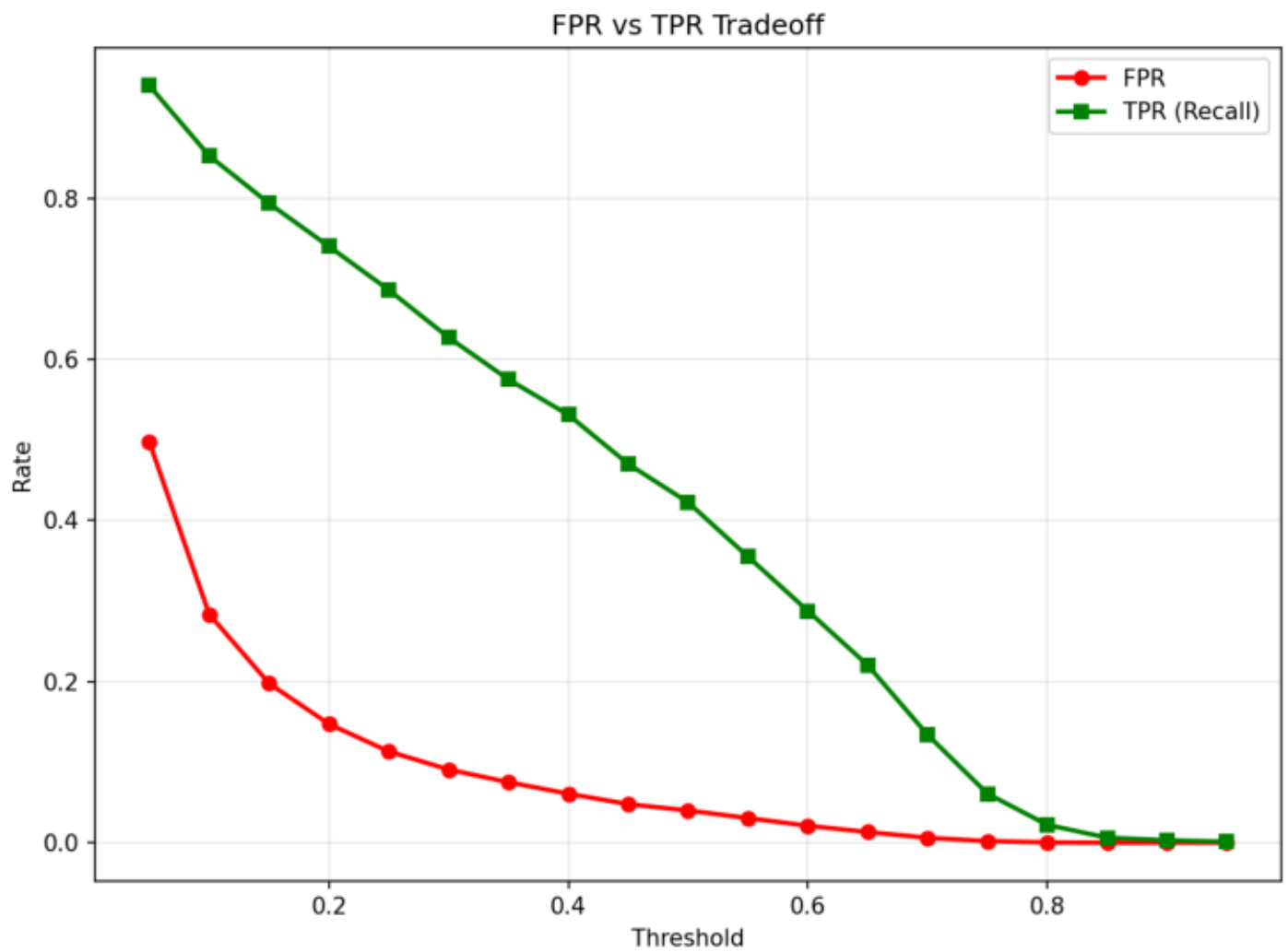
- At threshold 0.50, FPR=0.0401 (low); FP=338 out of 8426 negatives.
- At the same threshold, TPR=0.4225 with TP=665 and FN=909, showing the capture rate of positives.
- A threshold near 0.45 yields FPR $\approx$ 0.0479 with TPR $\approx$ 0.4701 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

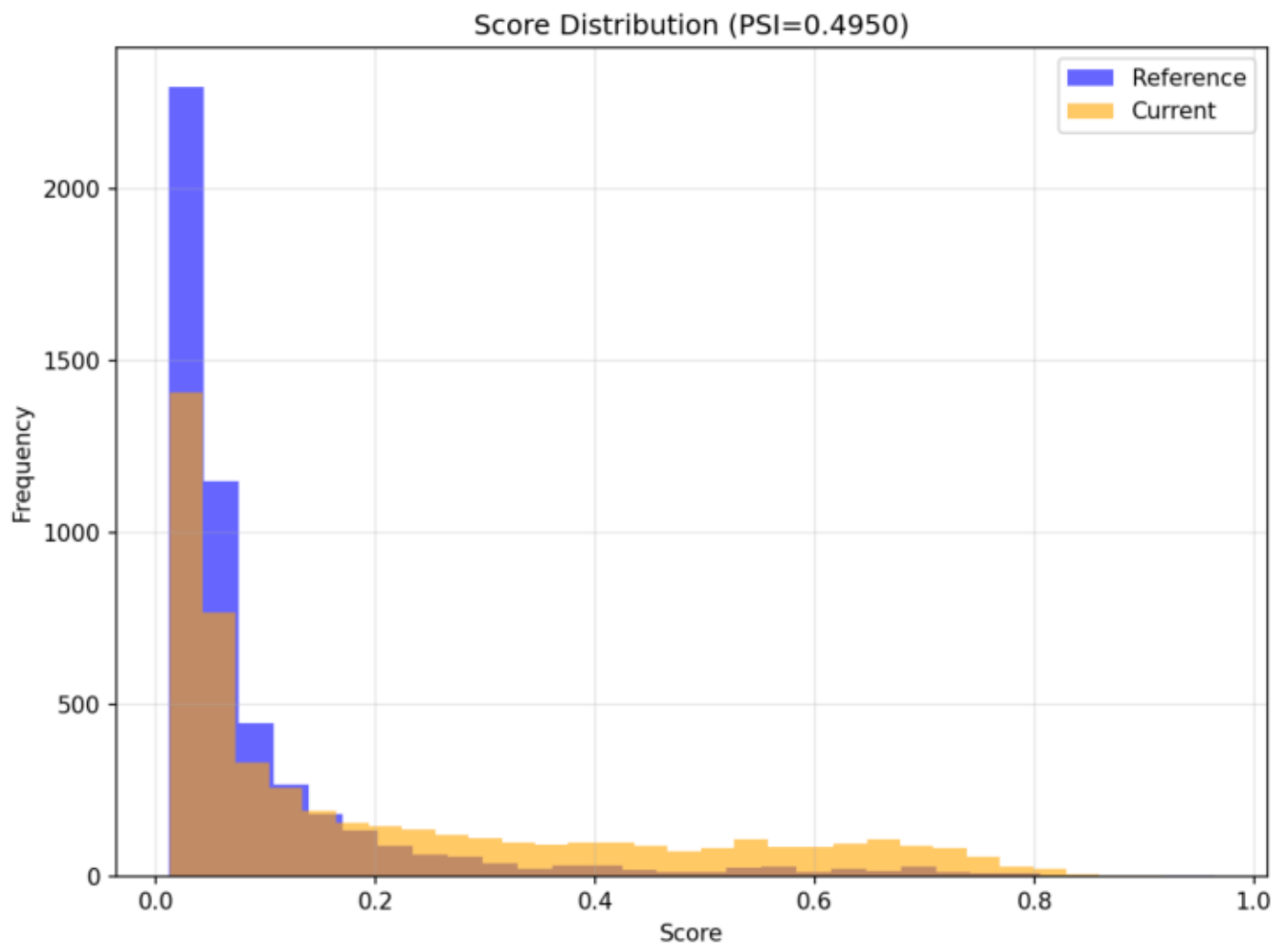


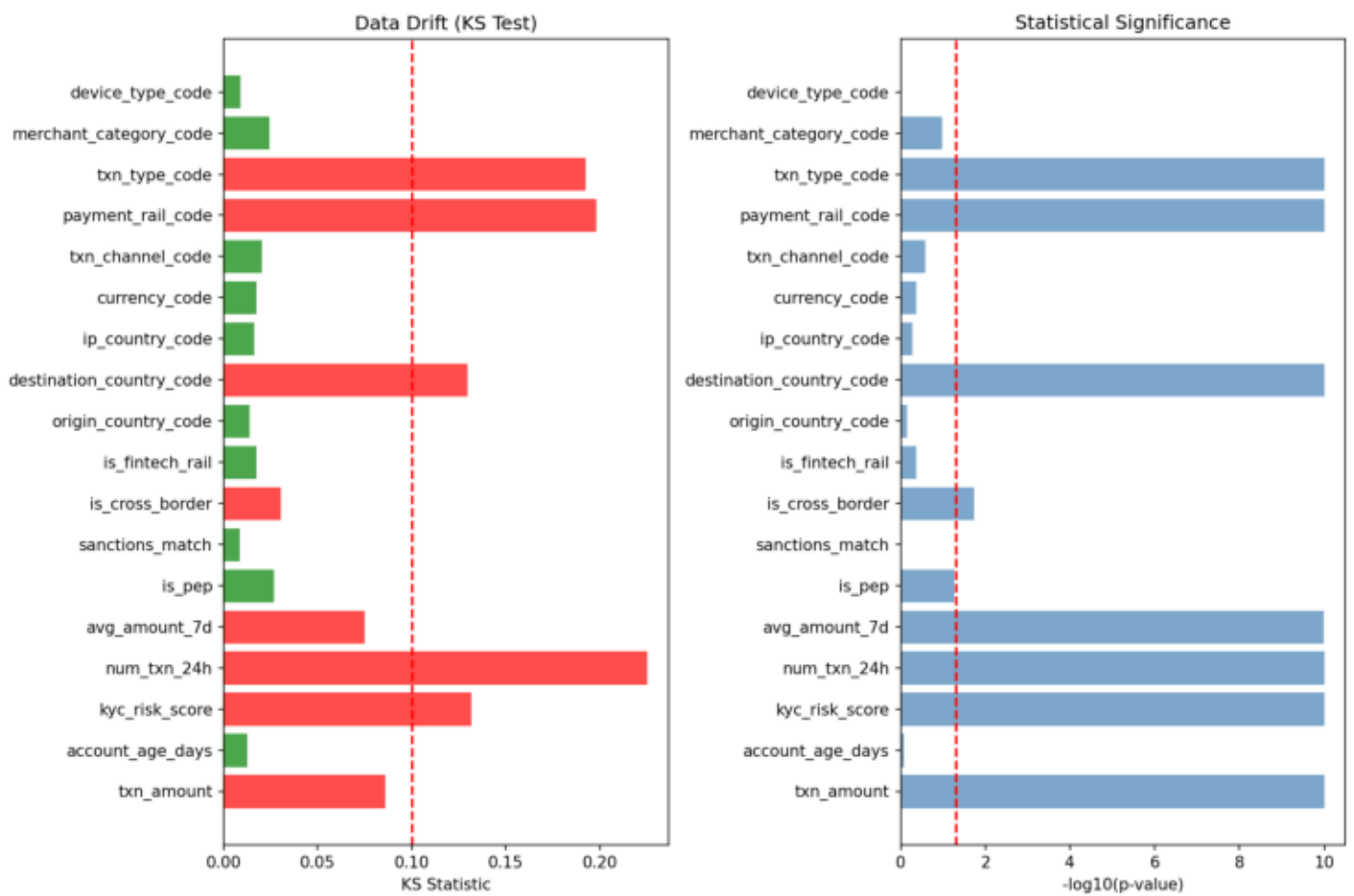
Overall - 3) Model Stability

- psi: 0.4950
- cv_auc_roc_mean: 0.8711
- cv_auc_roc_std: 0.0063
- cv_auc_pr_mean: 0.5992
- cv_auc_pr_std: 0.0250
- bootstrap_auc_roc_mean: 0.8512
- bootstrap_auc_roc_ci_lower: 0.8354
- bootstrap_auc_roc_ci_upper: 0.8630
- concept_drift_detected: False
- concept_drift_score: 0.0251

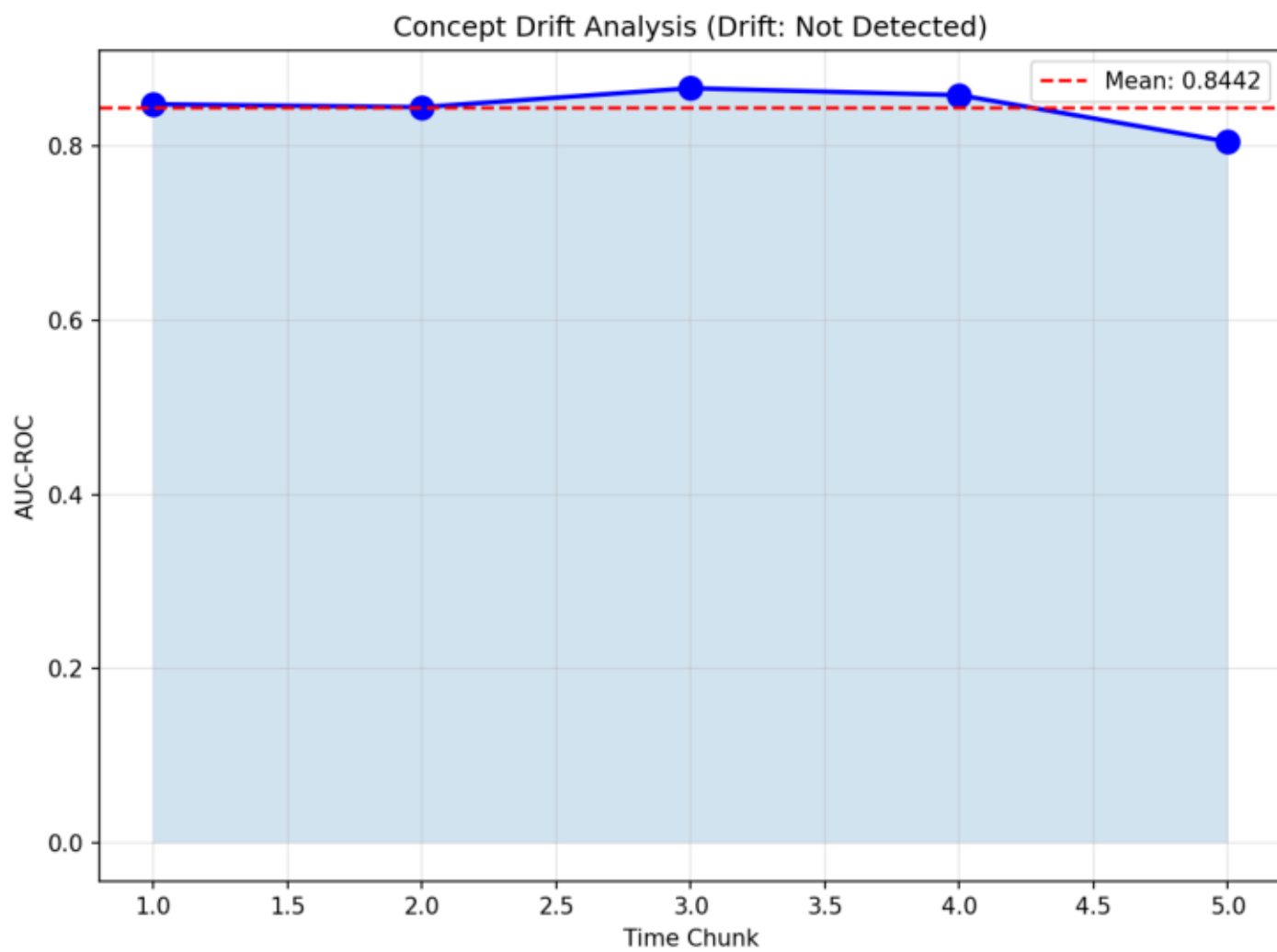
Overall - 3) Model Stability - Explanations

- PSI is 0.4950, indicating material score distribution shift between reference and current data.
- CV AUC-ROC is 0.8711 ± 0.0063 (stable across folds).
- CV AUC-PR is 0.5992 ± 0.0250 (moderate across folds).
- Bootstrap AUC-ROC mean is 0.8512 with 95% CI [0.8354, 0.8630] (width 0.0276).
- No concept drift detected (score 0.0251), suggesting stable performance over time.
- Data drift flagged 8/18 numeric features; top drift signals: ['num_txn_24h', 'payment_rail_code', 'txn_type_code'].

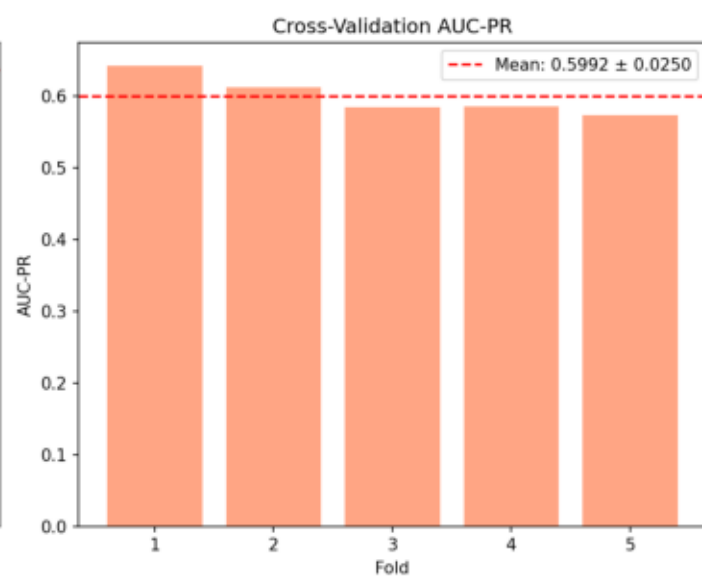
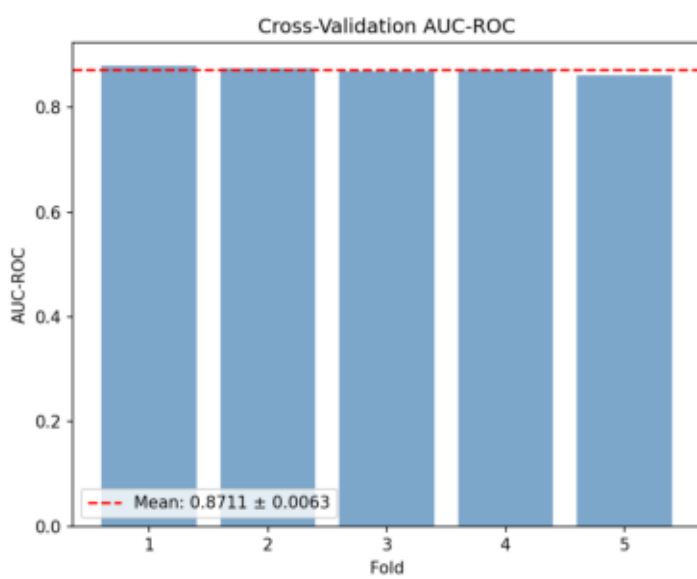




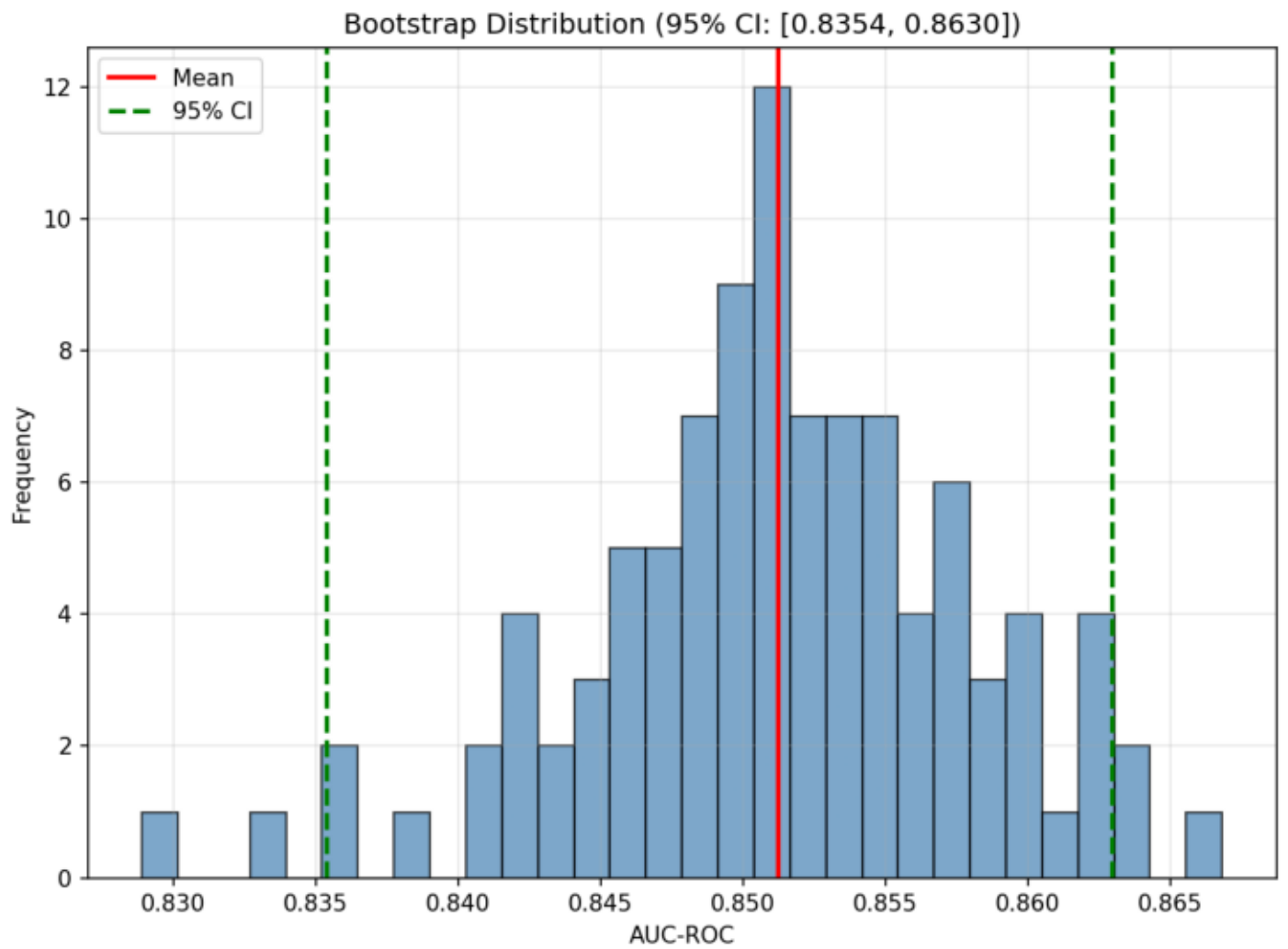
data_drift_heatmap: 8/18 features show drift; red bars highlight flagged features.



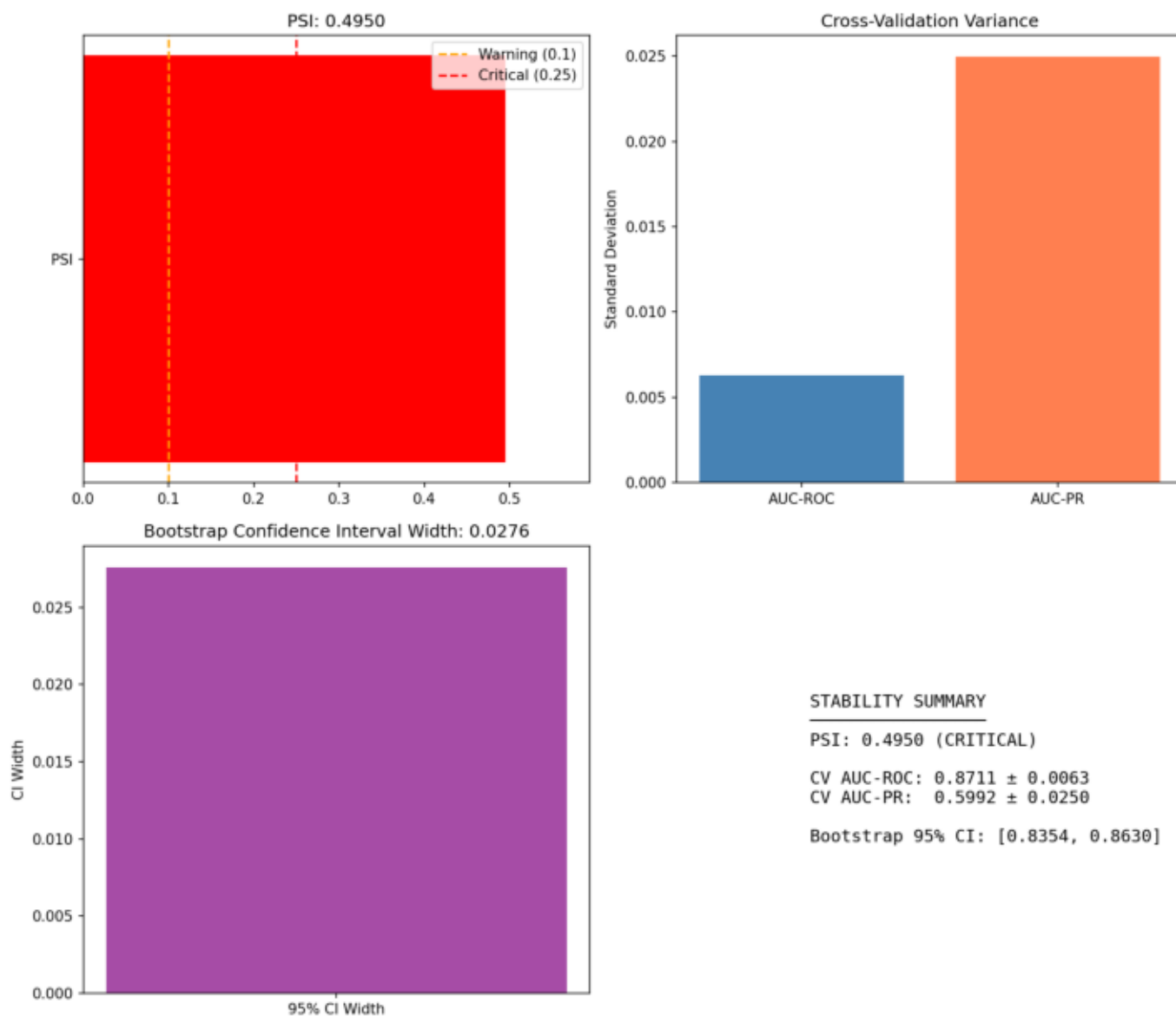
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8711 and PR 0.5992.



bootstrap_distribution: Bootstrap CI [0.8354, 0.8630] reflects performance stability.



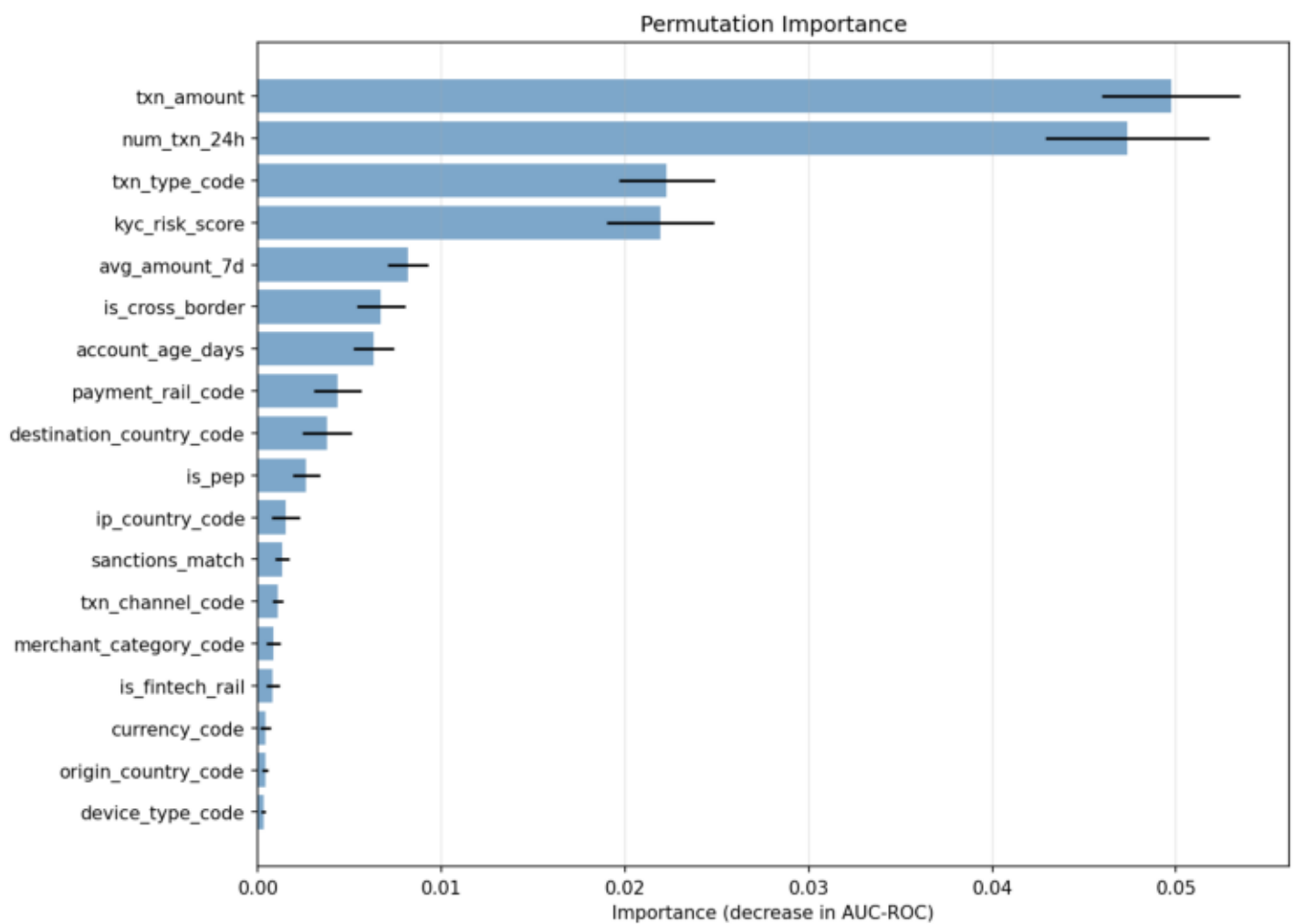
stability_summary: Dashboard summarizes PSI, CV variance, and bootstrap CI width.

Overall - 4) Model Interpretability

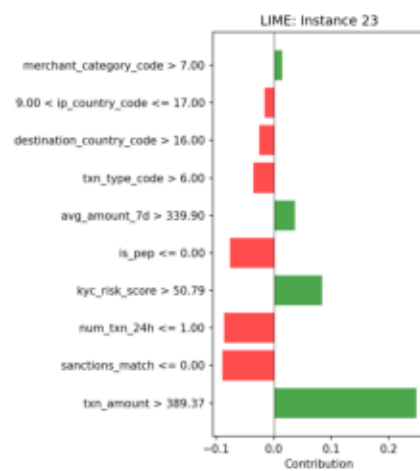
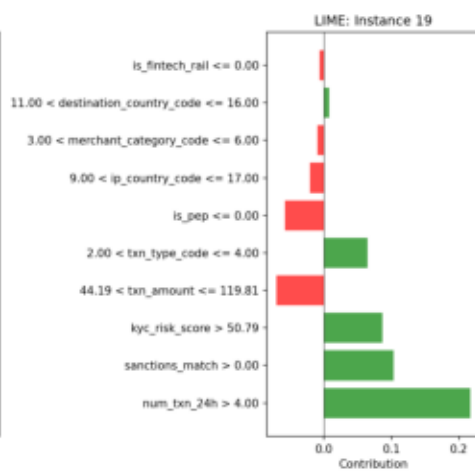
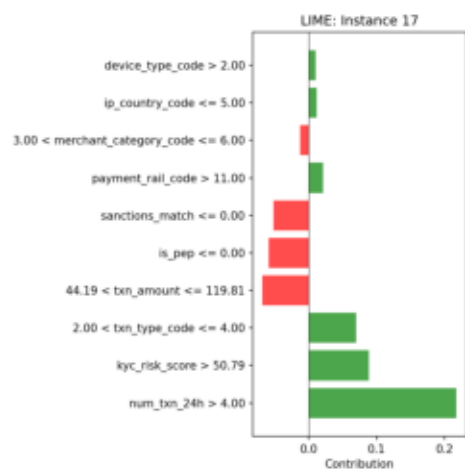
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'lime']
- shap_status: unavailable
- shap_reason: No module named 'shap'
- perm_top_features: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score', 'avg_amount_7d']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

Overall - 4) Model Interpretability - Explanations

- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score', 'avg_amount_7d'].
- SHAP plots not generated because SHAP is unavailable (No module named 'shap').
- LIME generated local explanations for 3 instances; top contributors for instance_17: ['num_txn_24h > 4.00', 'kyc_risk_score > 50.79', '2.00 < txn_type_code <= 4.00', '44.19 < txn_amount <= 119.81', 'is_pep <= 0.00'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'origin_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score'].

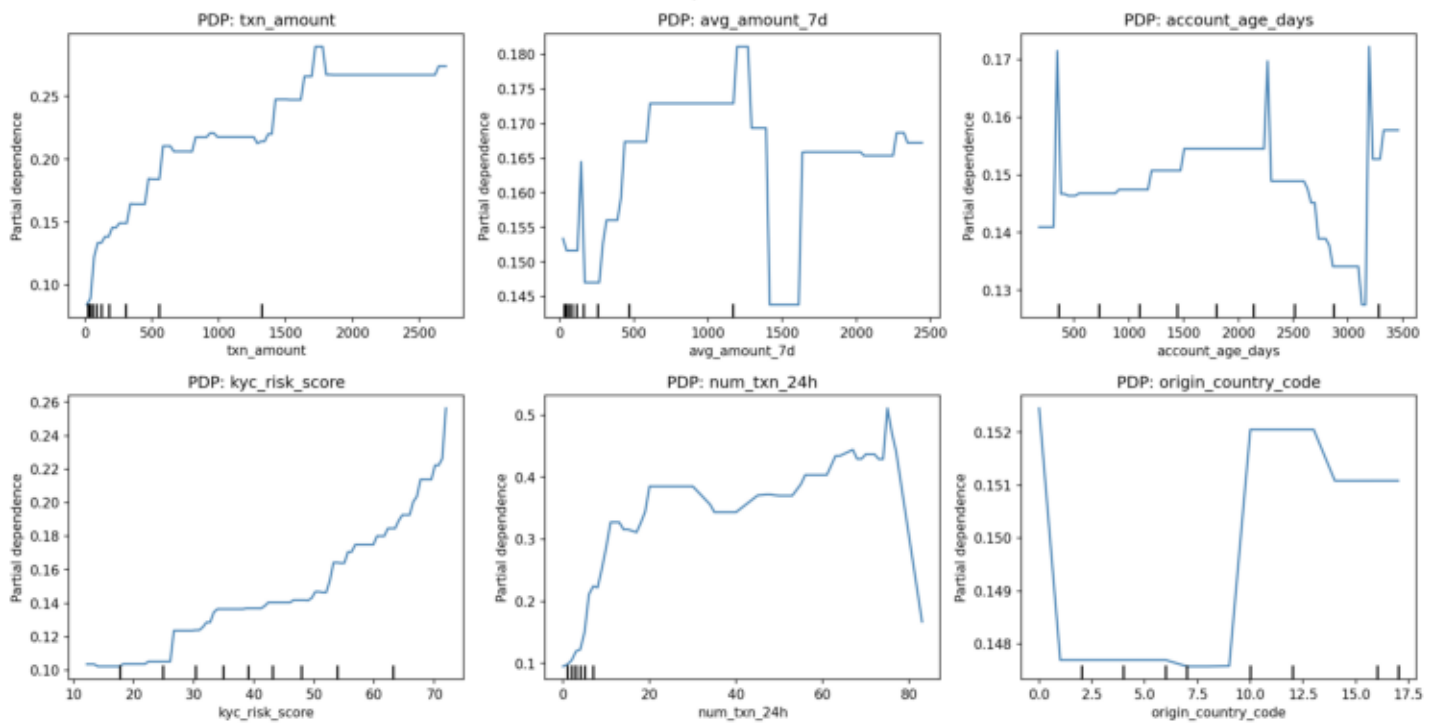


permutation_importance: Bars show how much AUC drops when each feature is permuted.



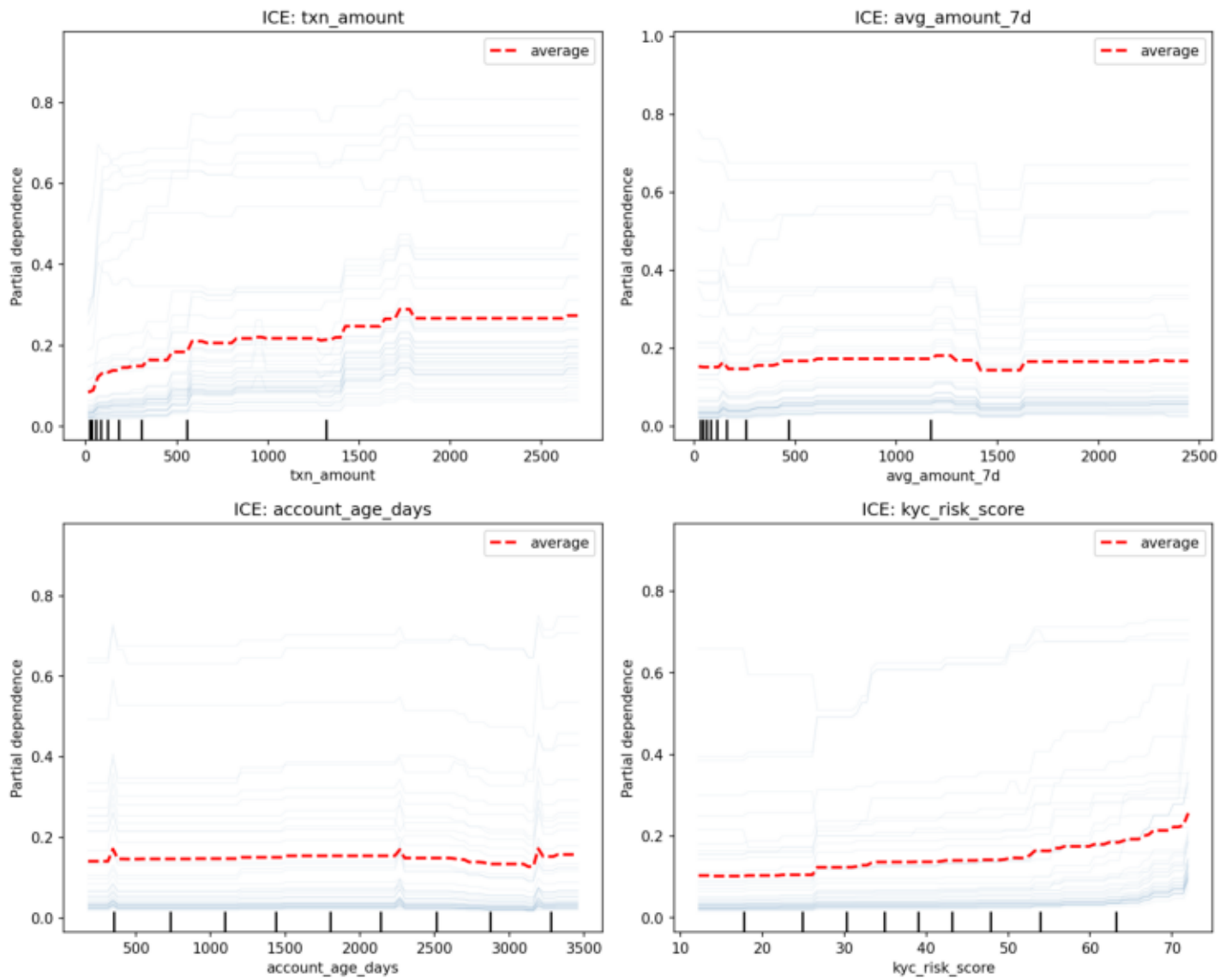
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



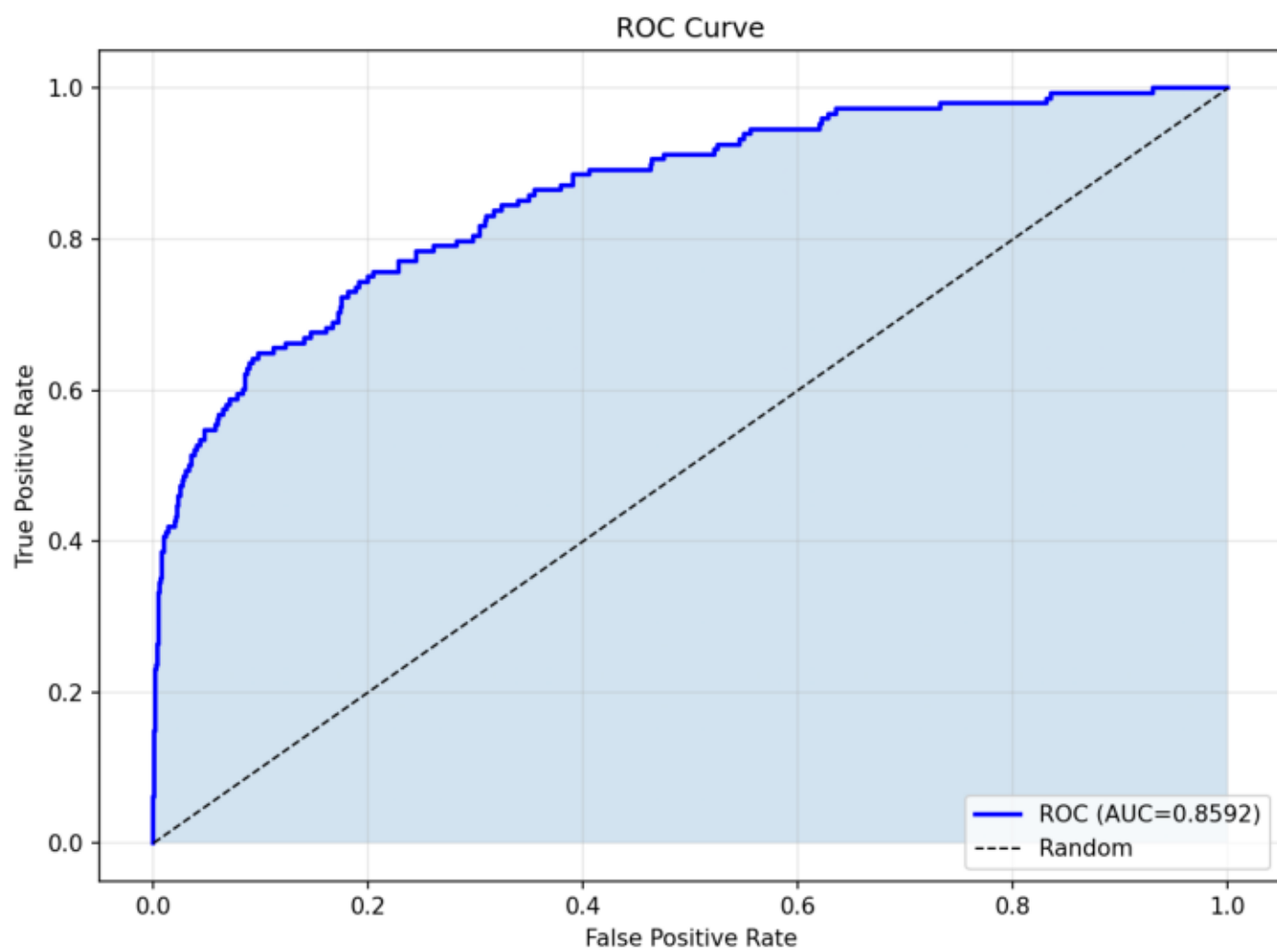
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q1 - 1) Model Effectiveness

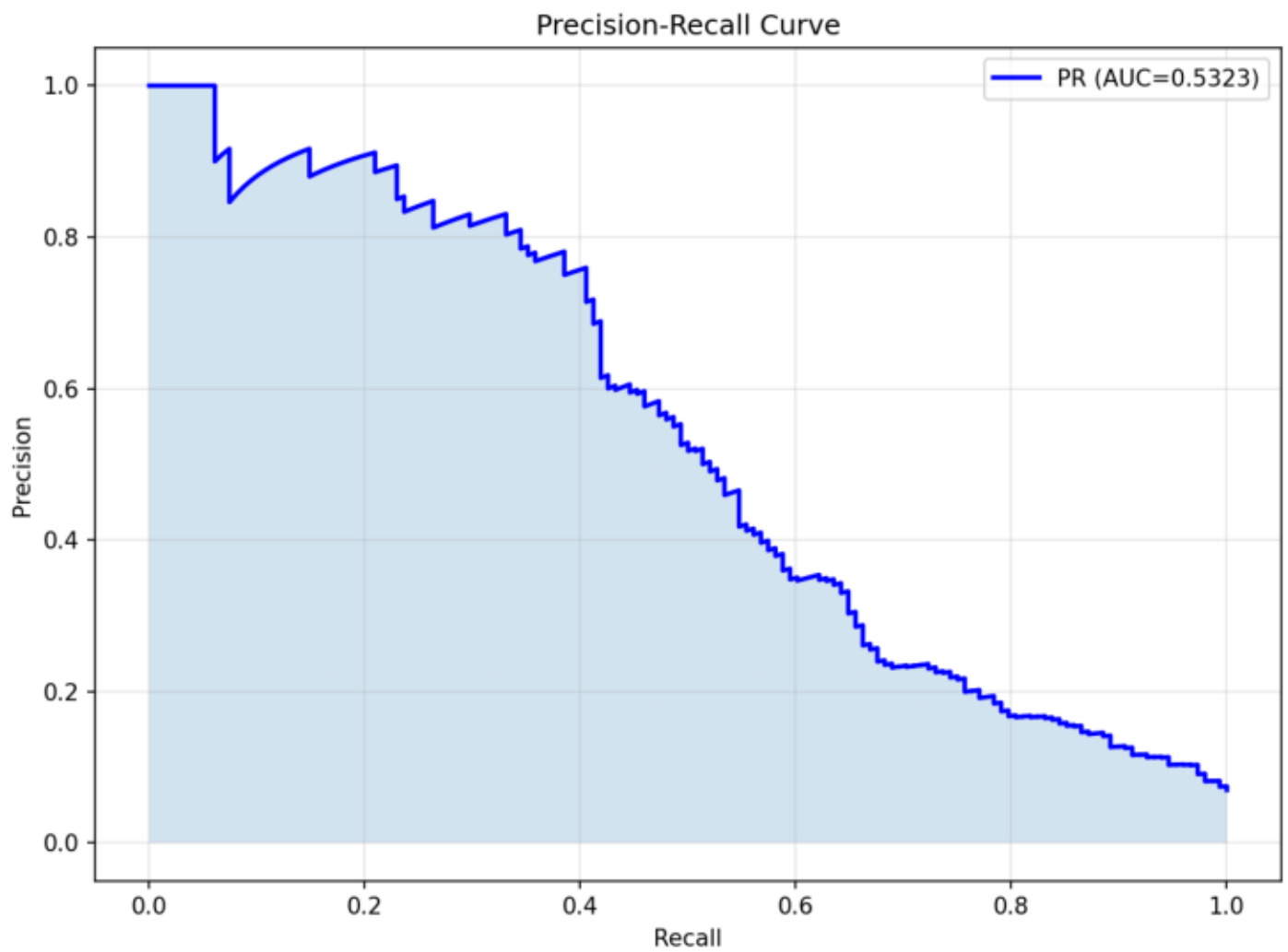
- auc_roc: 0.8592
- auc_pr: 0.5323
- precision: 0.8305
- recall: 0.3311
- f1: 0.4734
- ks_statistic: 0.5499
- ks_threshold: 0.1400
- confusion_matrix.TN: 1964
- confusion_matrix.FP: 10
- confusion_matrix.FN: 99
- confusion_matrix.TP: 49
- precision_at_k.10: 0.9000
- precision_at_k.50: 0.8200
- precision_at_k.100: 0.6200
- precision_at_k.200: 0.4150
- precision_at_k.500: 0.2200
- recall_at_k.10: 0.0608
- recall_at_k.50: 0.2770
- recall_at_k.100: 0.4189
- recall_at_k.200: 0.5608
- recall_at_k.500: 0.7432

value_date_quarter=2024Q1 - 1) Model Effectiveness - Explanations

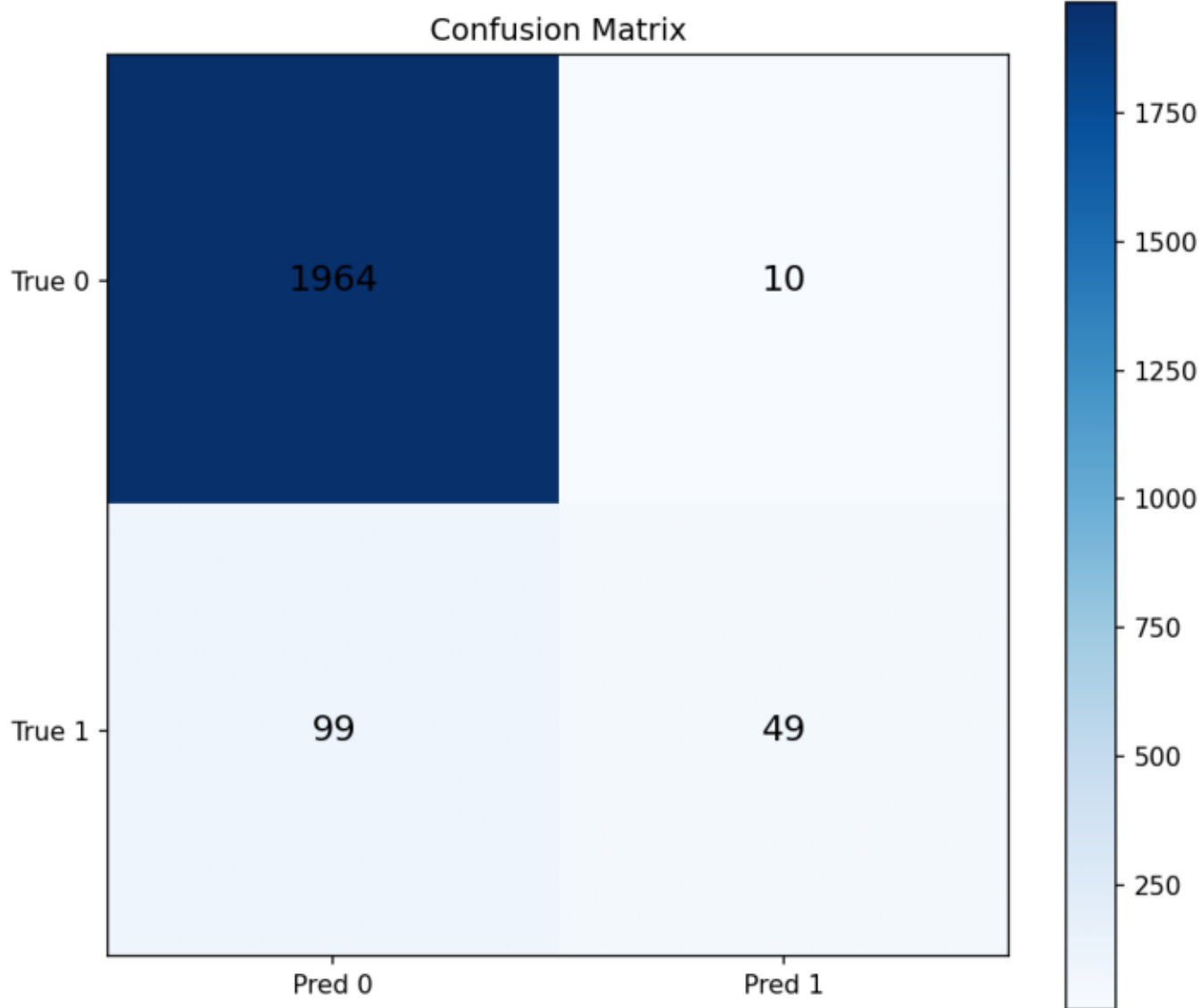
- Positive rate is 6.97%; AUC-ROC is 0.8592 (strong ranking).
- AUC-PR is 0.5323, above the base rate 6.97%, indicating meaningful lift.
- At threshold 0.50, precision=0.8305, recall=0.3311, F1=0.4734 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=49, FP=10, FN=99, TN=1964 (FPR=0.51%, FNR=66.89%).
- KS is 0.5499 at threshold 0.14, indicating very strong separation.
- Top-10: precision=0.9000, recall=0.0608 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.2200, recall=0.7432 shows how much coverage you get with a larger queue.



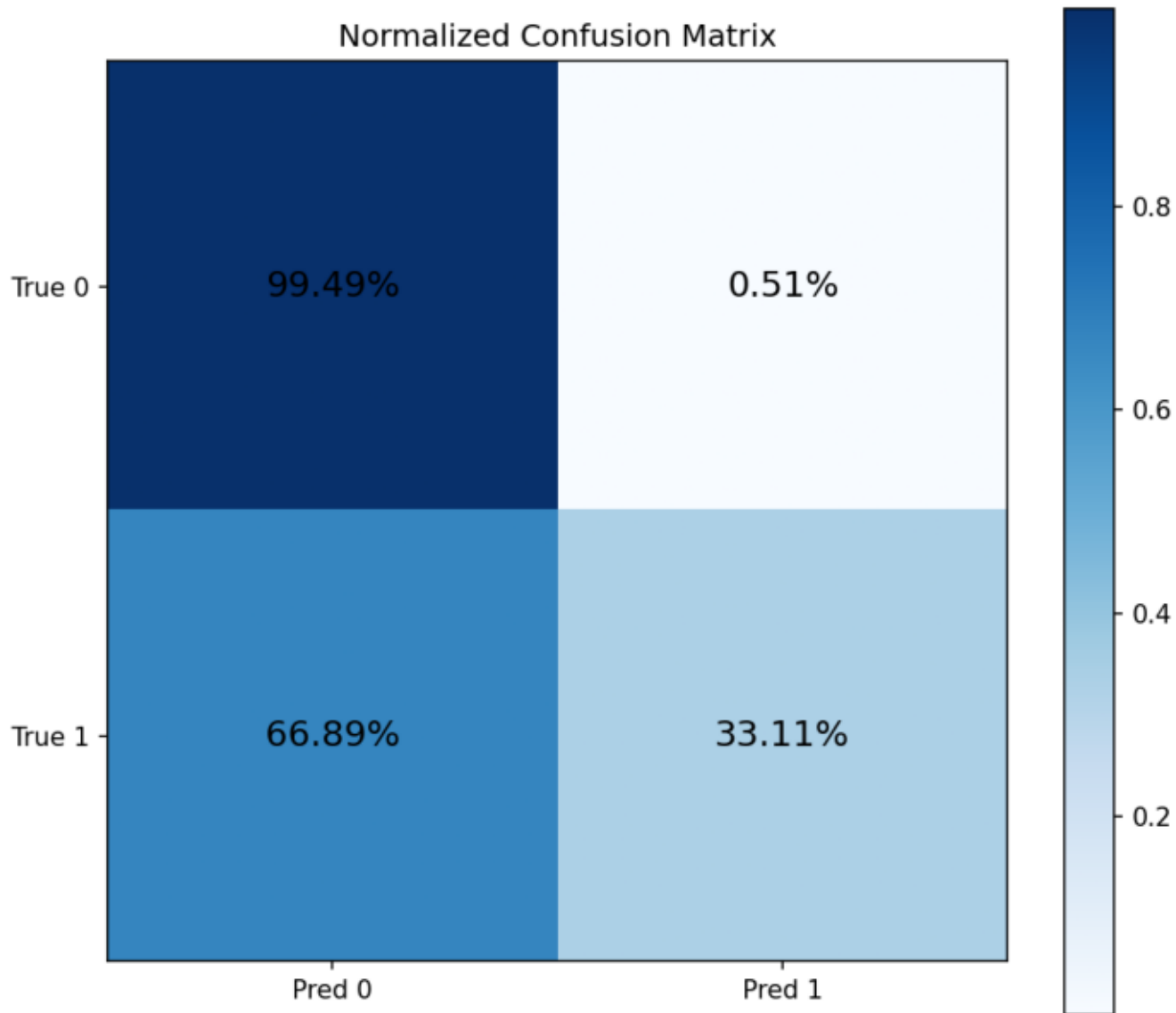
roc_curve: AUC-ROC=0.8592; curve above diagonal indicates strong separation.



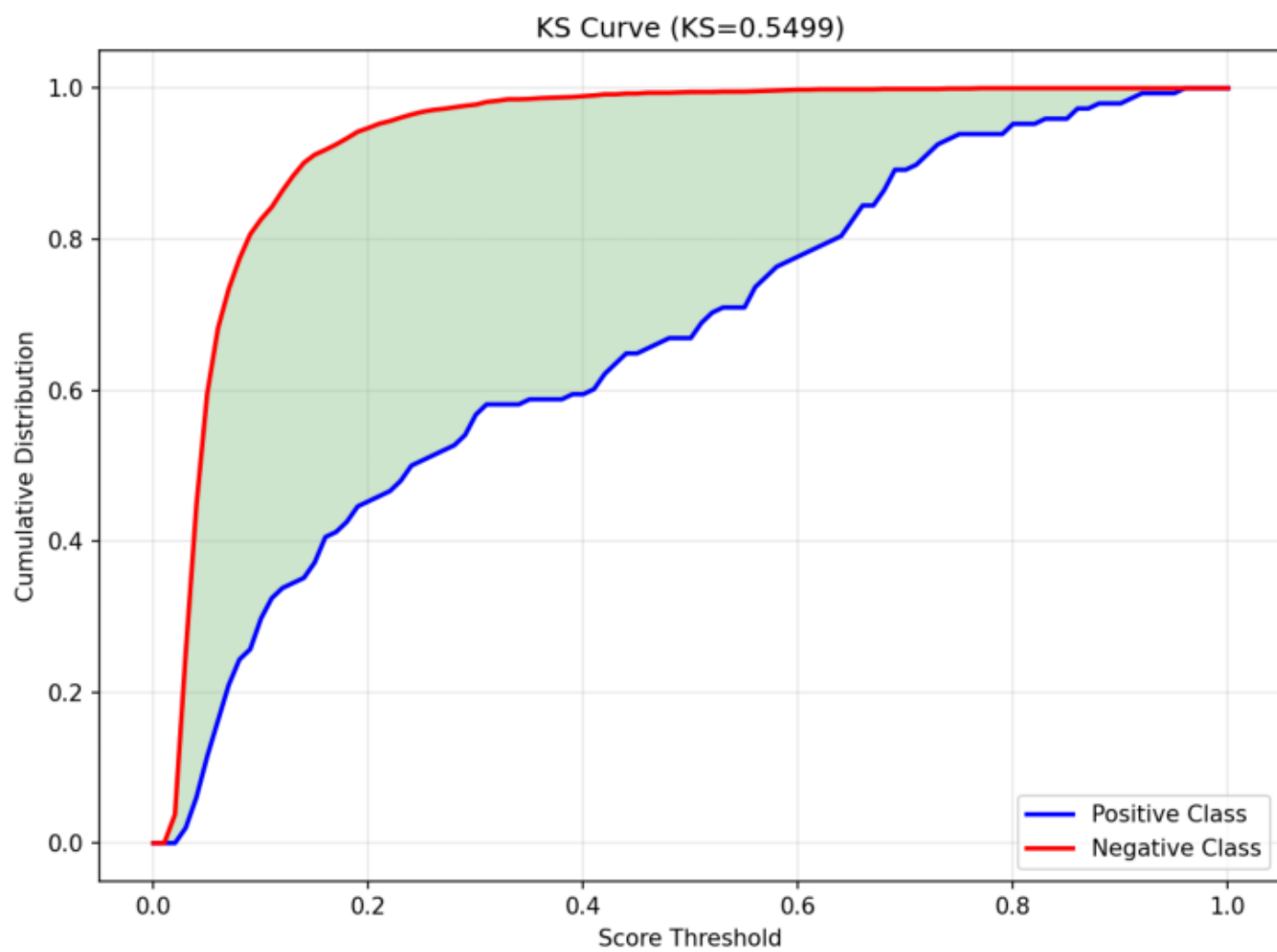
pr_curve: AUC-PR=0.5323 vs baseline 6.97%; higher curve implies better precision at recall.



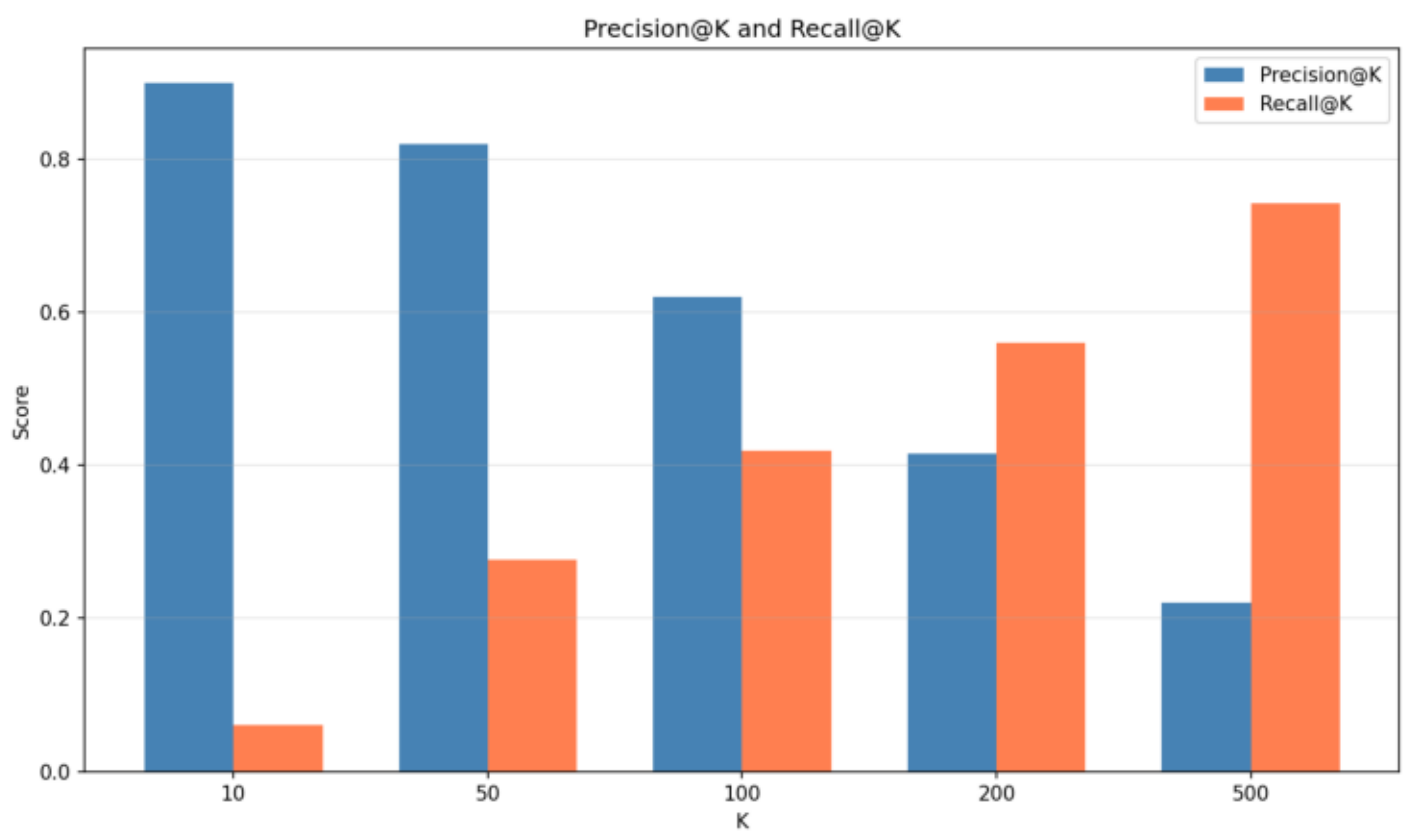
confusion_matrix: At threshold 0.50: TP=49, FP=10, FN=99, TN=1964.



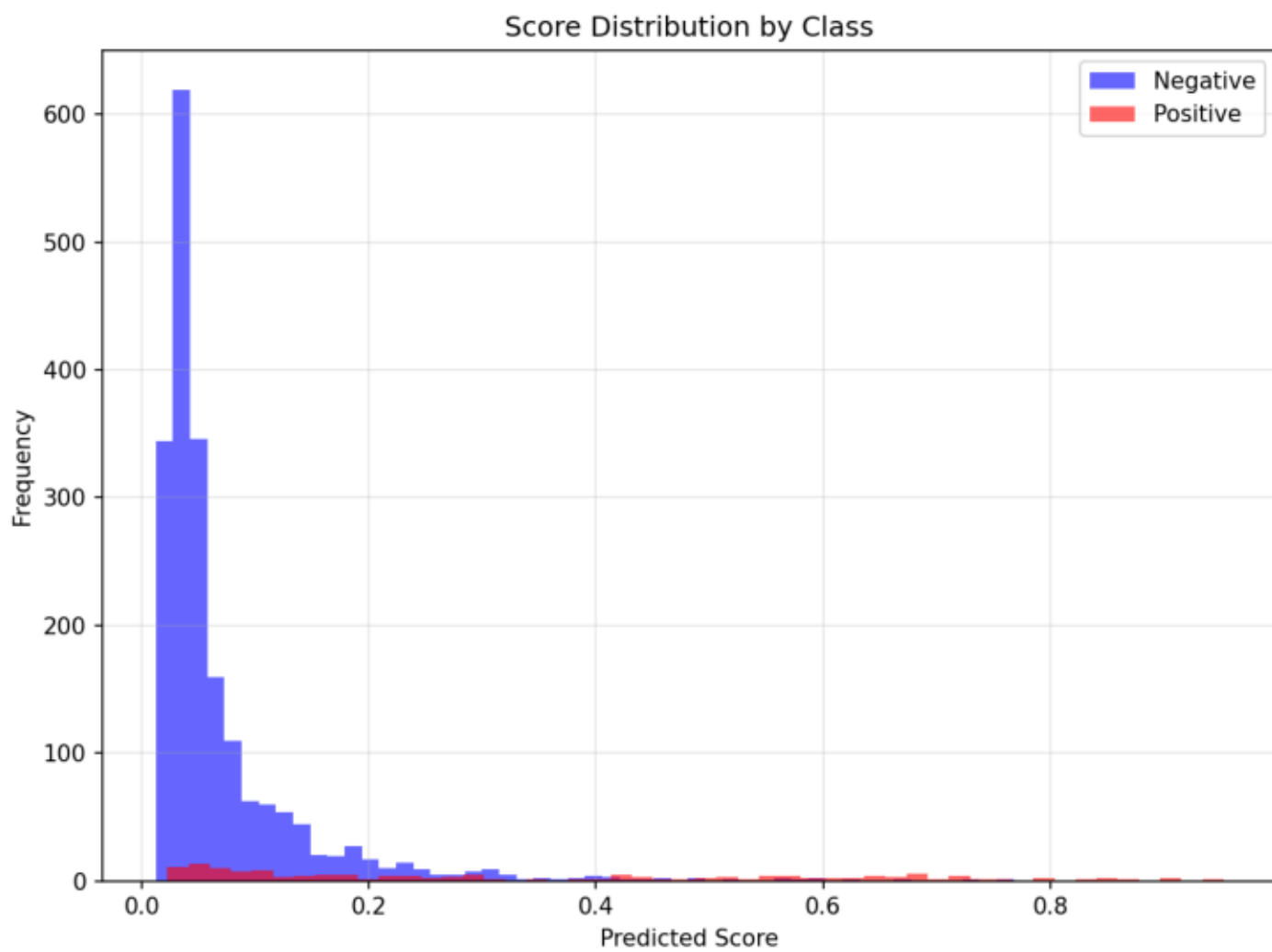
confusion_matrix_norm: Normalized rates: TPR=33.11%, TNR=99.49%.



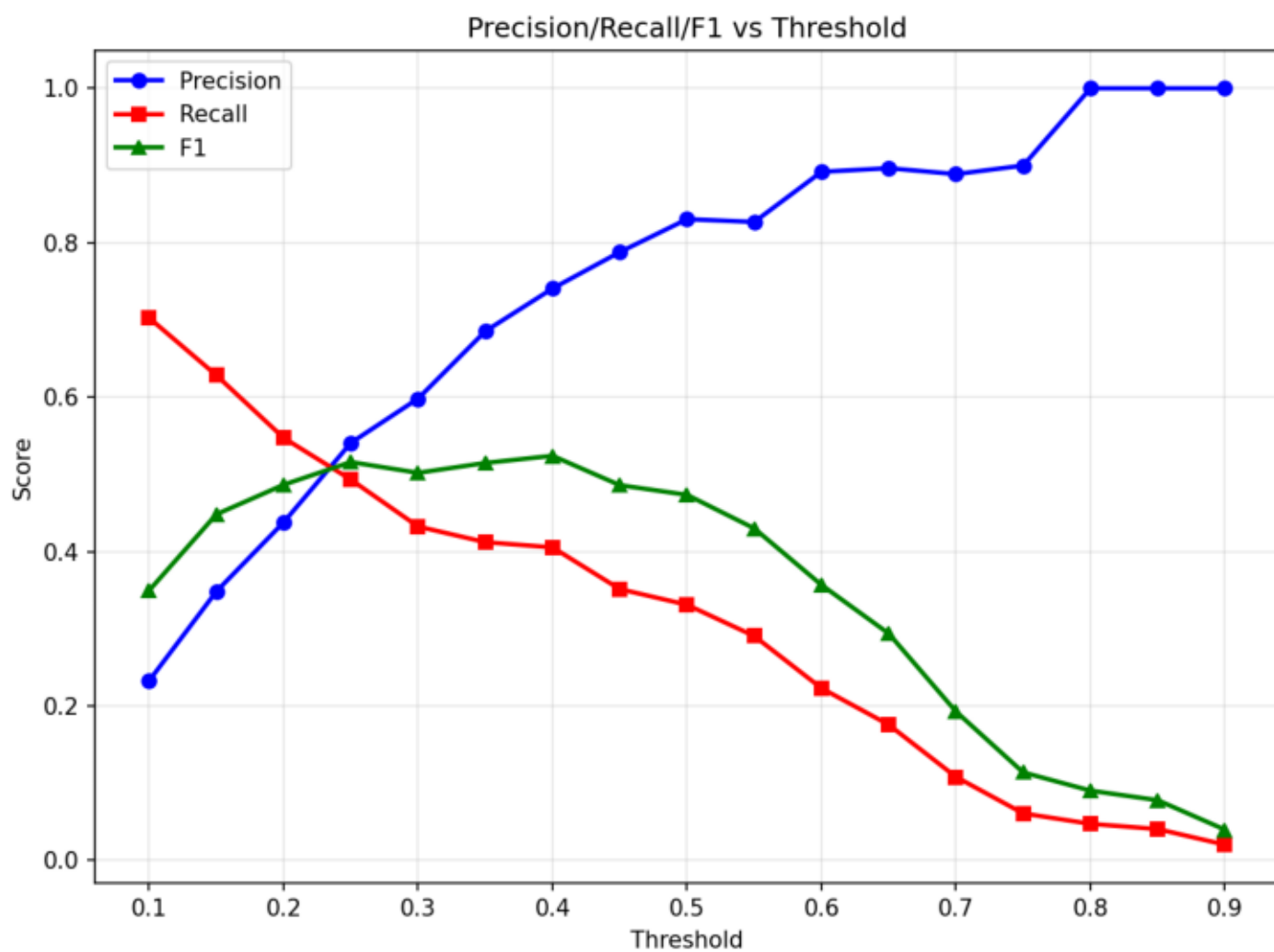
ks_curve: Maximum separation KS=0.5499 at threshold 0.14.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.332, negative=0.069 (larger gap is better).



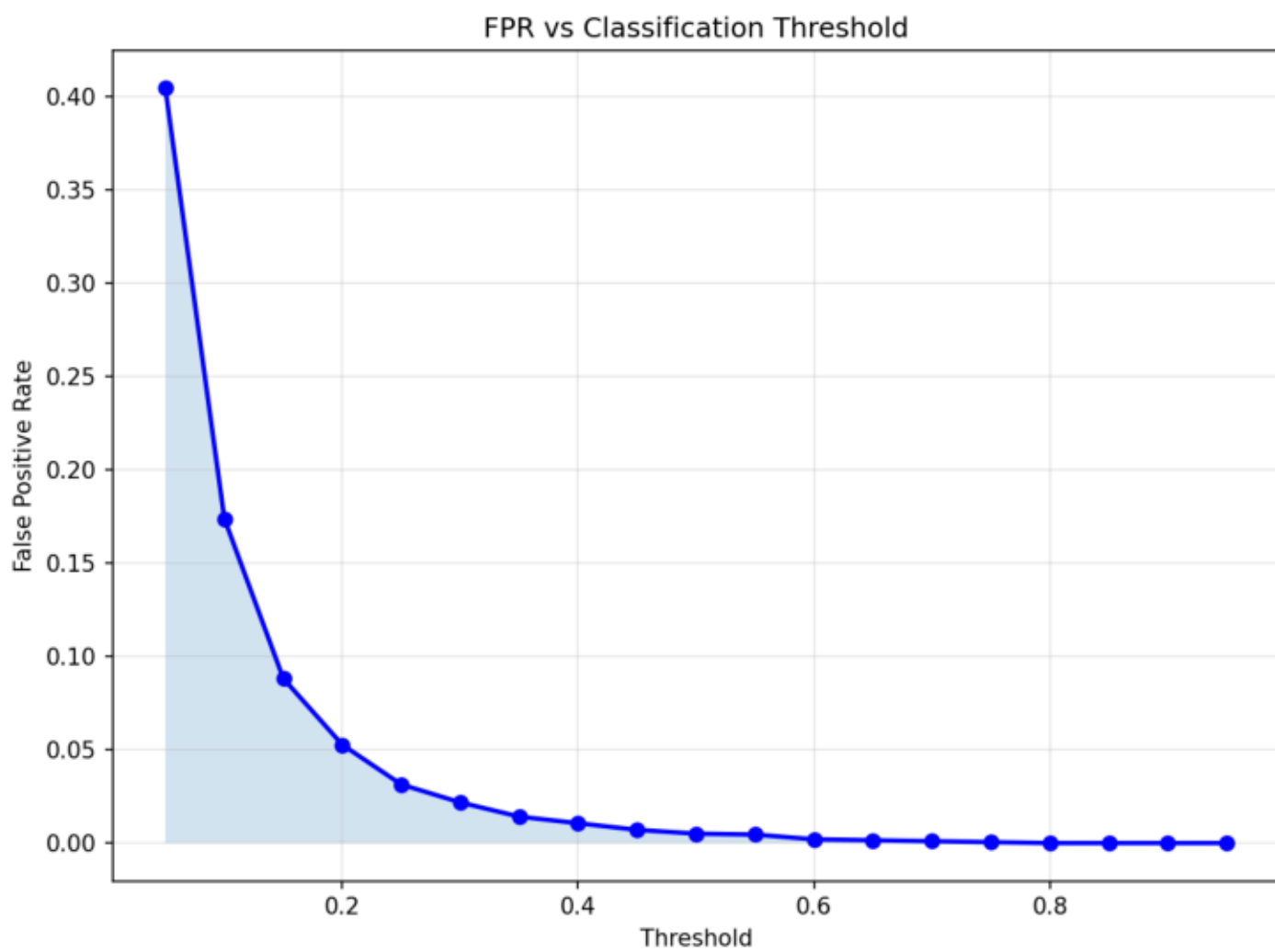
threshold_analysis: Best F1 in tested grid is 0.5240 at threshold 0.40.

value_date_quarter=2024Q1 - 2) Model Efficiency

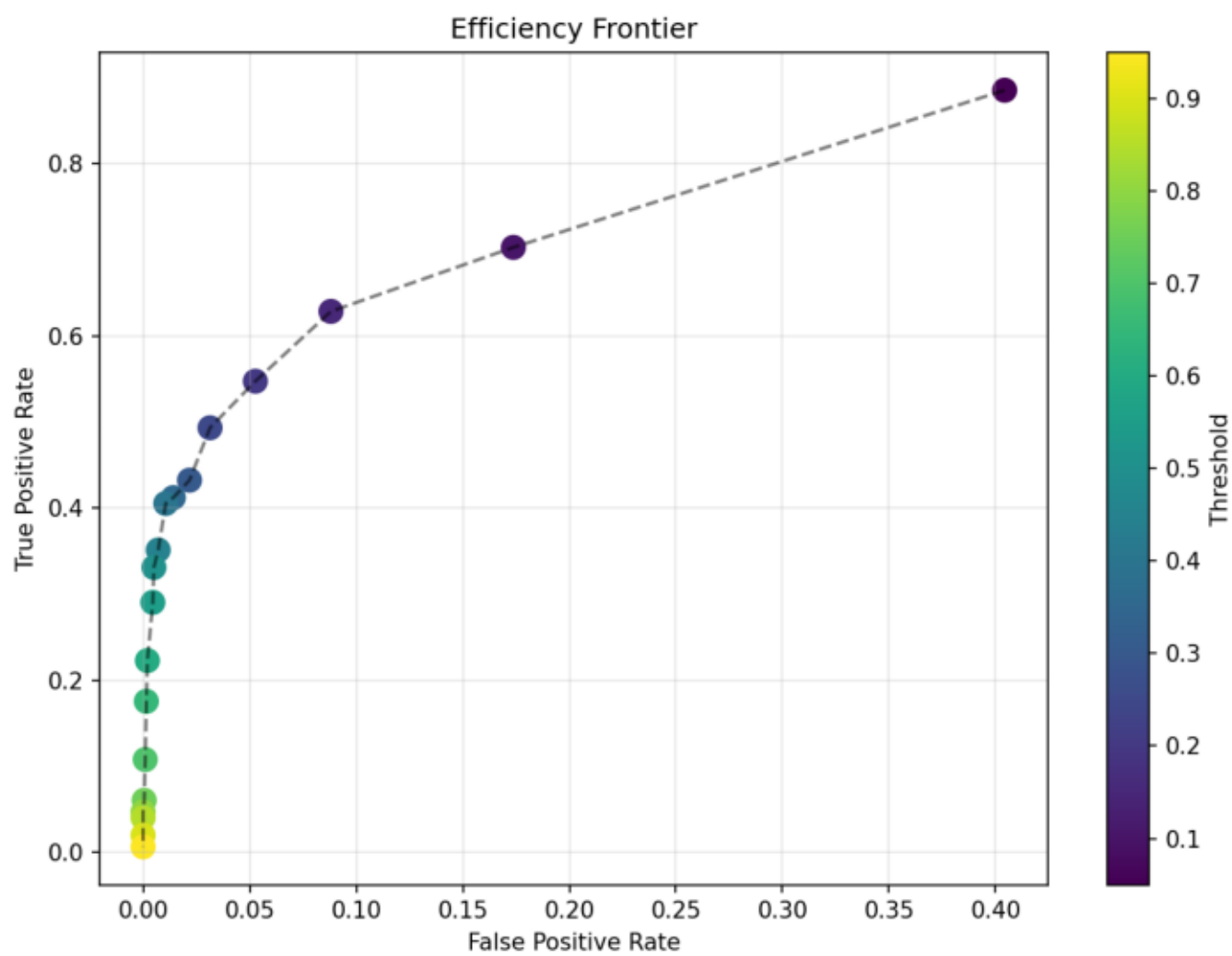
- fpr: 0.0051
- tn: 1964
- fp: 10
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.4043
- fpr_at_thresholds.t_0.10: 0.1738
- fpr_at_thresholds.t_0.15: 0.0881
- fpr_at_thresholds.t_0.20: 0.0527
- fpr_at_thresholds.t_0.25: 0.0314
- fpr_at_thresholds.t_0.30: 0.0218
- fpr_at_thresholds.t_0.35: 0.0142
- fpr_at_thresholds.t_0.40: 0.0106
- fpr_at_thresholds.t_0.45: 0.0071
- fpr_at_thresholds.t_0.50: 0.0051
- fpr_at_thresholds.t_0.55: 0.0046
- fpr_at_thresholds.t_0.60: 0.0020
- fpr_at_thresholds.t_0.65: 0.0015
- fpr_at_thresholds.t_0.70: 0.0010
- fpr_at_thresholds.t_0.75: 0.0005
- fpr_at_thresholds.t_0.80: 0.0000
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q1 - 2) Model Efficiency - Explanations

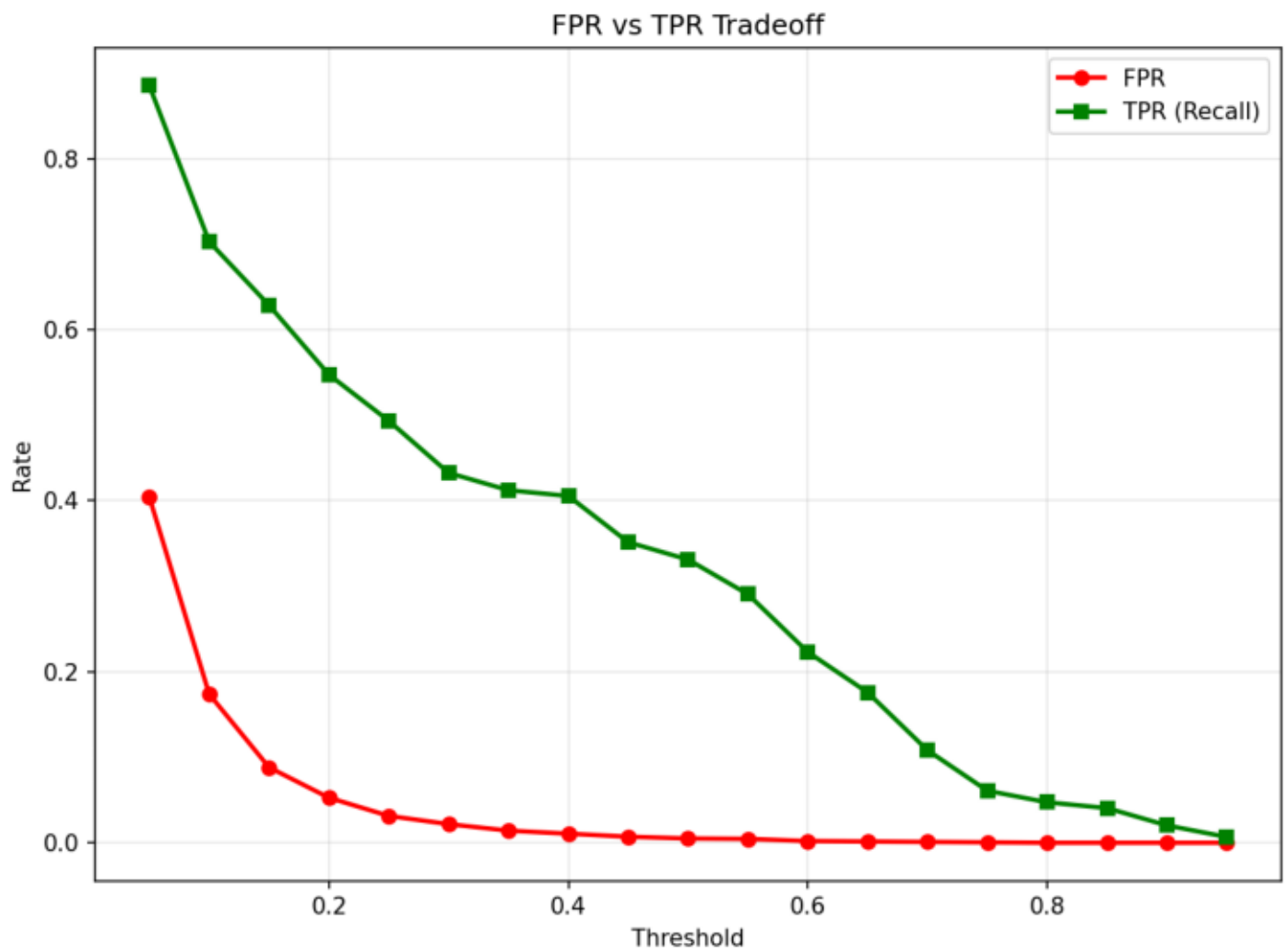
- At threshold 0.50, FPR=0.0051 (low); FP=10 out of 1974 negatives.
- At the same threshold, TPR=0.3311 with TP=49 and FN=99, showing the capture rate of positives.
- A threshold near 0.20 yields FPR $\approx$ 0.0527 with TPR $\approx$ 0.5473 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

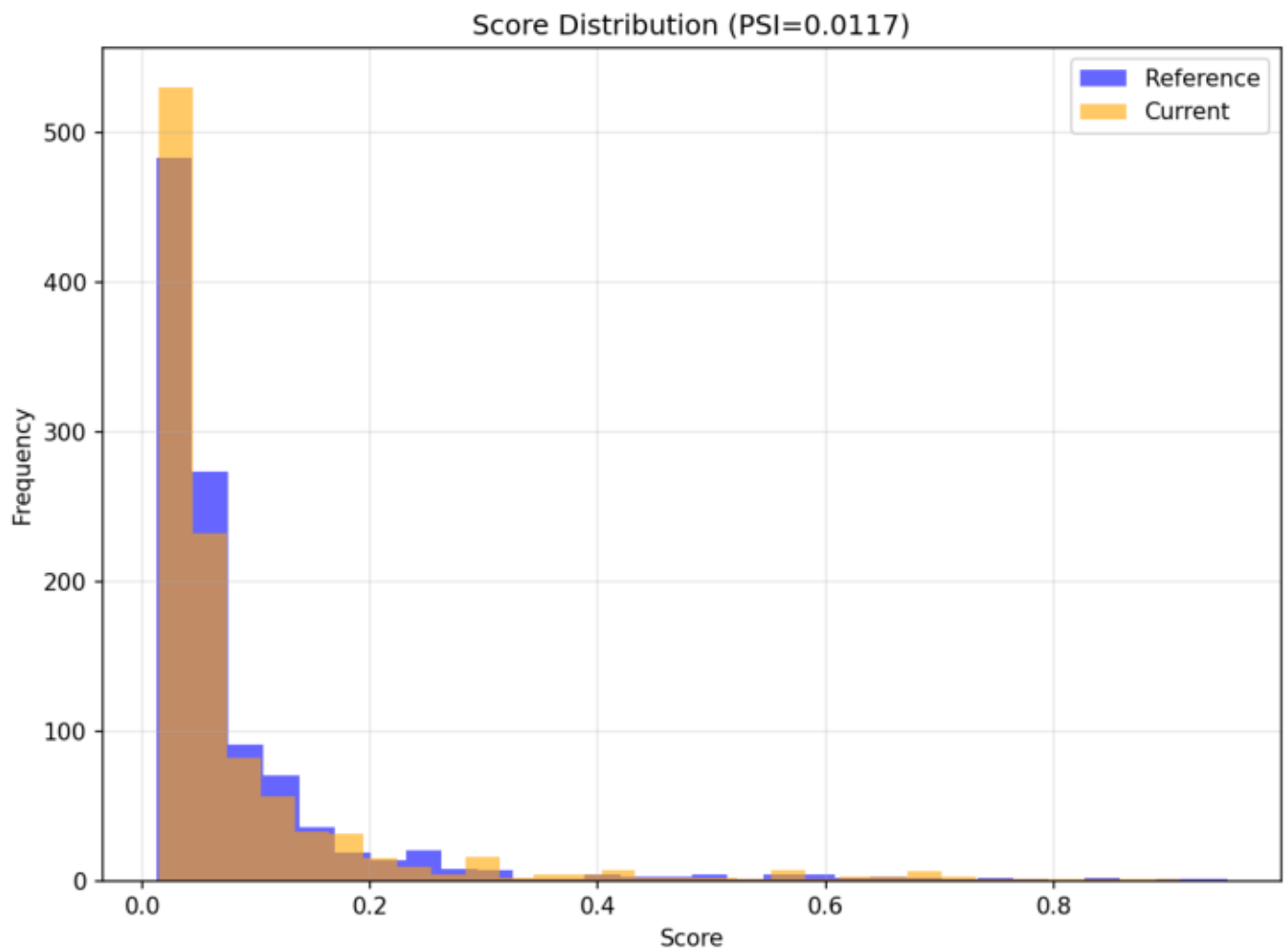


value_date_quarter=2024Q1 - 3) Model Stability

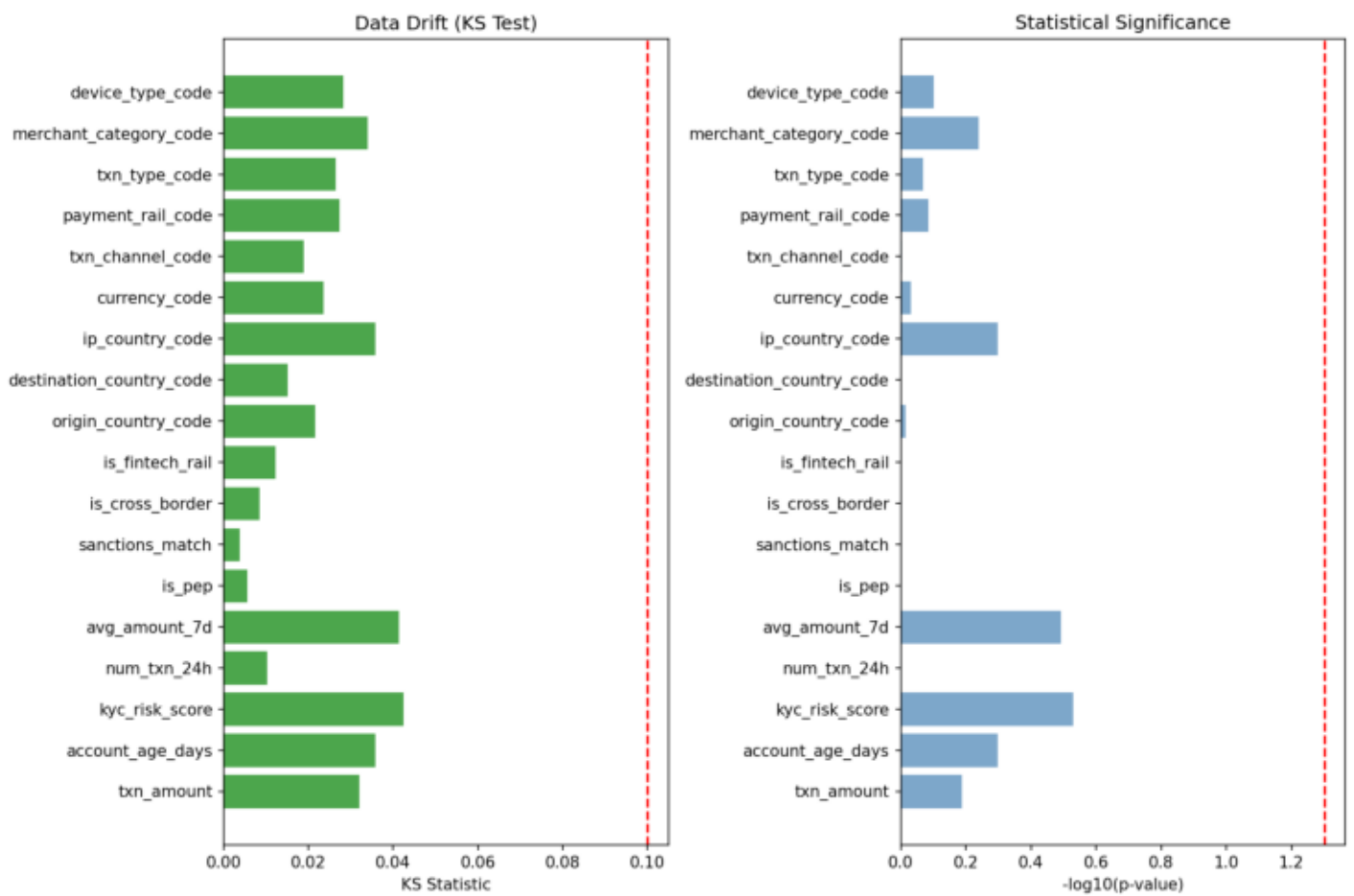
- psi: 0.0117
- cv_auc_roc_mean: 0.8587
- cv_auc_roc_std: 0.0369
- cv_auc_pr_mean: 0.5311
- cv_auc_pr_std: 0.1497
- bootstrap_auc_roc_mean: 0.8605
- bootstrap_auc_roc_ci_lower: 0.8226
- bootstrap_auc_roc_ci_upper: 0.9054
- concept_drift_detected: False
- concept_drift_score: 0.0441

value_date_quarter=2024Q1 - 3) Model Stability - Explanations

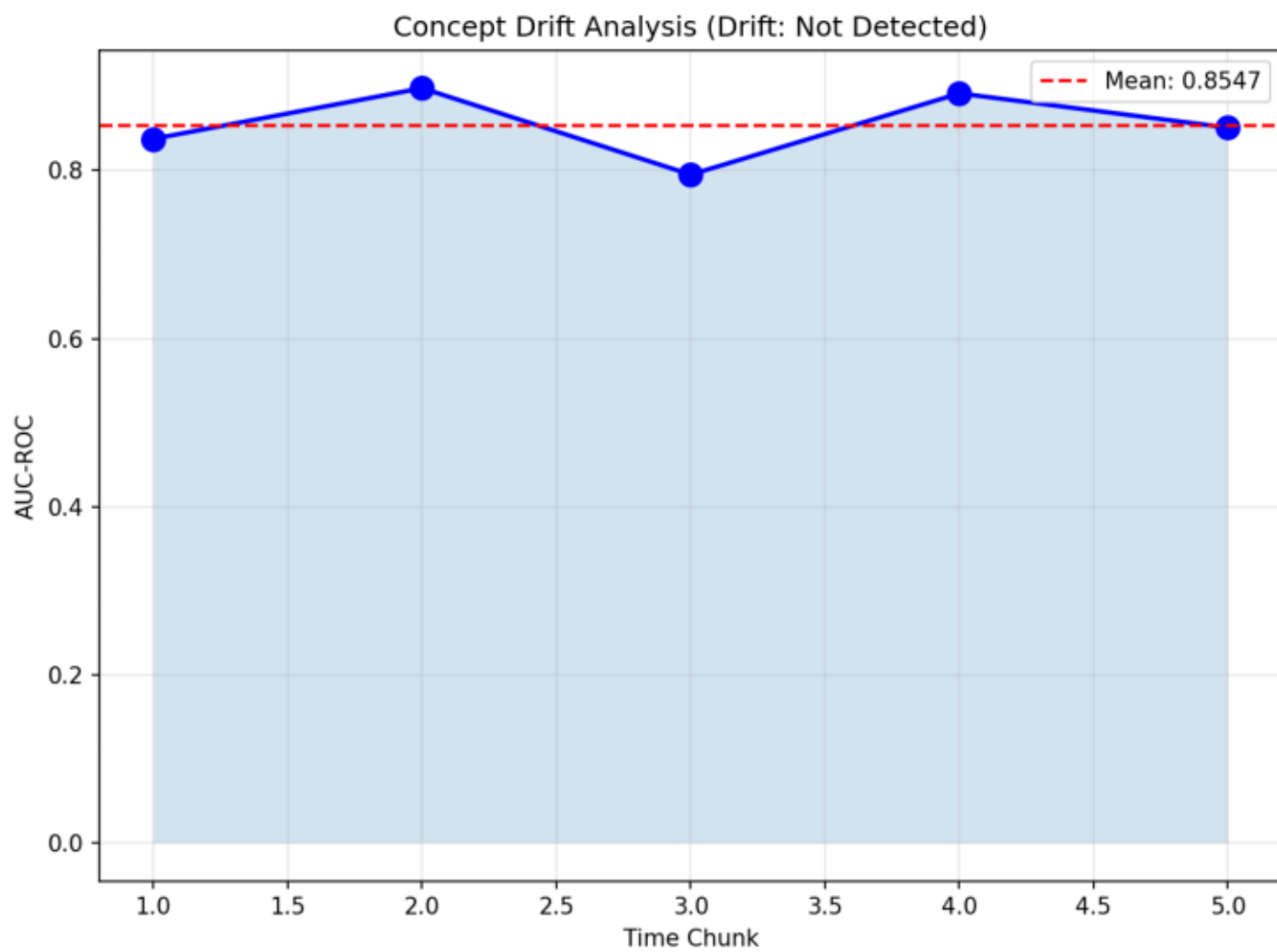
- PSI is 0.0117, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8587 ± 0.0369 (moderate across folds).
- CV AUC-PR is 0.5311 ± 0.1497 (variable across folds).
- Bootstrap AUC-ROC mean is 0.8605 with 95% CI [0.8226, 0.9054] (width 0.0827).
- No concept drift detected (score 0.0441), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['kyc_risk_score', 'avg_amount_7d', 'account_age_days'].



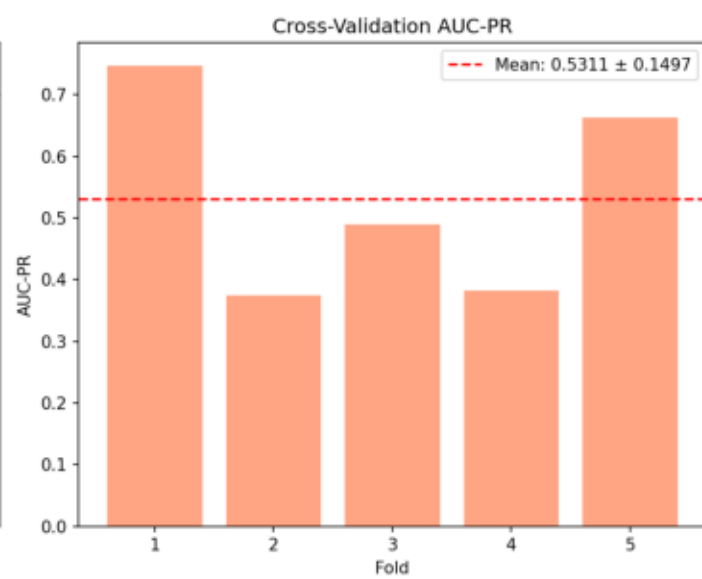
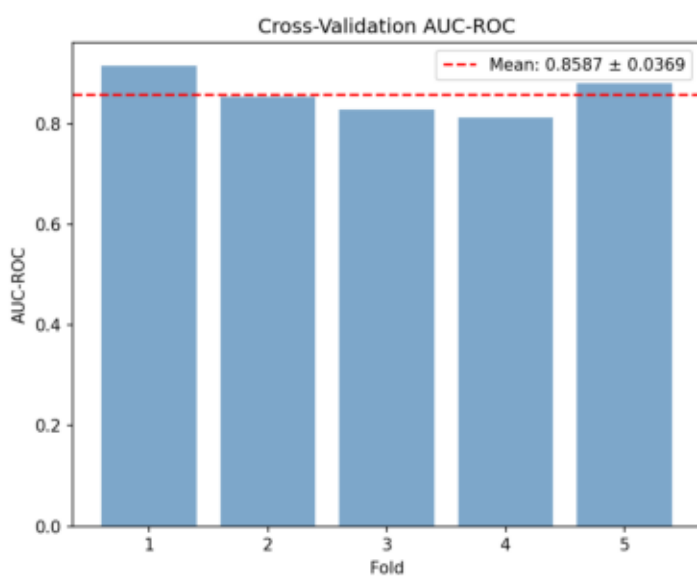
psi_distribution: Score distributions overlap consistent with PSI=0.0117.



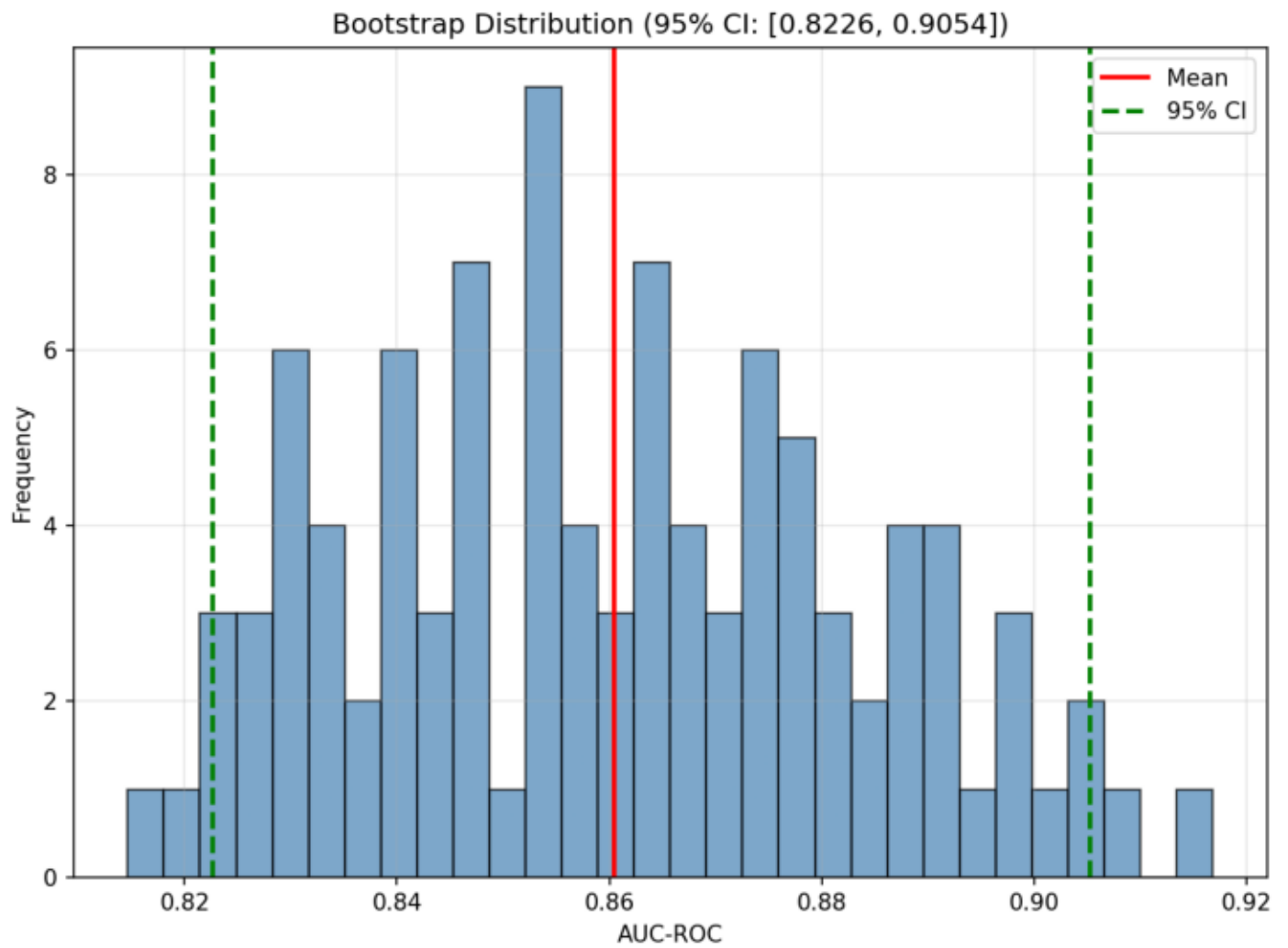
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



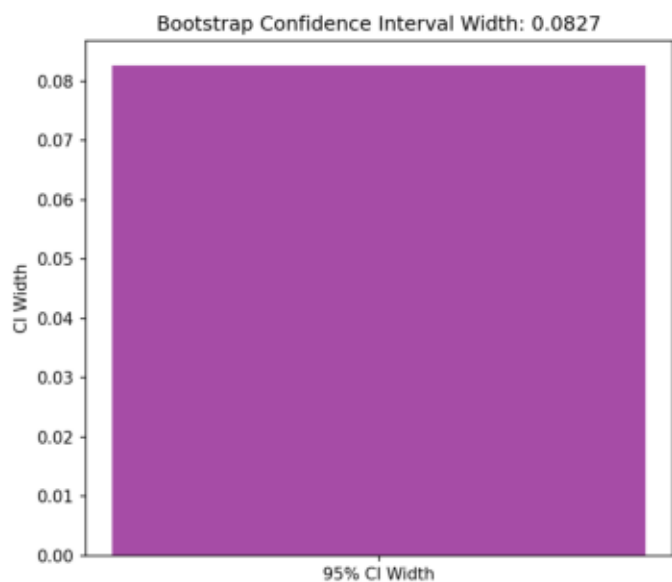
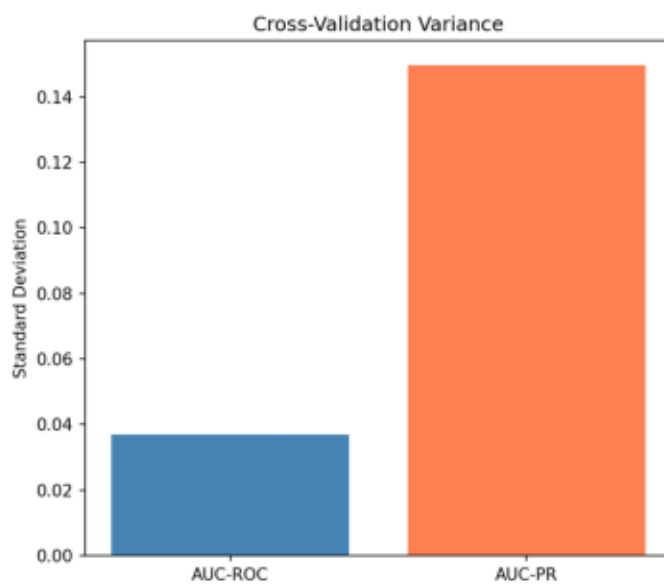
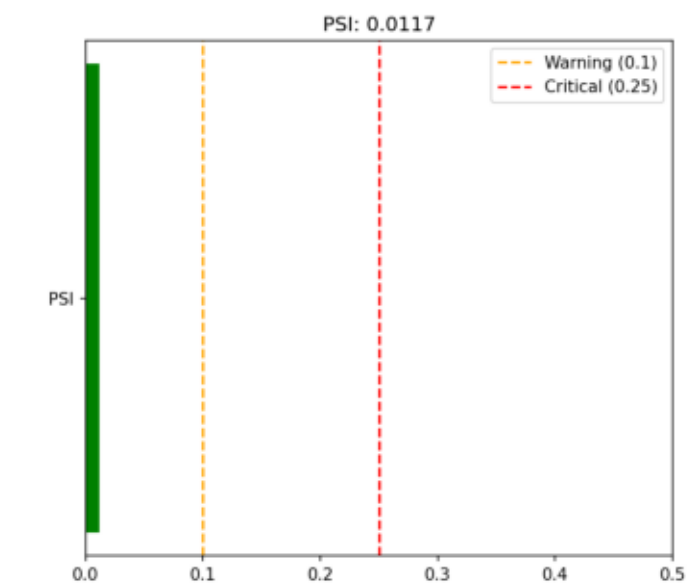
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8587 and PR 0.5311.



bootstrap_distribution: Bootstrap CI [0.8226, 0.9054] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0117 (OK)

CV AUC-ROC: 0.8587 ± 0.0369

CV AUC-PR: 0.5311 ± 0.1497

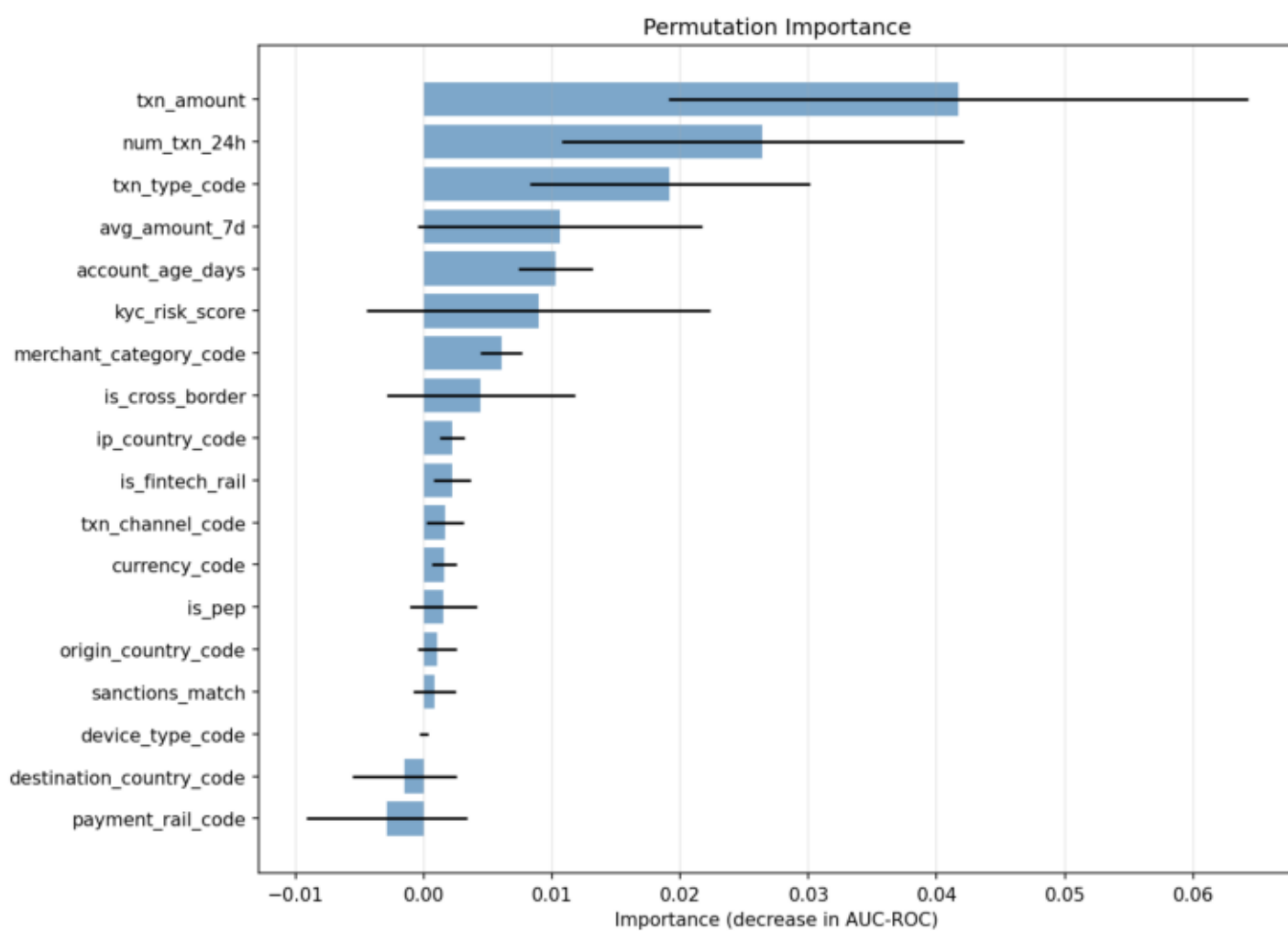
Bootstrap 95% CI: [0.8226, 0.9054]

value_date_quarter=2024Q1 - 4) Model Interpretability

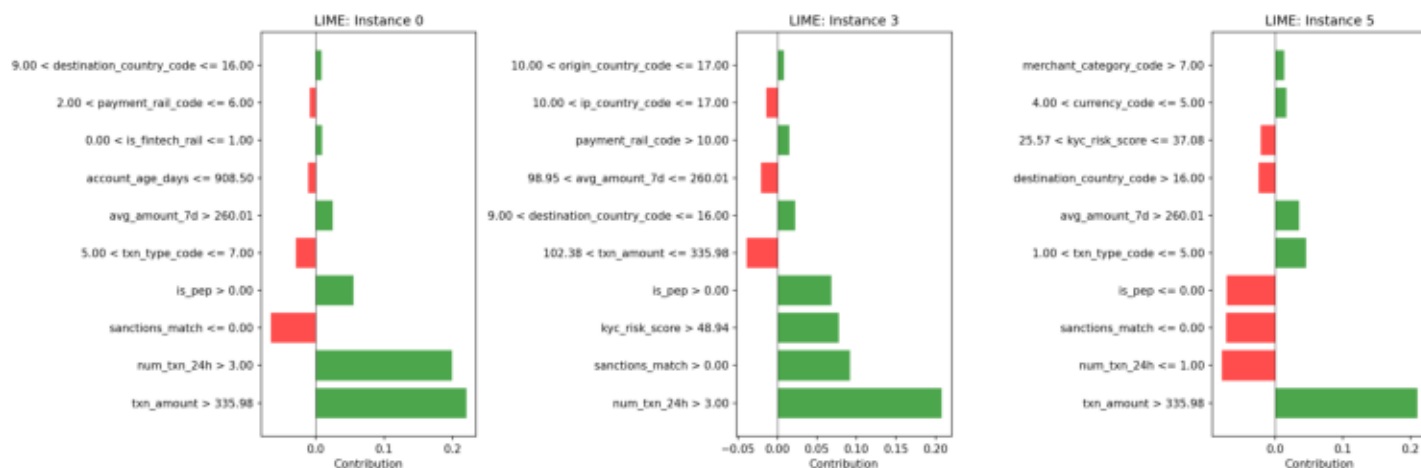
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'lime']
- shap_status: unavailable
- shap_reason: No module named 'shap'
- perm_top_features: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'avg_amount_7d', 'account_age_days']...
- lime_instances: 3
- pdp_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q1 - 4) Model Interpretability - Explanations

- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'avg_amount_7d', 'account_age_days'].
- SHAP plots not generated because SHAP is unavailable (No module named 'shap').
- LIME generated local explanations for 3 instances; top contributors for instance_0: ['txn_amount > 335.98', 'num_txn_24h > 3.00', 'sanctions_match <= 0.00', 'is_pep > 0.00', '5.00 < txn_type_code <= 7.00'].
- PDP plots show average effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'destination_country_code'].
- ICE plots show individual-level effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score'].

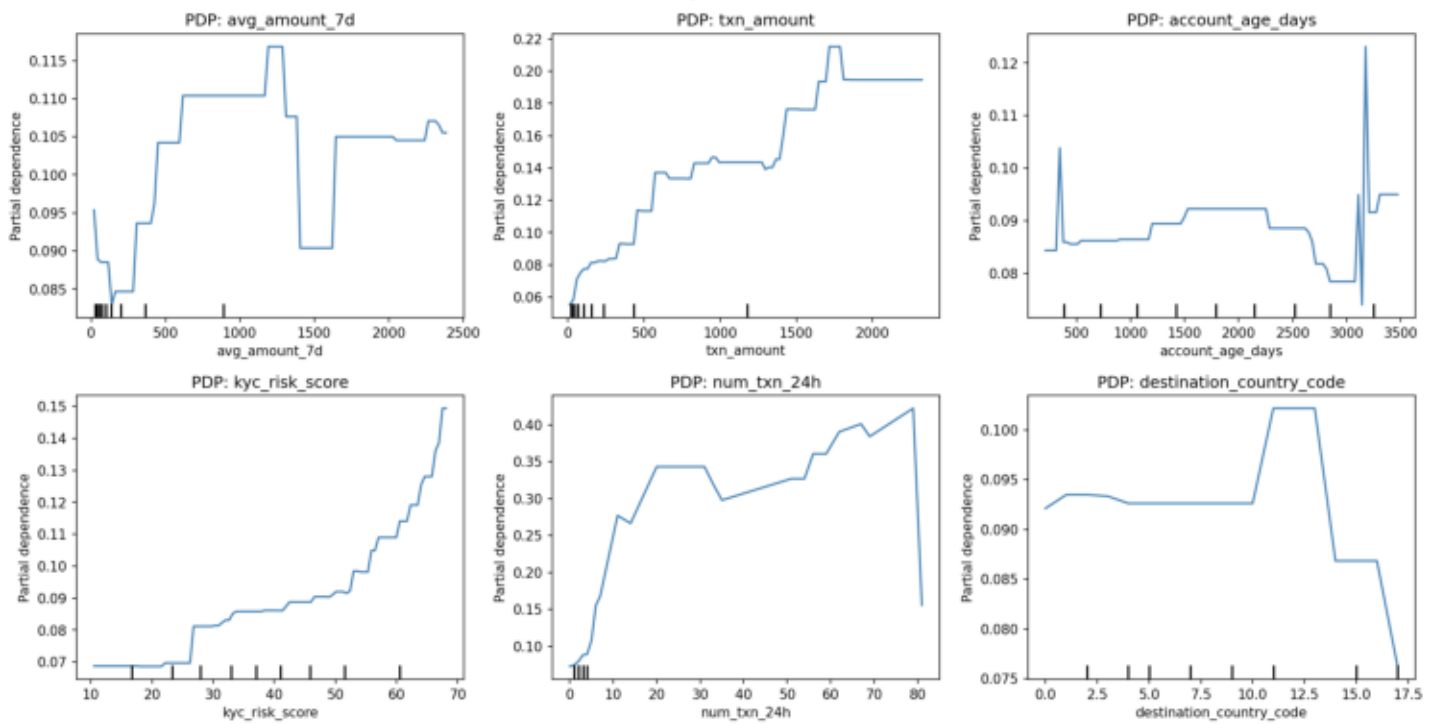


permutation_importance: Bars show how much AUC drops when each feature is permuted.



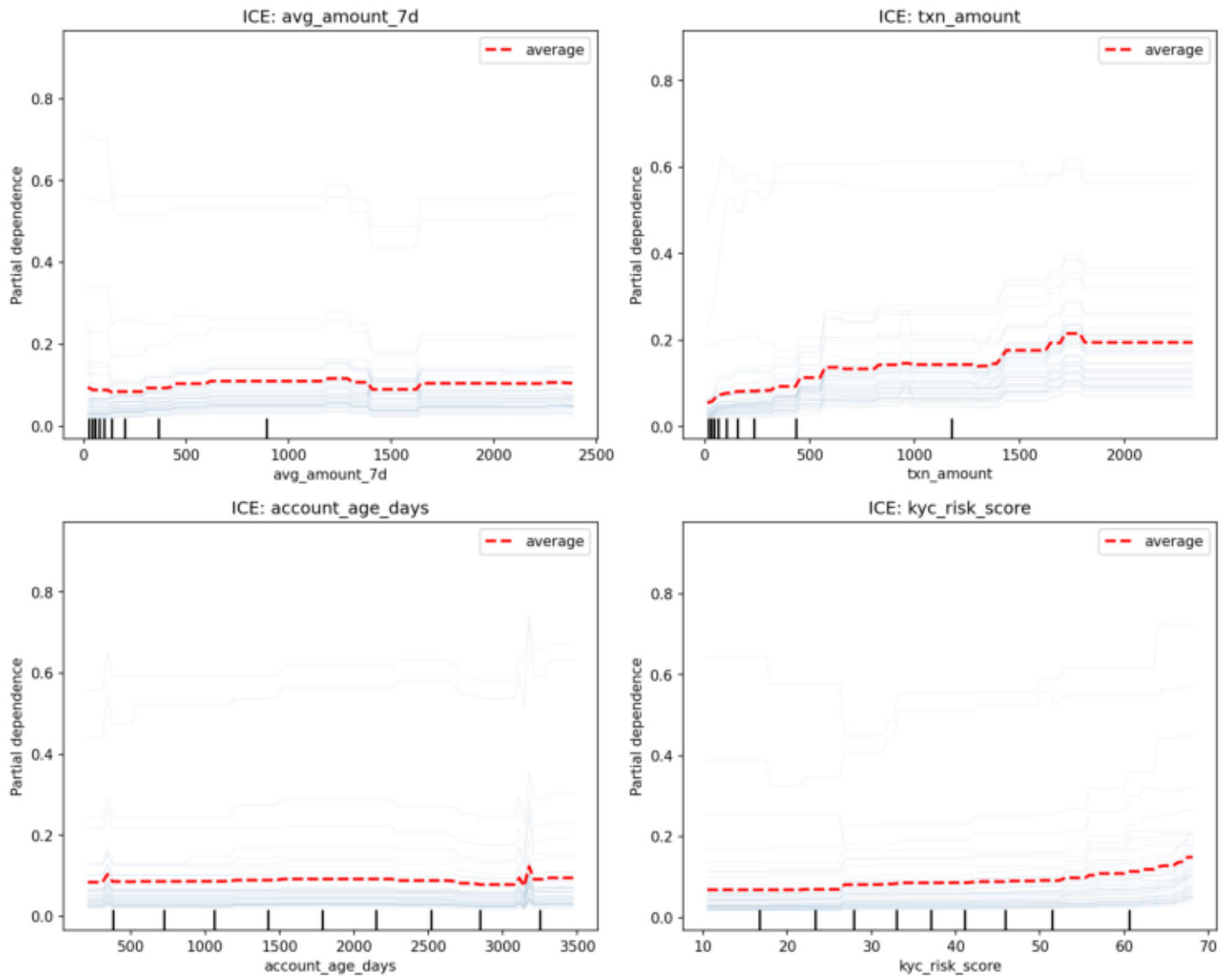
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



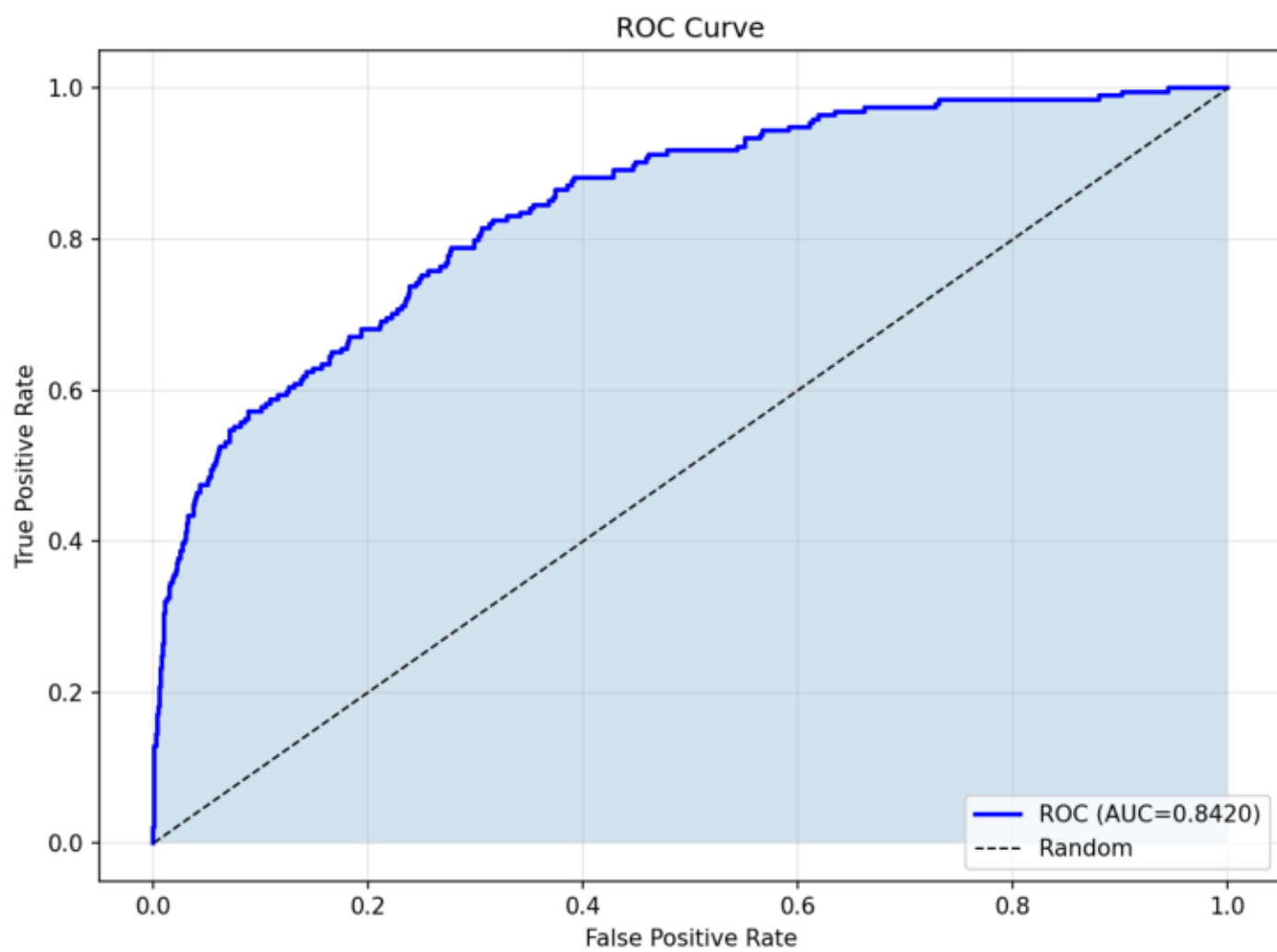
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q2 - 1) Model Effectiveness

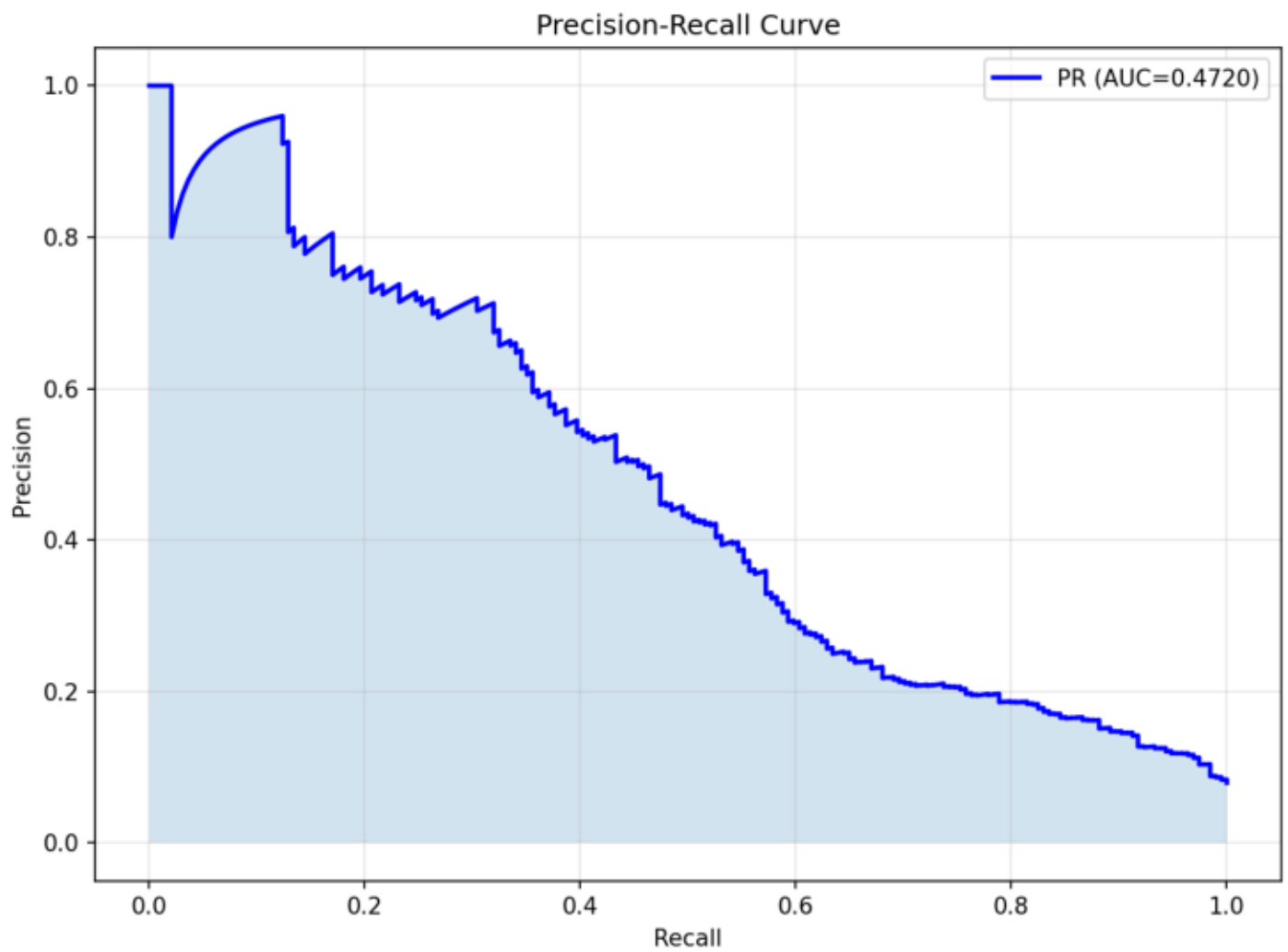
- auc_roc: 0.8420
- auc_pr: 0.4720
- precision: 0.7188
- recall: 0.2371
- f1: 0.3566
- ks_statistic: 0.5052
- ks_threshold: 0.0700
- confusion_matrix.TN: 2236
- confusion_matrix.FP: 18
- confusion_matrix.FN: 148
- confusion_matrix.TP: 46
- precision_at_k.10: 0.9000
- precision_at_k.50: 0.7600
- precision_at_k.100: 0.6600
- precision_at_k.200: 0.4600
- precision_at_k.500: 0.2500
- recall_at_k.10: 0.0464
- recall_at_k.50: 0.1959
- recall_at_k.100: 0.3402
- recall_at_k.200: 0.4742
- recall_at_k.500: 0.6443

value_date_quarter=2024Q2 - 1) Model Effectiveness - Explanations

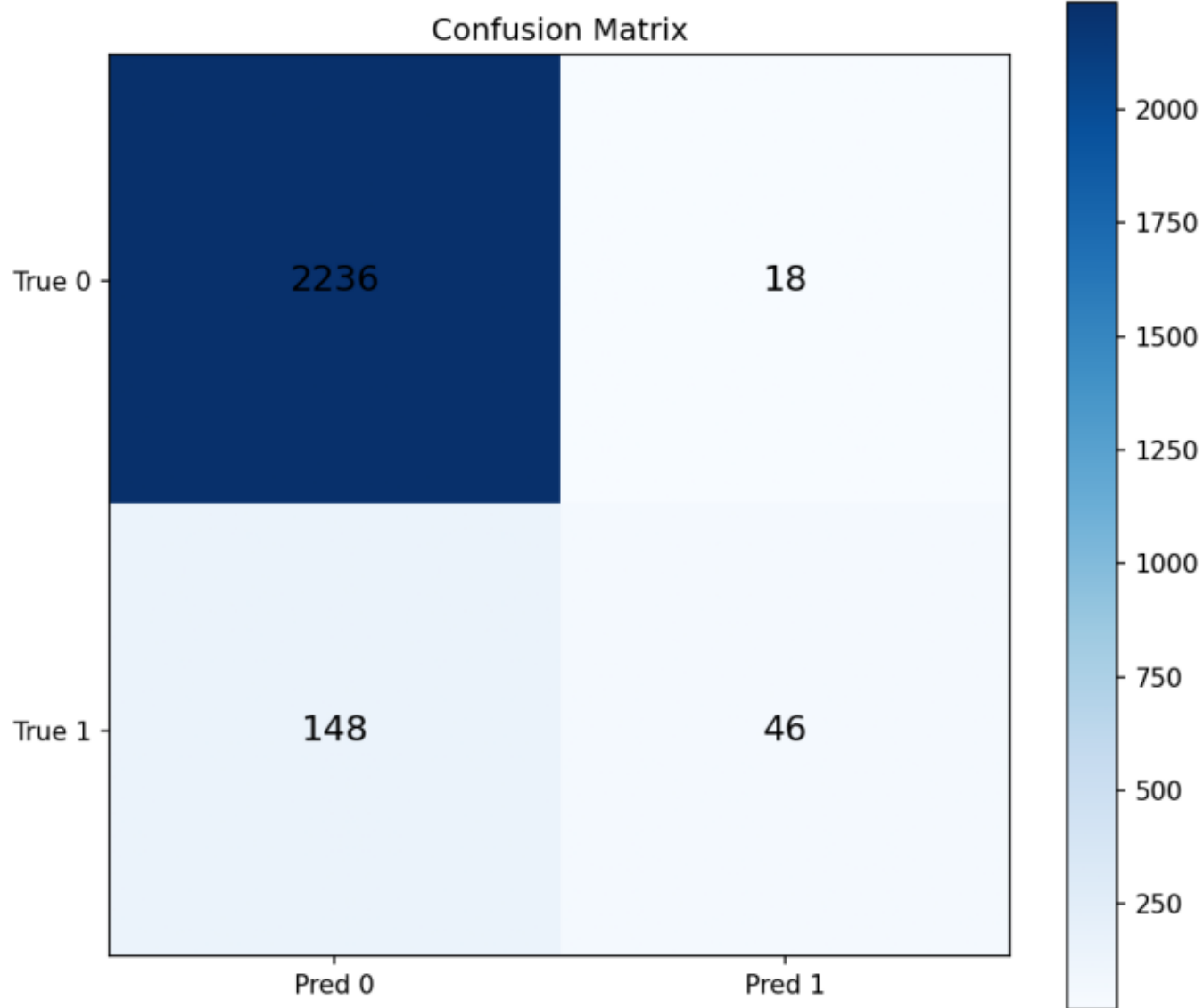
- Positive rate is 7.92%; AUC-ROC is 0.8420 (strong ranking).
- AUC-PR is 0.4720, above the base rate 7.92%, indicating meaningful lift.
- At threshold 0.50, precision=0.7188, recall=0.2371, F1=0.3566 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=46, FP=18, FN=148, TN=2236 (FPR=0.80%, FNR=76.29%).
- KS is 0.5052 at threshold 0.07, indicating very strong separation.
- Top-10: precision=0.9000, recall=0.0464 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.2500, recall=0.6443 shows how much coverage you get with a larger queue.



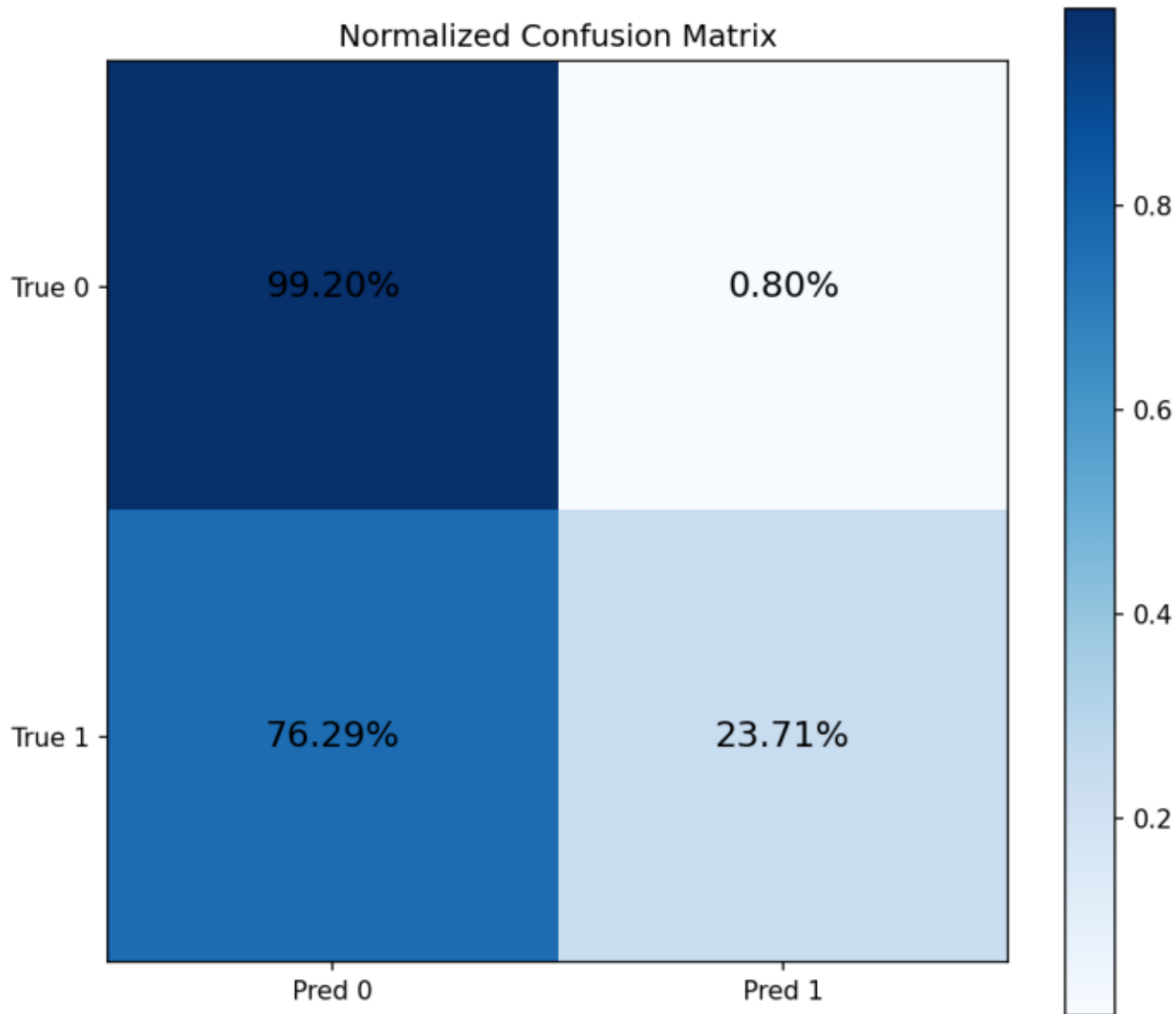
roc_curve: AUC-ROC=0.8420; curve above diagonal indicates strong separation.



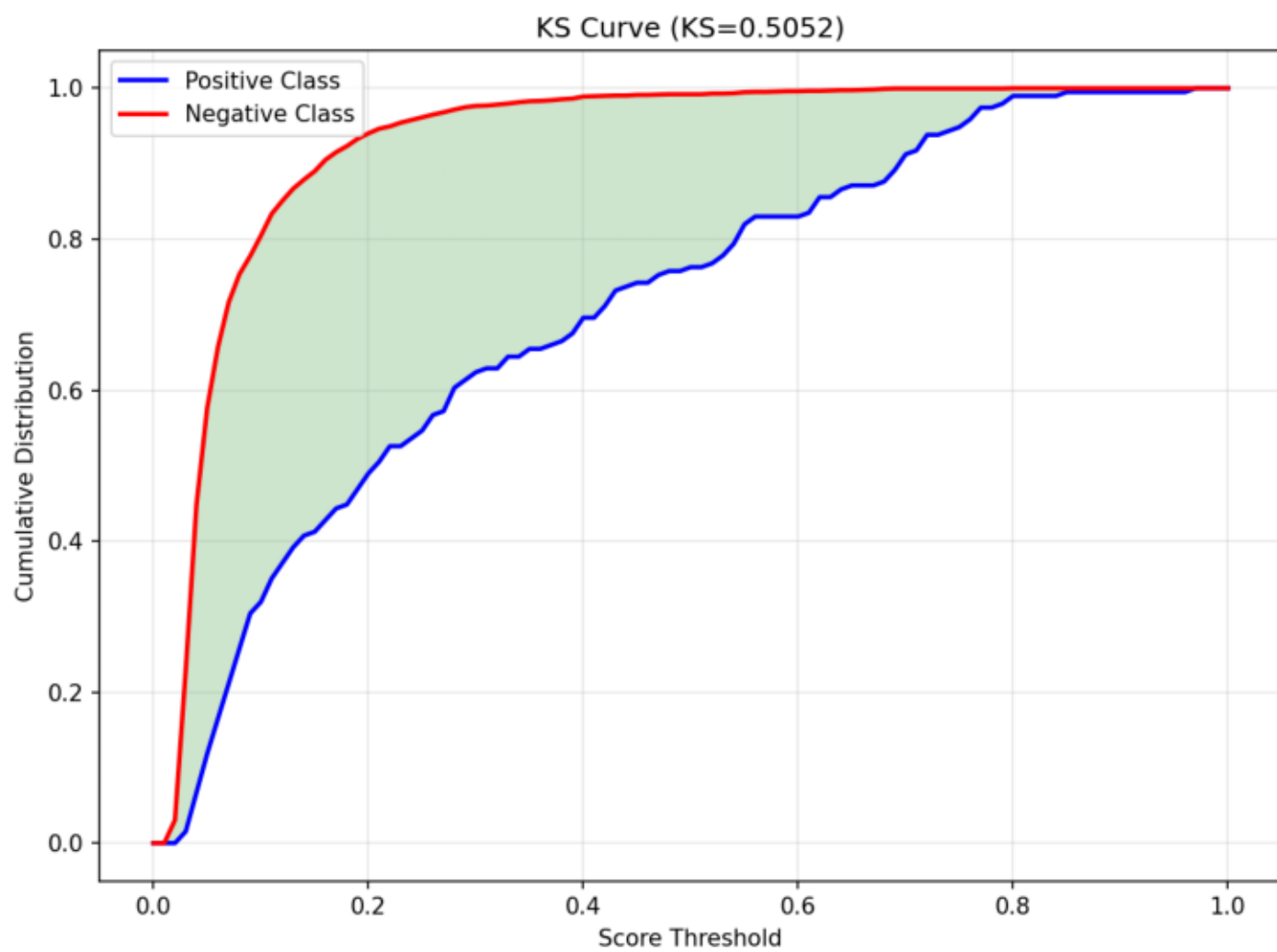
pr_curve: AUC-PR=0.4720 vs baseline 7.92%; higher curve implies better precision at recall.



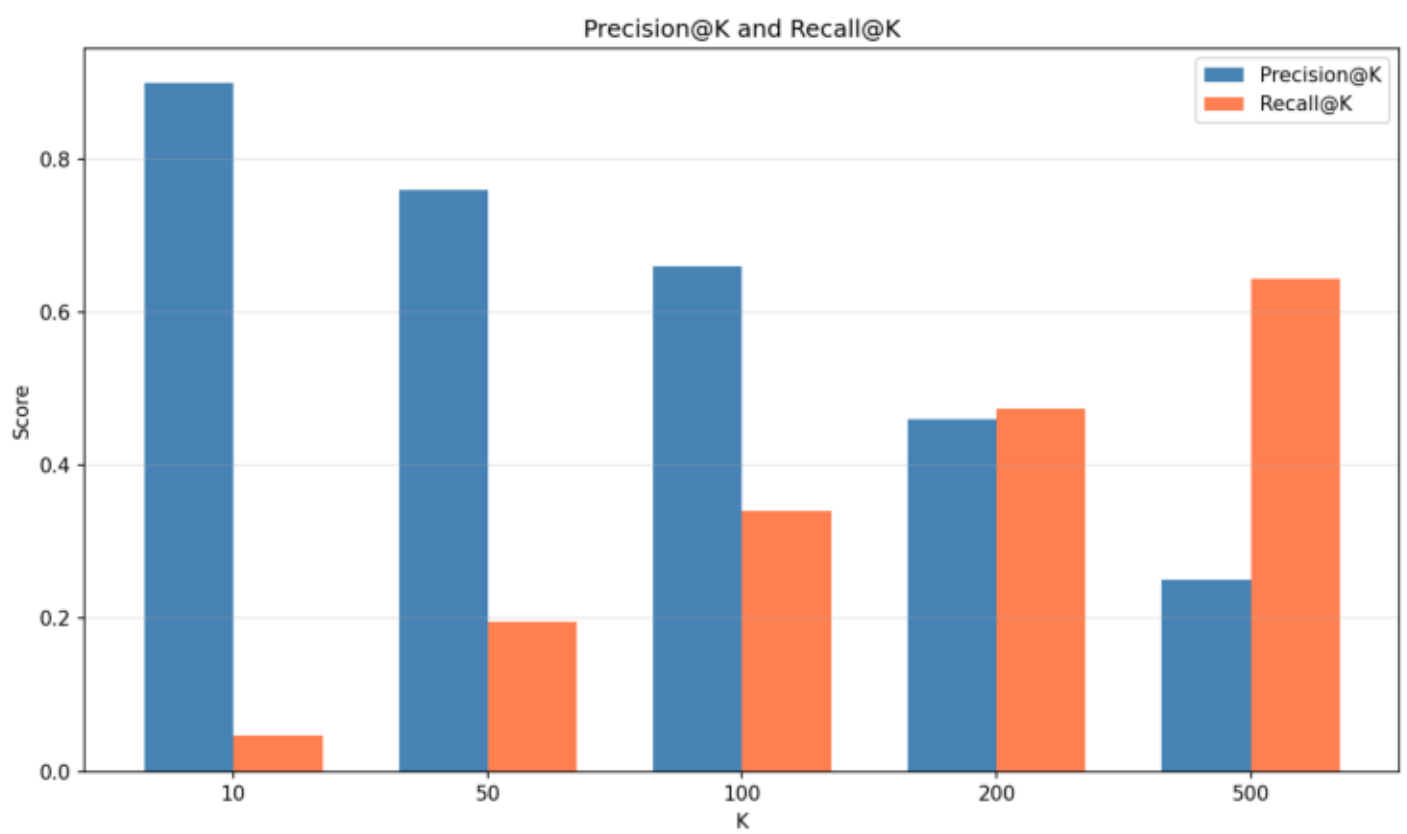
confusion_matrix: At threshold 0.50: TP=46, FP=18, FN=148, TN=2236.



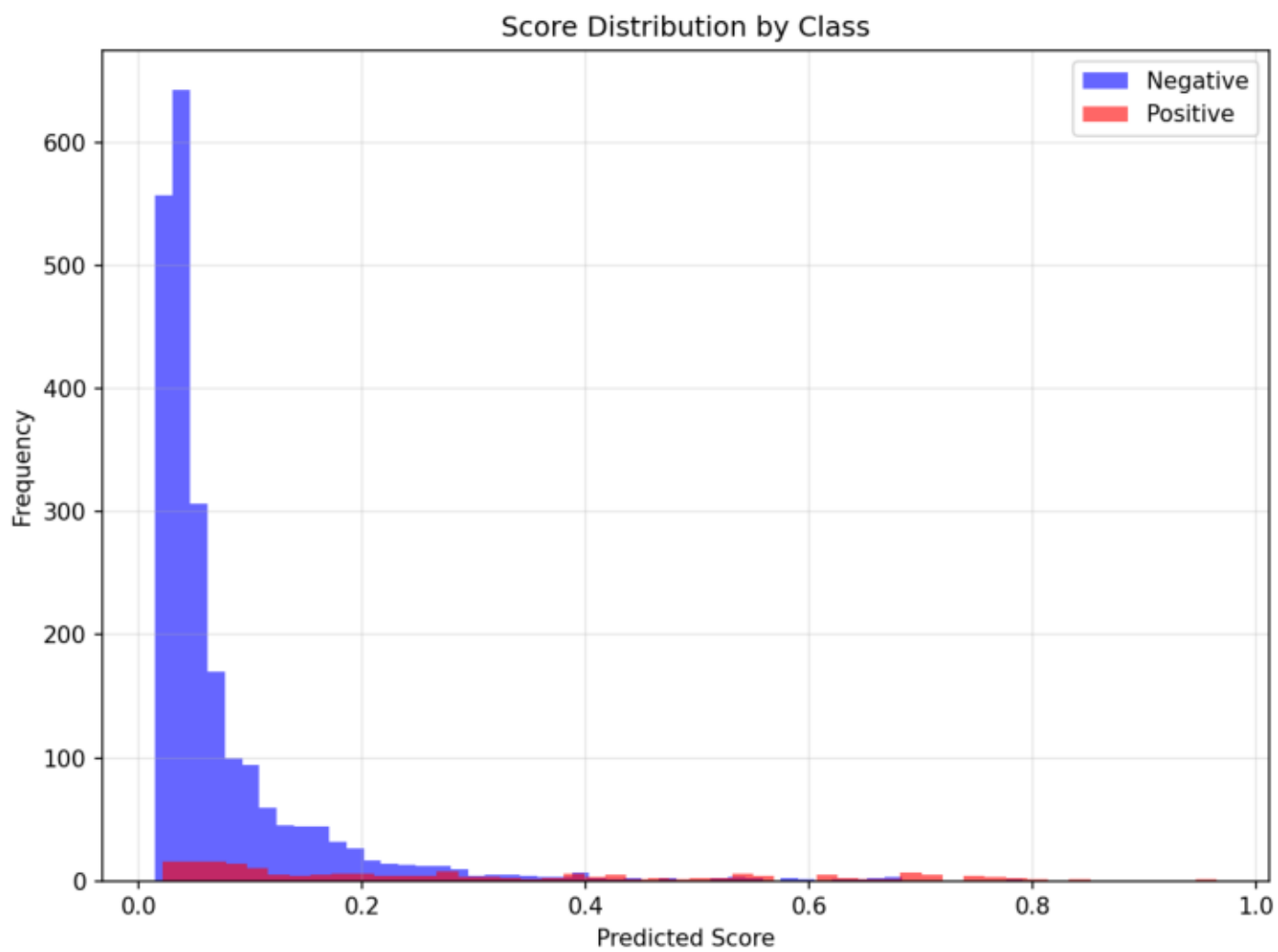
confusion_matrix_norm: Normalized rates: TPR=23.71%, TNR=99.20%.



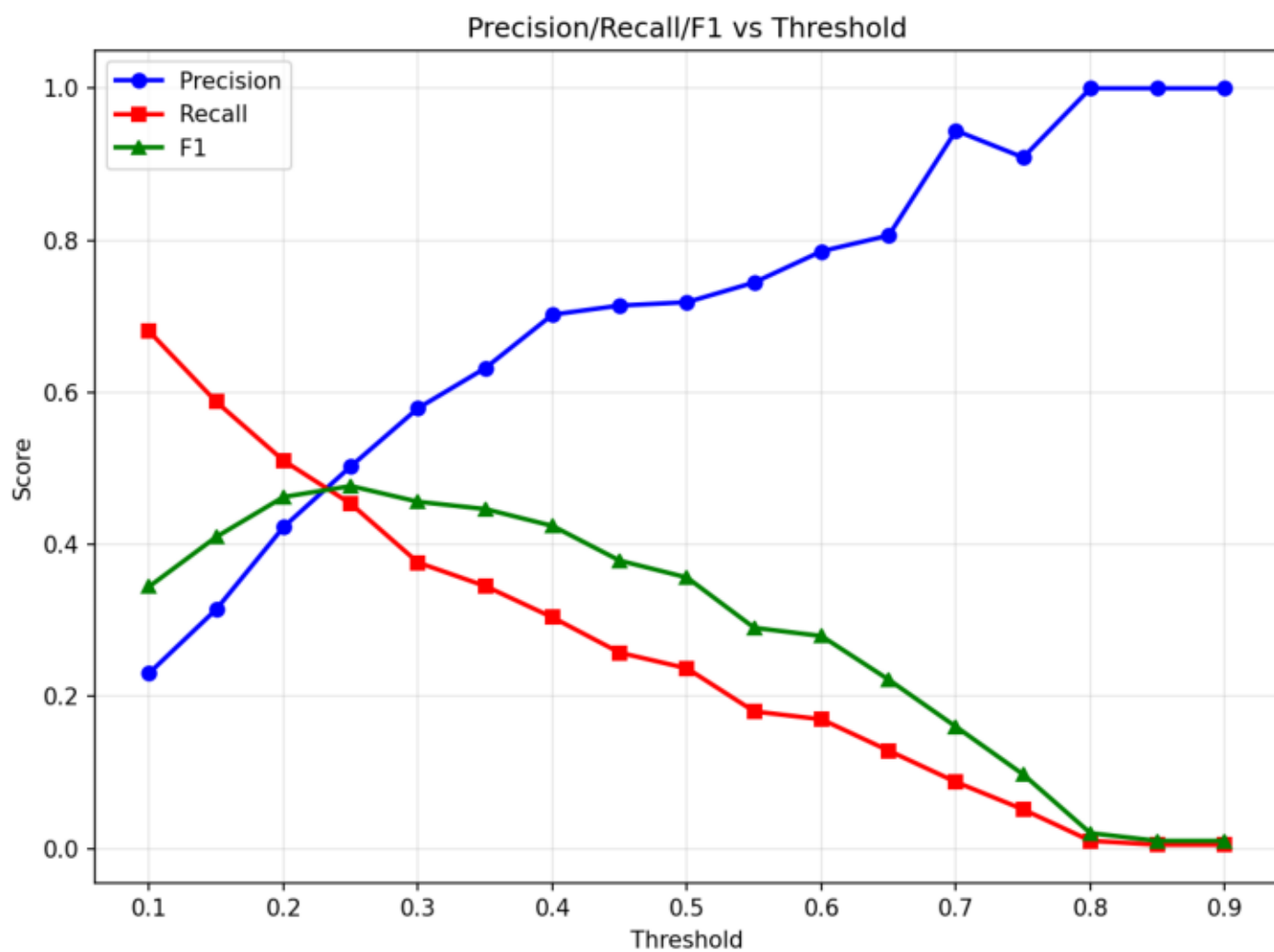
ks_curve: Maximum separation KS=0.5052 at threshold 0.07.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.291, negative=0.073 (larger gap is better).



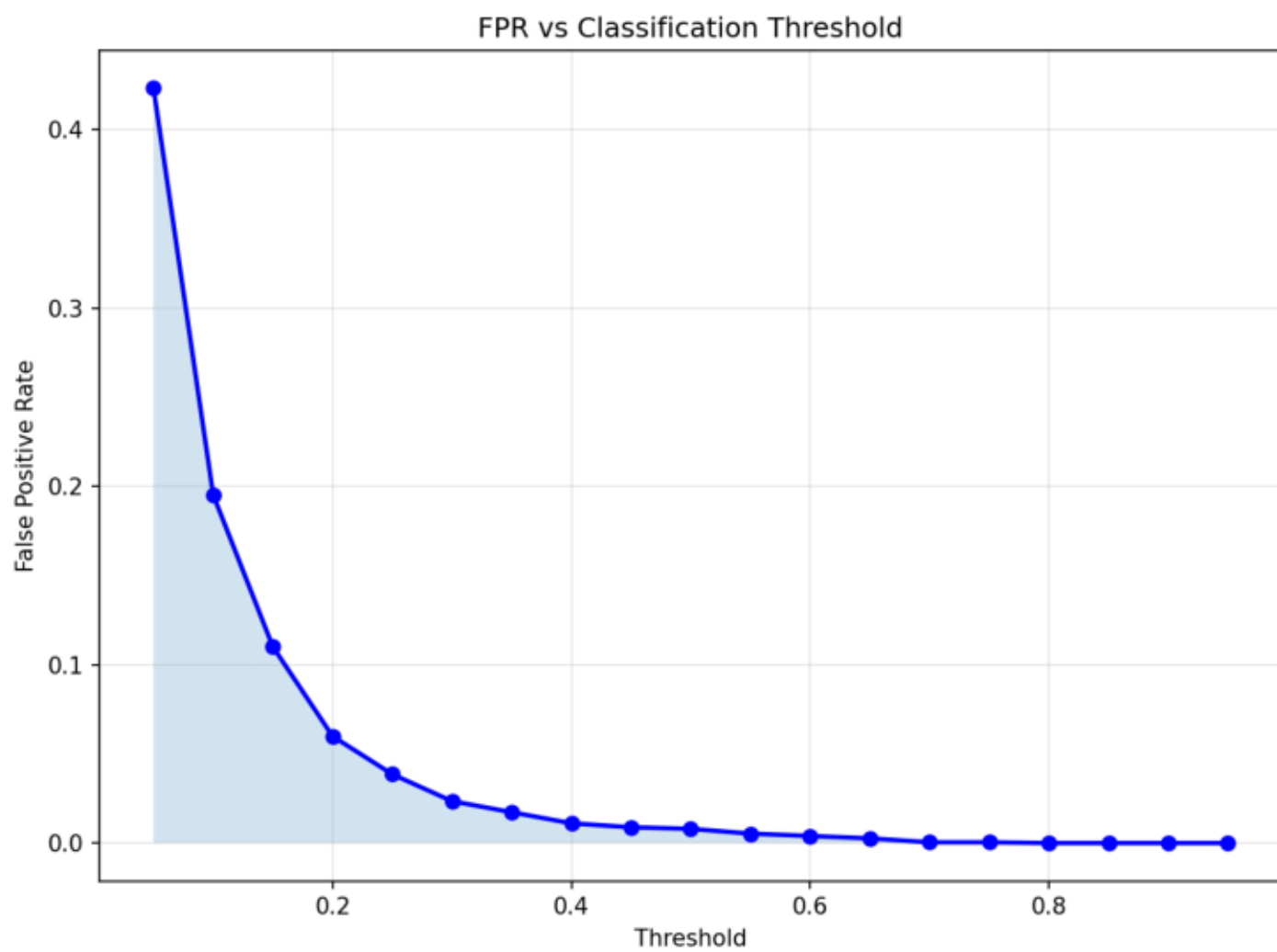
threshold_analysis: Best F1 in tested grid is 0.4770 at threshold 0.25.

value_date_quarter=2024Q2 - 2) Model Efficiency

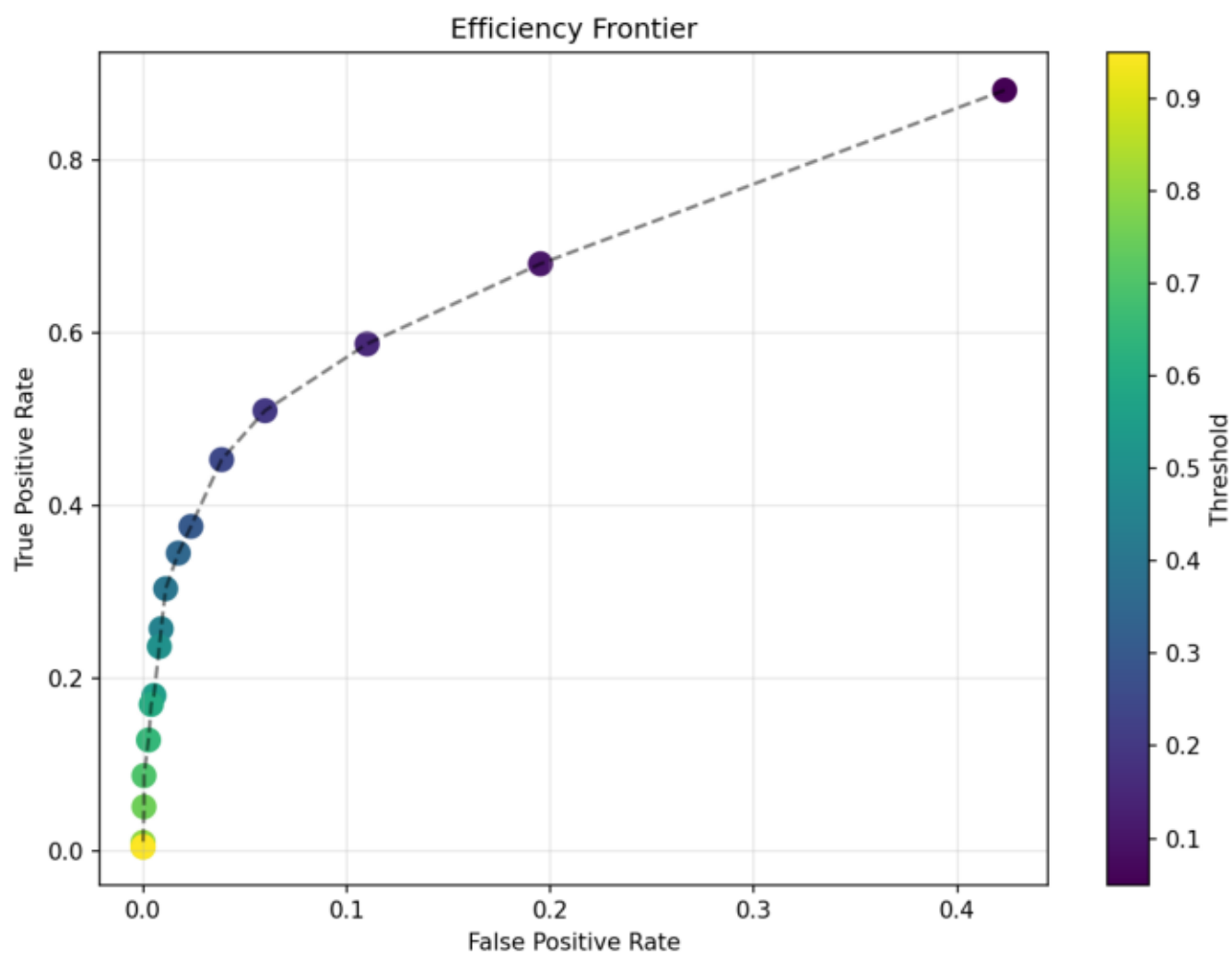
- fpr: 0.0080
- tn: 2236
- fp: 18
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.4232
- fpr_at_thresholds.t_0.10: 0.1952
- fpr_at_thresholds.t_0.15: 0.1100
- fpr_at_thresholds.t_0.20: 0.0599
- fpr_at_thresholds.t_0.25: 0.0386
- fpr_at_thresholds.t_0.30: 0.0235
- fpr_at_thresholds.t_0.35: 0.0173
- fpr_at_thresholds.t_0.40: 0.0111
- fpr_at_thresholds.t_0.45: 0.0089
- fpr_at_thresholds.t_0.50: 0.0080
- fpr_at_thresholds.t_0.55: 0.0053
- fpr_at_thresholds.t_0.60: 0.0040
- fpr_at_thresholds.t_0.65: 0.0027
- fpr_at_thresholds.t_0.70: 0.0004
- fpr_at_thresholds.t_0.75: 0.0004
- fpr_at_thresholds.t_0.80: 0.0000
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q2 - 2) Model Efficiency - Explanations

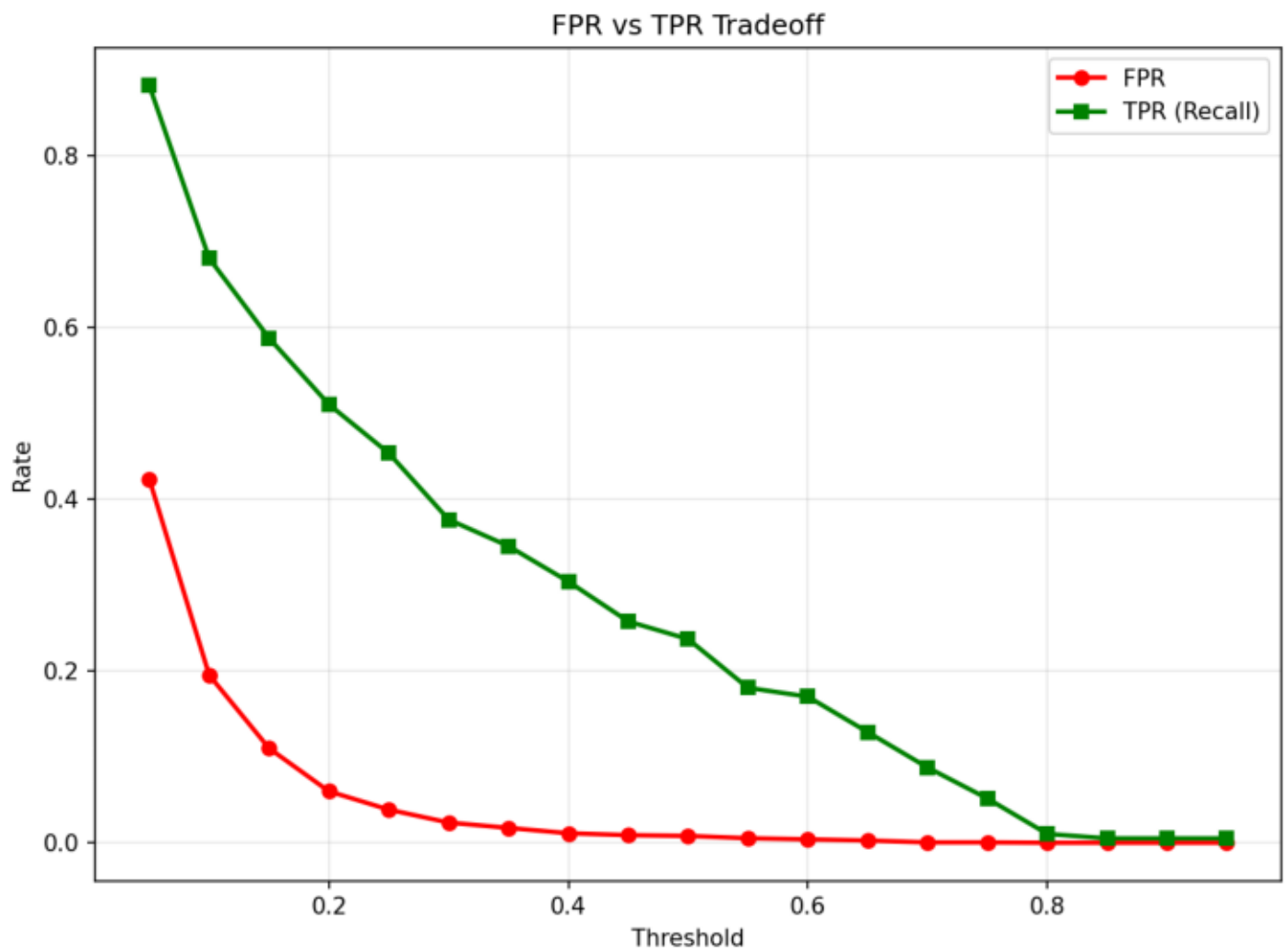
- At threshold 0.50, FPR=0.0080 (low); FP=18 out of 2254 negatives.
- At the same threshold, TPR=0.2371 with TP=46 and FN=148, showing the capture rate of positives.
- A threshold near 0.20 yields FPR $\approx$ 0.0599 with TPR $\approx$ 0.5103 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.



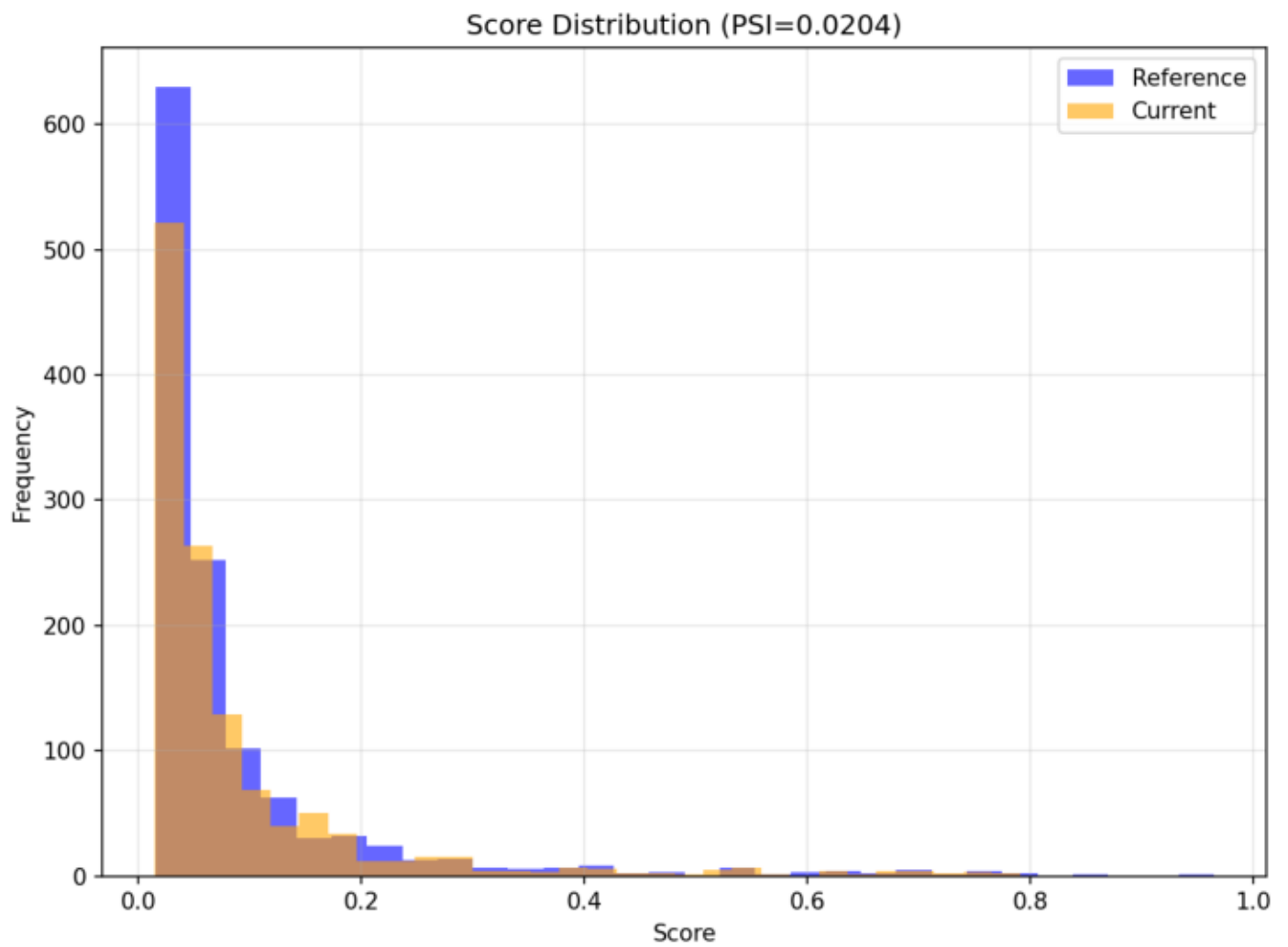
fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

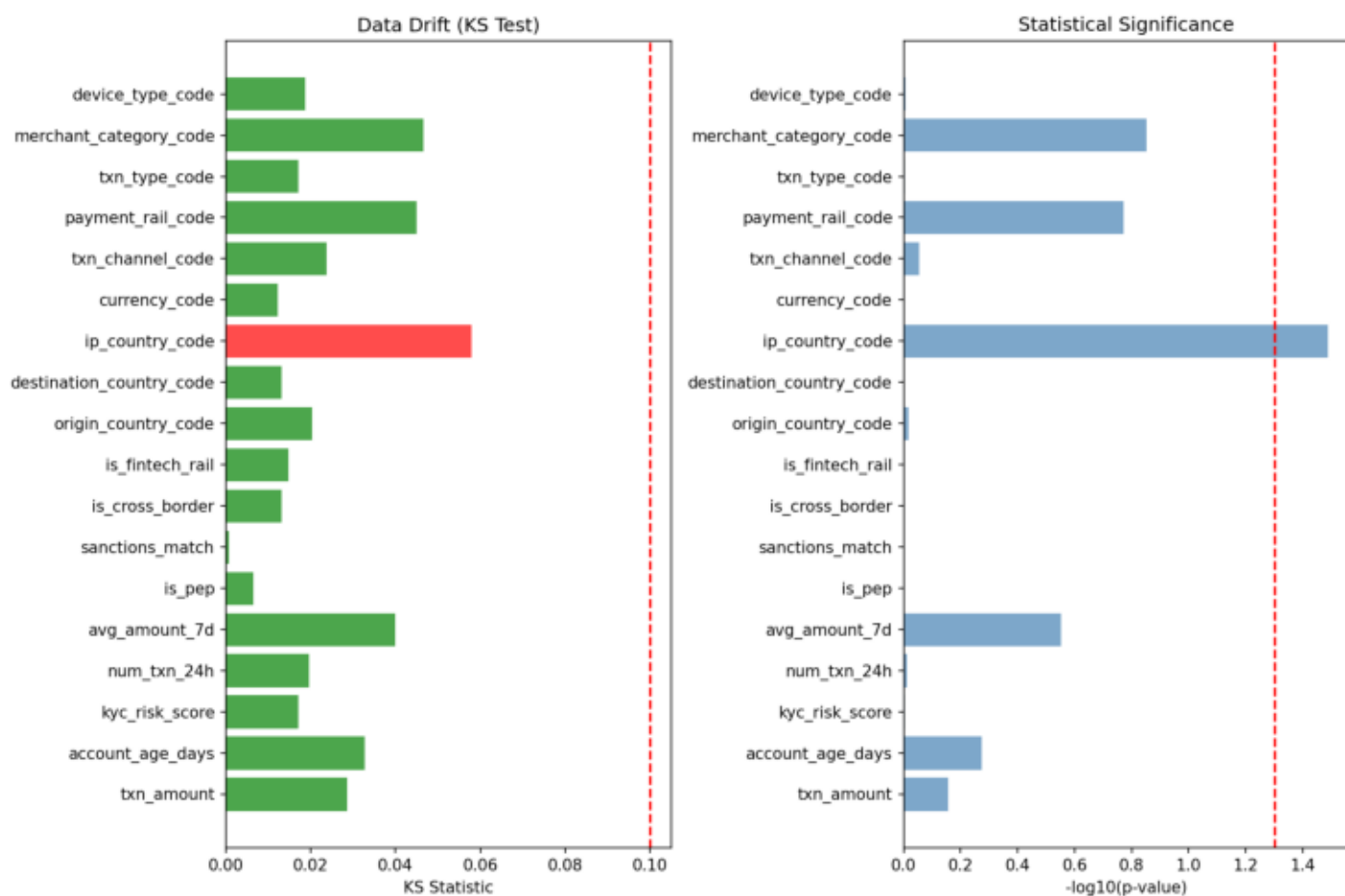
value_date_quarter=2024Q2 - 3) Model Stability

- psi: 0.0204
- cv_auc_roc_mean: 0.8446
- cv_auc_roc_std: 0.0062
- cv_auc_pr_mean: 0.4895
- cv_auc_pr_std: 0.0338
- bootstrap_auc_roc_mean: 0.8540
- bootstrap_auc_roc_ci_lower: 0.8242
- bootstrap_auc_roc_ci_upper: 0.8914
- concept_drift_detected: False
- concept_drift_score: 0.0206

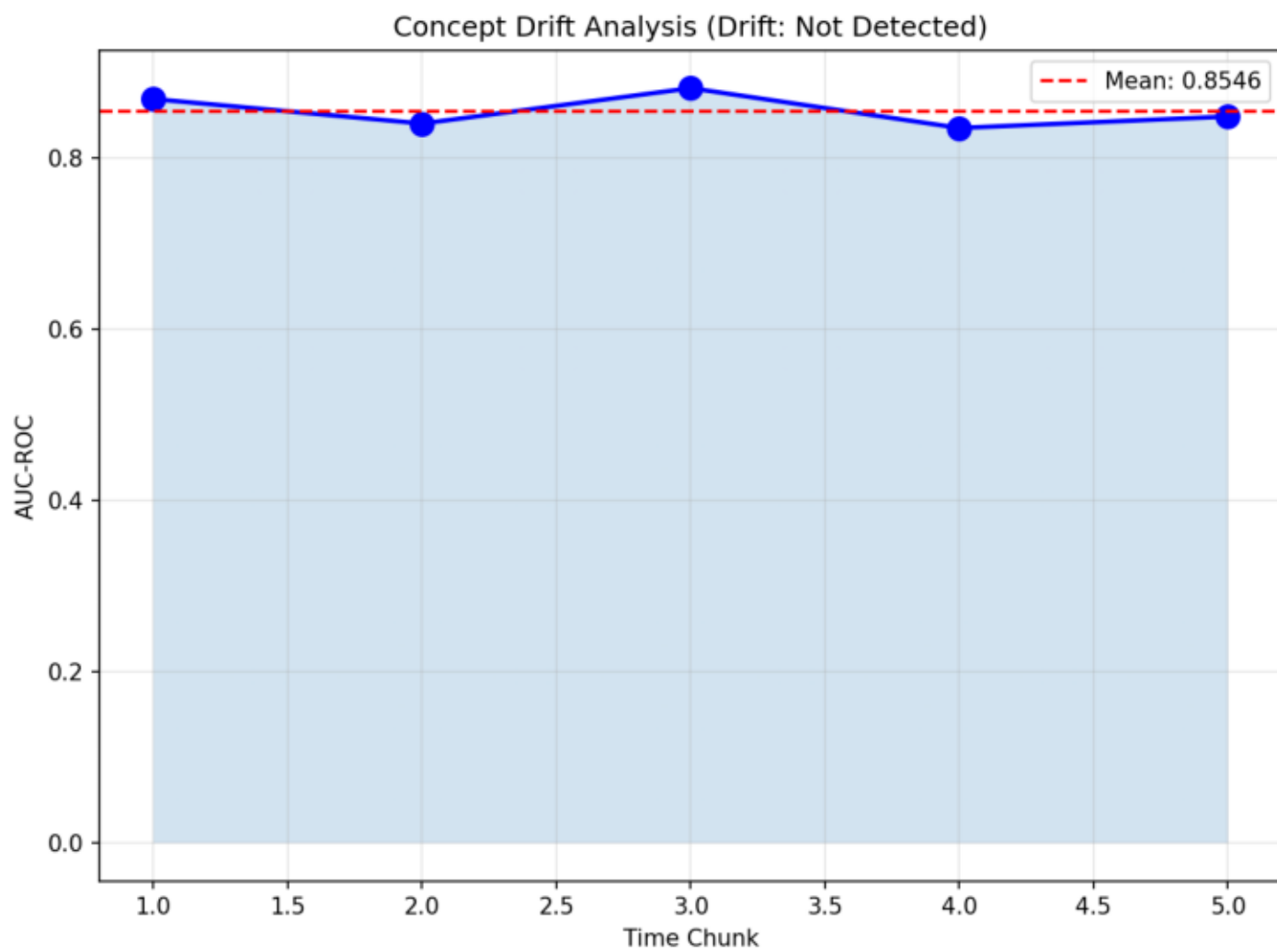
value_date_quarter=2024Q2 - 3) Model Stability - Explanations

- PSI is 0.0204, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8446 ± 0.0062 (stable across folds).
- CV AUC-PR is 0.4895 ± 0.0338 (variable across folds).
- Bootstrap AUC-ROC mean is 0.8540 with 95% CI [0.8242, 0.8914] (width 0.0672).
- No concept drift detected (score 0.0206), suggesting stable performance over time.
- Data drift flagged 1/18 numeric features; top drift signals: ['ip_country_code', 'merchant_category_code', 'payment_rail_code'].

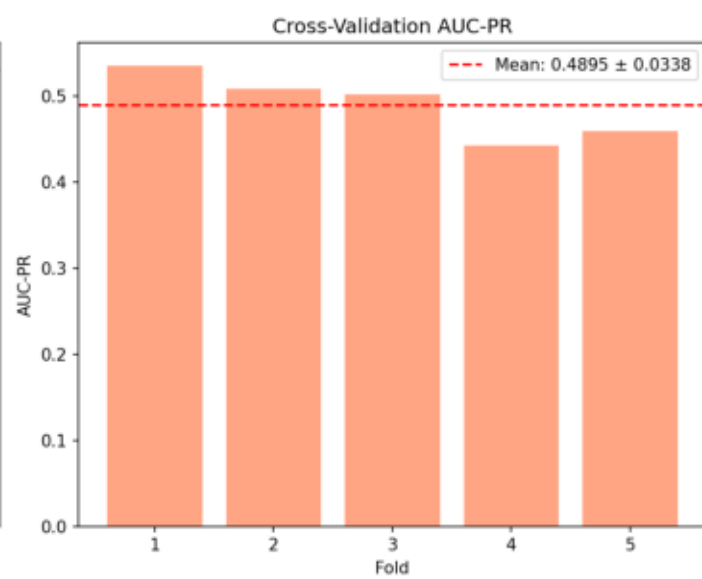
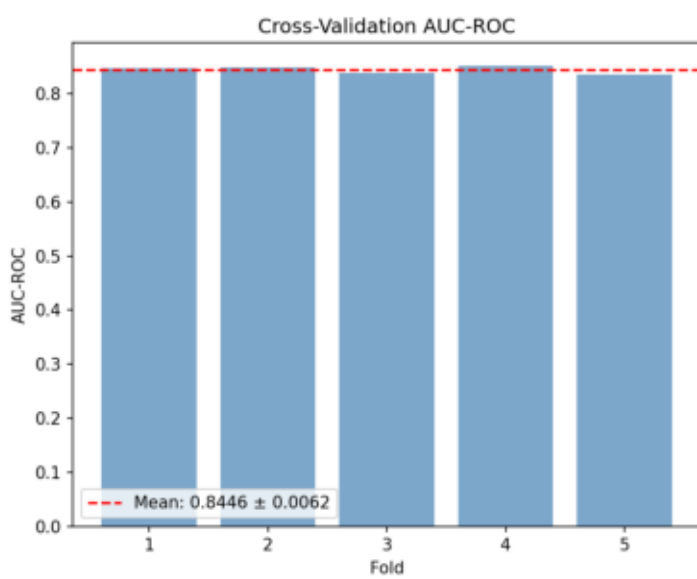




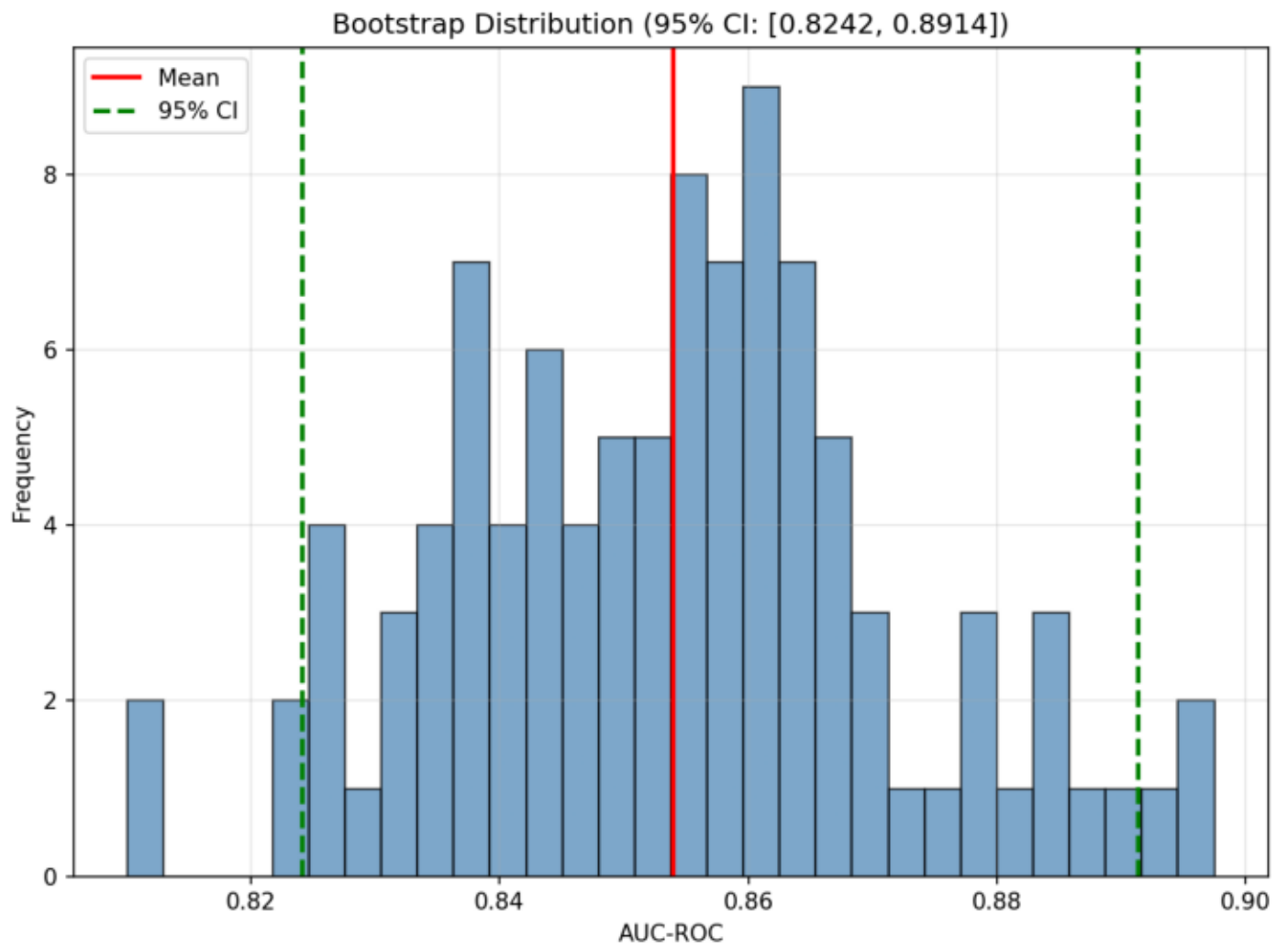
data_drift_heatmap: 1/18 features show drift; red bars highlight flagged features.



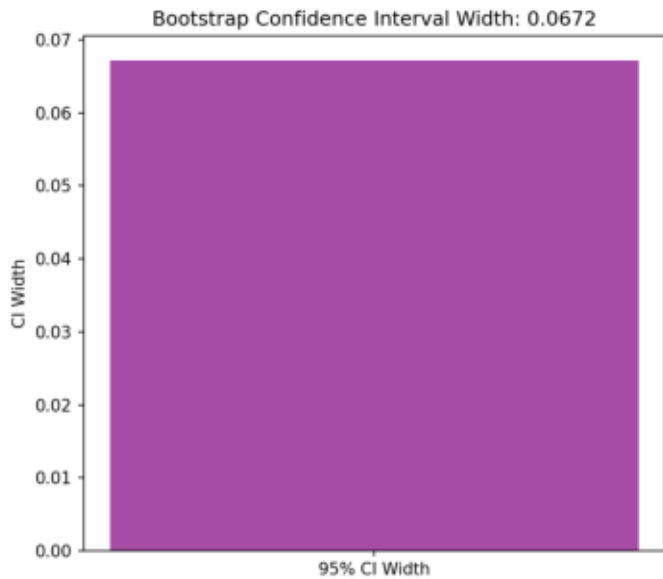
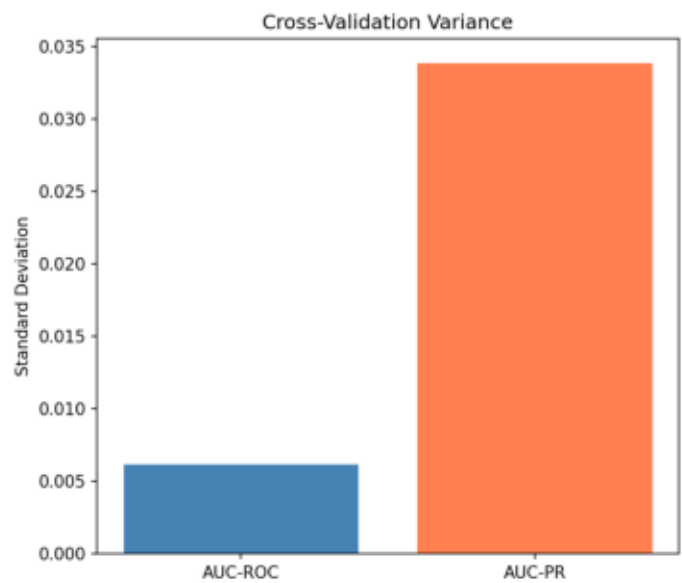
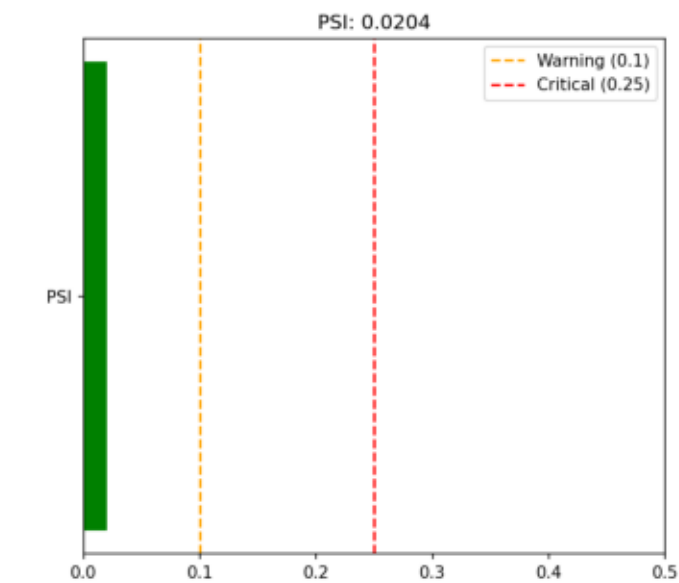
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8446 and PR 0.4895.



bootstrap_distribution: Bootstrap CI [0.8242, 0.8914] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0204 (OK)

CV AUC-ROC: 0.8446 $\pm$ 0.0062

CV AUC-PR: 0.4895 $\pm$ 0.0338

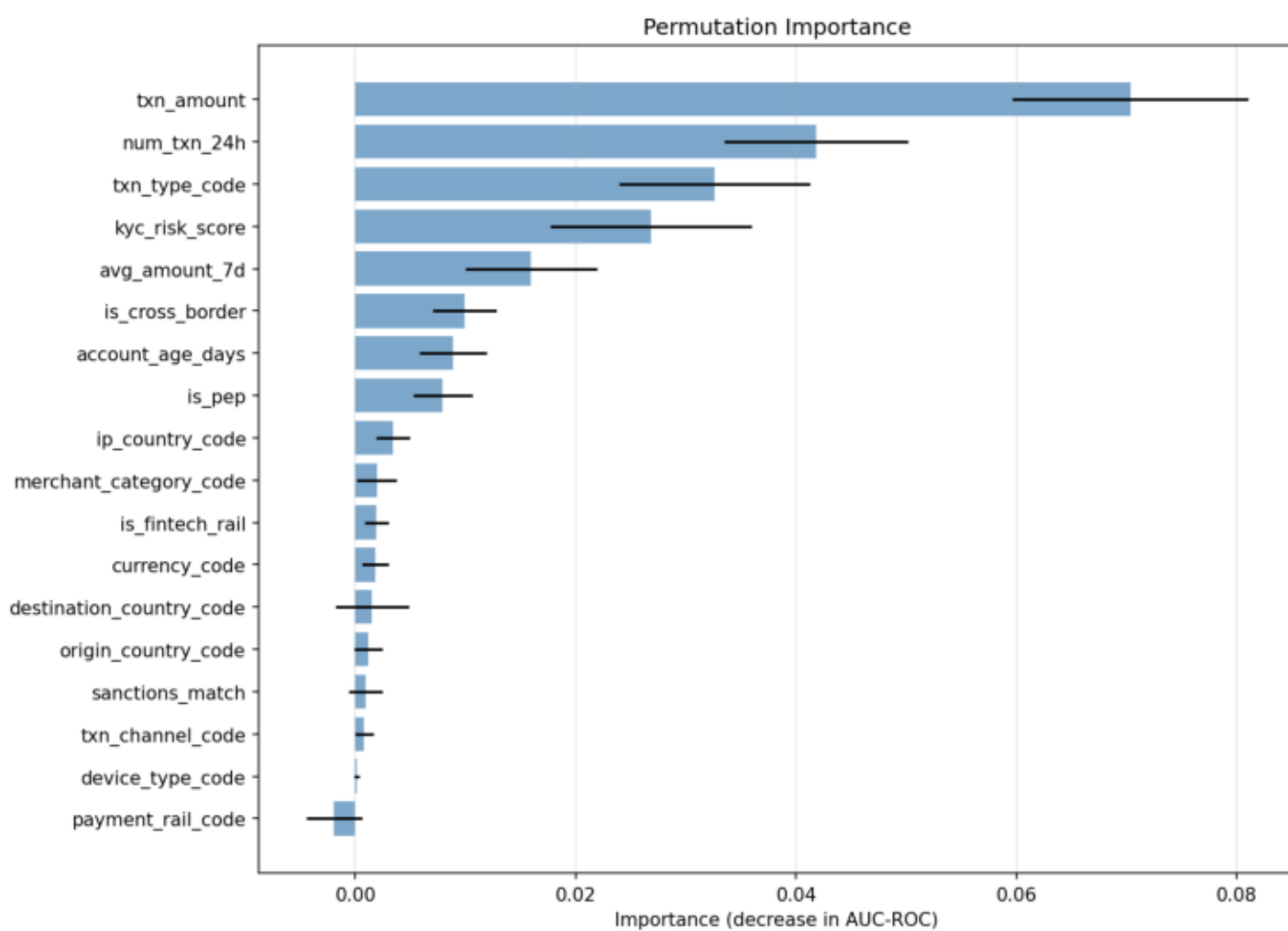
Bootstrap 95% CI: [0.8242, 0.8914]

value_date_quarter=2024Q2 - 4) Model Interpretability

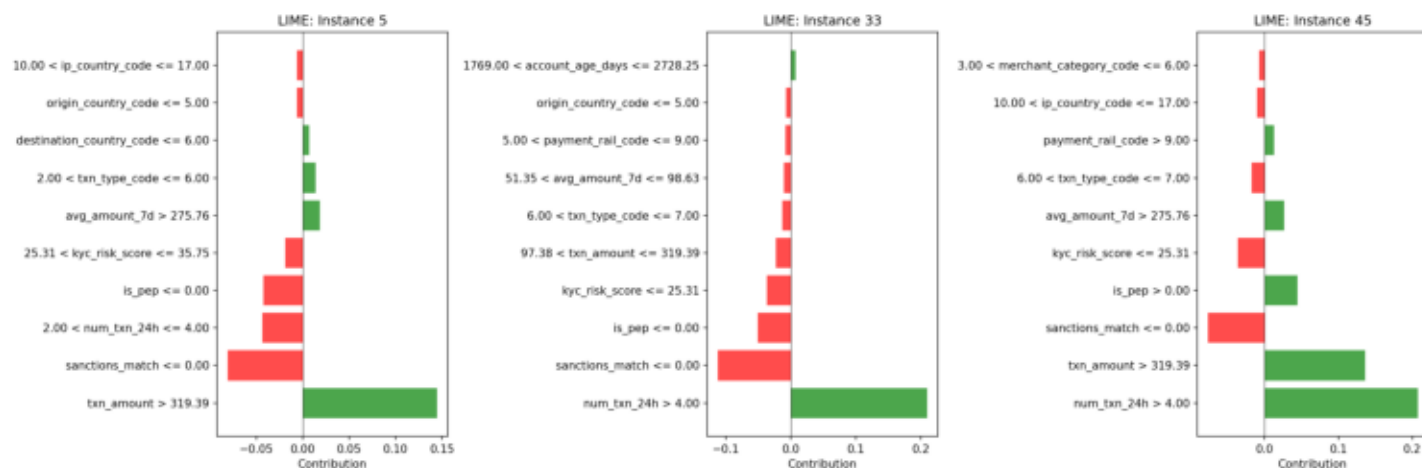
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'lime']
- shap_status: unavailable
- shap_reason: No module named 'shap'
- perm_top_features: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score', 'avg_amount_7d']...
- lime_instances: 3
- pdp_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q2 - 4) Model Interpretability - Explanations

- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score',
- 'avg_amount_7d'].
- SHAP plots not generated because SHAP is unavailable (No module named 'shap').
- LIME generated local explanations for 3 instances; top contributors for instance_5: ['txn_amount > 319.39',
- 'sanctions_match <= 0.00', '2.00 < num_txn_24h <= 4.00', 'is_pep <= 0.00', '25.31 < kyc_risk_score <= 35.75'].
- PDP plots show average effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score',
- 'num_txn_24h', 'origin_country_code'].
- ICE plots show individual-level effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days',
- 'kyc_risk_score'].

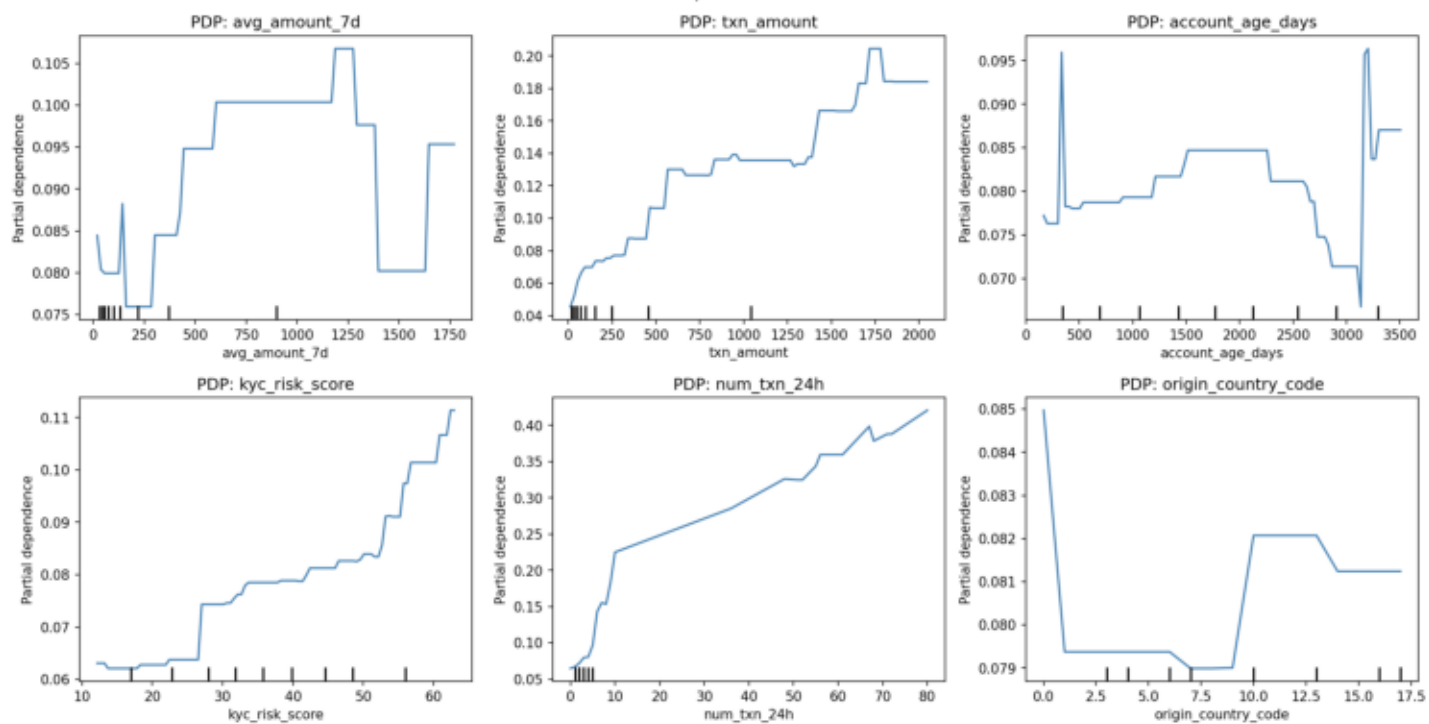


permutation_importance: Bars show how much AUC drops when each feature is permuted.



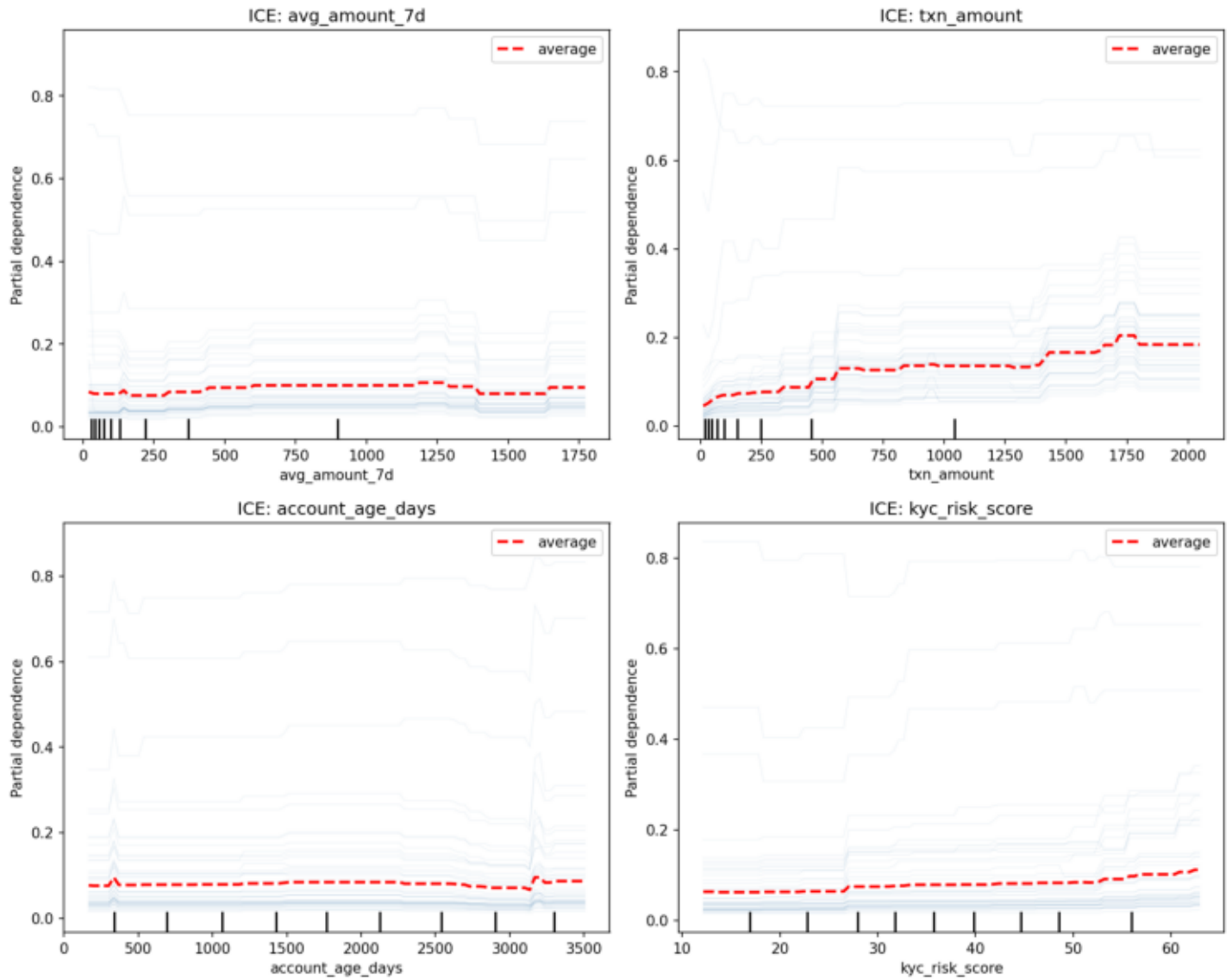
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



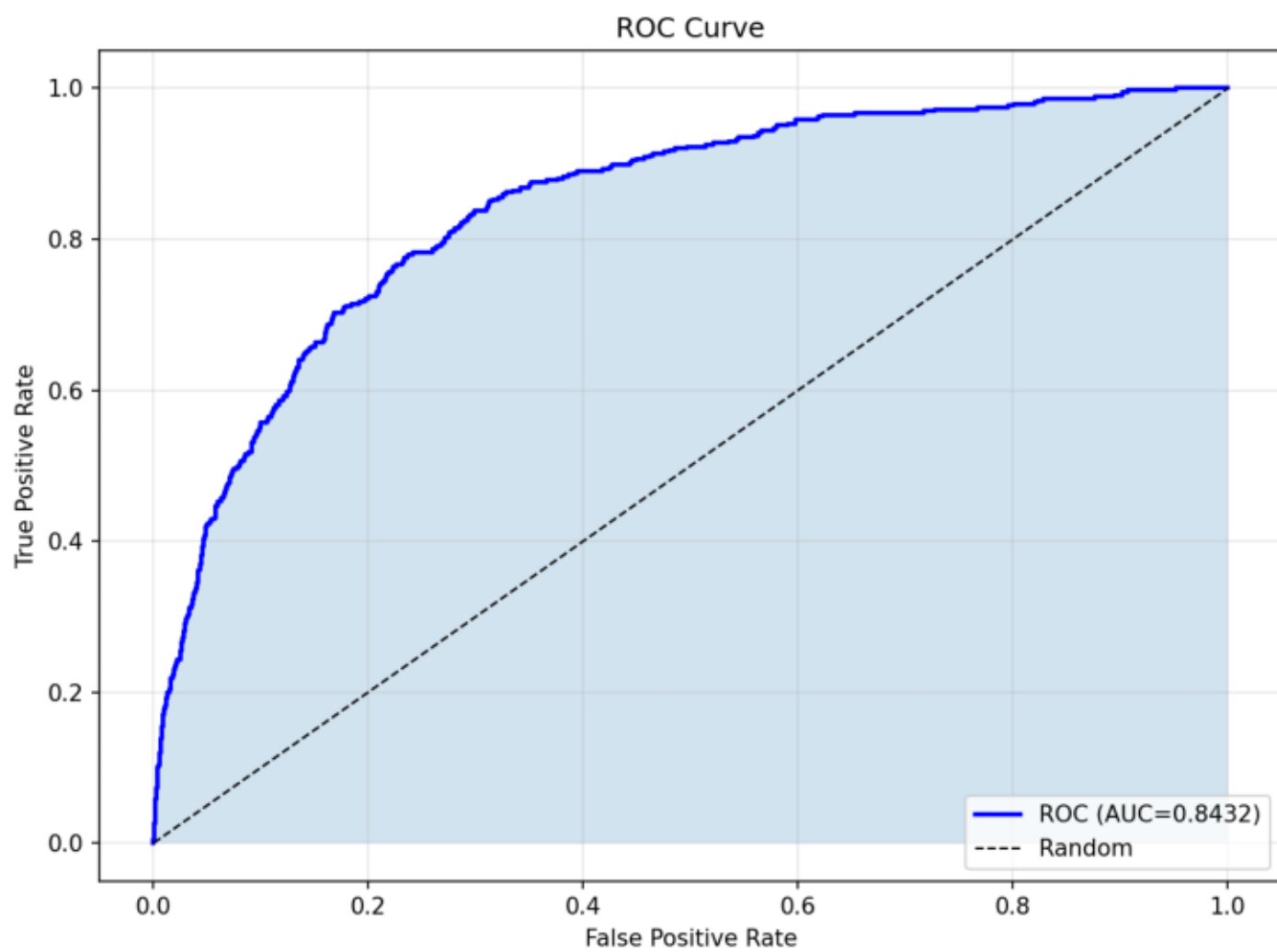
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q3 - 1) Model Effectiveness

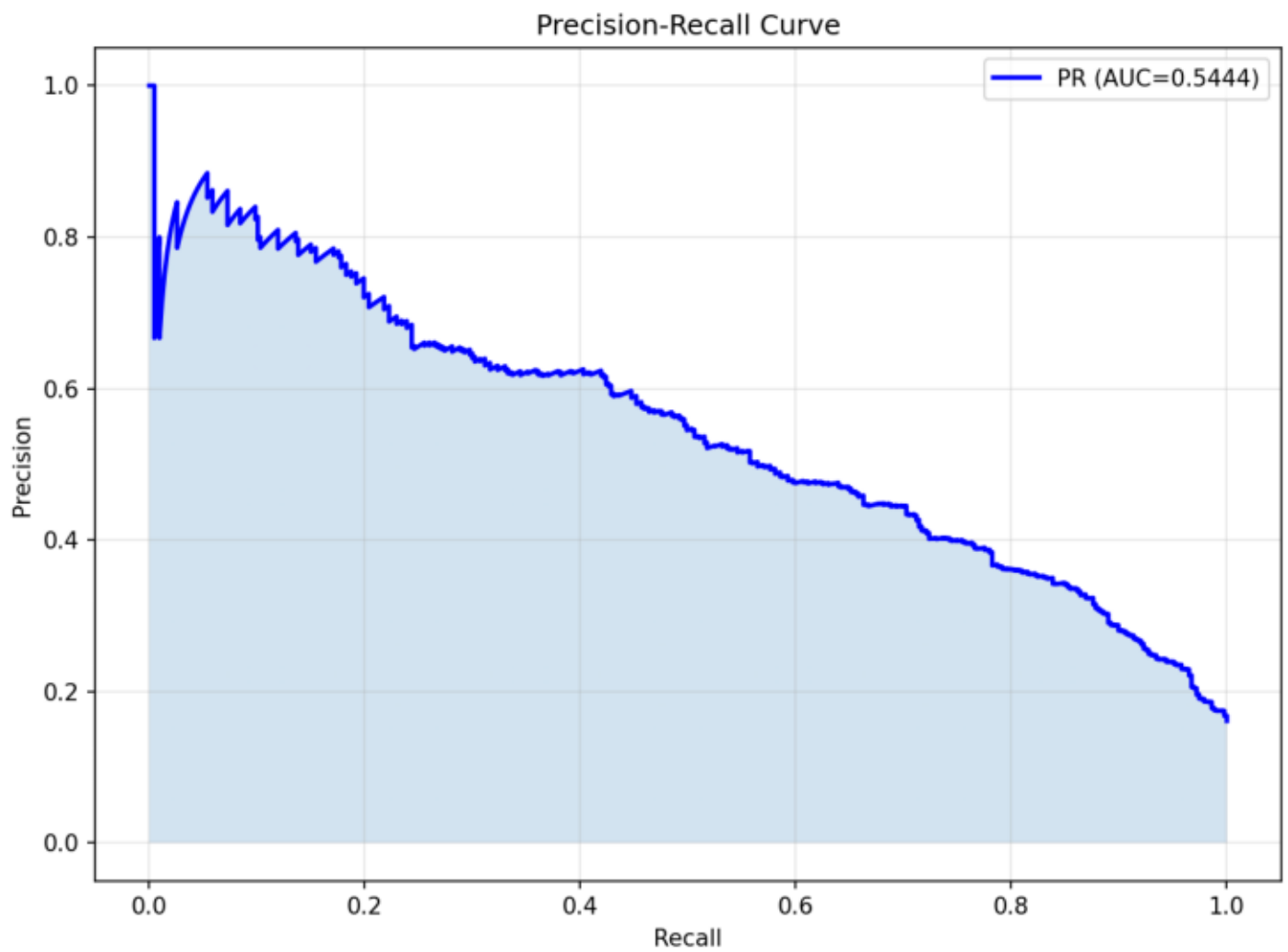
- auc_roc: 0.8432
- auc_pr: 0.5444
- precision: 0.6301
- recall: 0.3232
- f1: 0.4272
- ks_statistic: 0.5416
- ks_threshold: 0.1600
- confusion_matrix.TN: 2144
- confusion_matrix.FP: 81
- confusion_matrix.FN: 289
- confusion_matrix.TP: 138
- precision_at_k.10: 0.8000
- precision_at_k.50: 0.8400
- precision_at_k.100: 0.7600
- precision_at_k.200: 0.6400
- precision_at_k.500: 0.4940
- recall_at_k.10: 0.0187
- recall_at_k.50: 0.0984
- recall_at_k.100: 0.1780
- recall_at_k.200: 0.2998
- recall_at_k.500: 0.5785

value_date_quarter=2024Q3 - 1) Model Effectiveness - Explanations

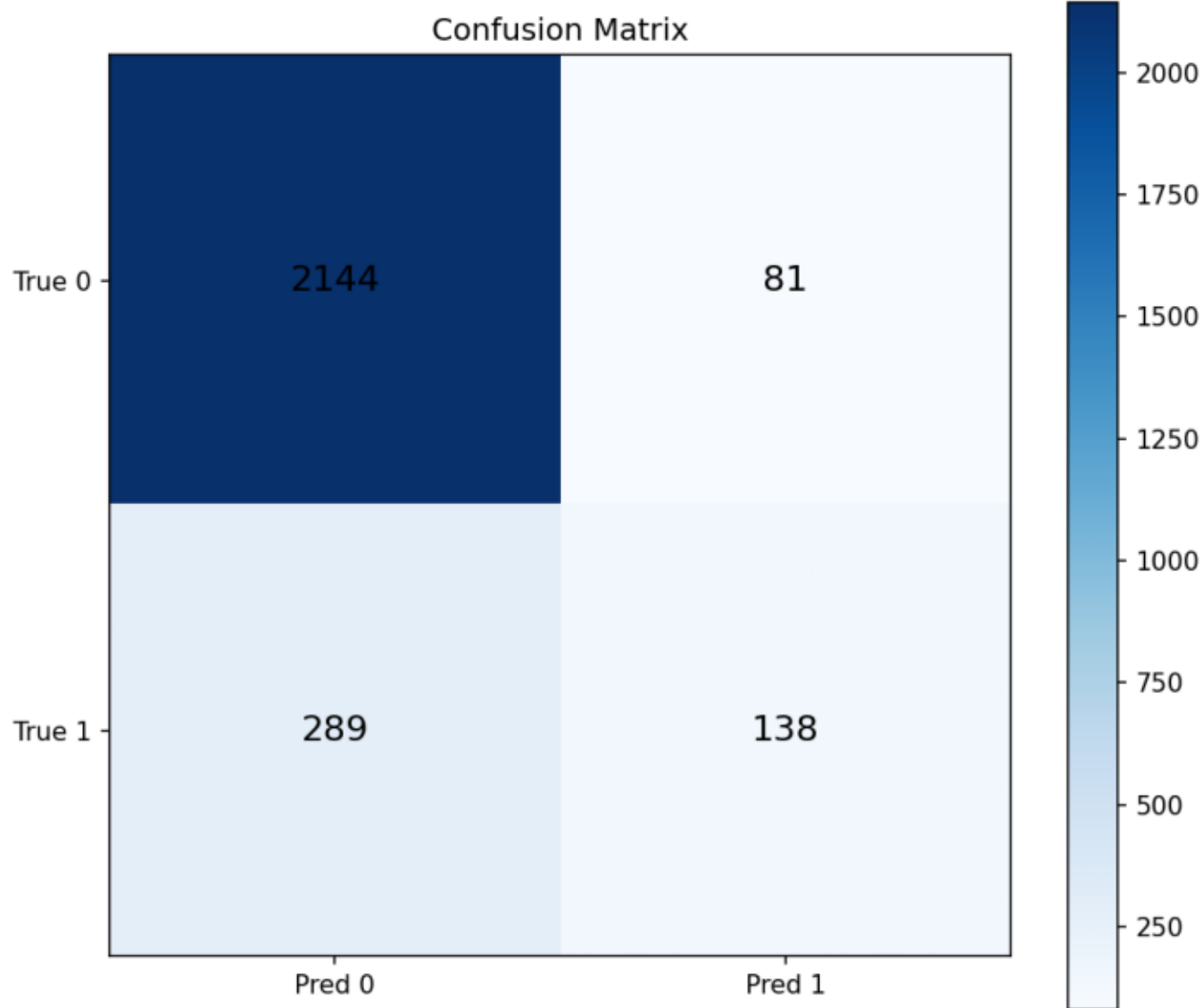
- Positive rate is 16.10%; AUC-ROC is 0.8432 (strong ranking).
- AUC-PR is 0.5444, above the base rate 16.10%, indicating meaningful lift.
- At threshold 0.50, precision=0.6301, recall=0.3232, F1=0.4272 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=138, FP=81, FN=289, TN=2144 (FPR=3.64%, FNR=67.68%).
- KS is 0.5416 at threshold 0.16, indicating very strong separation.
- Top-10: precision=0.8000, recall=0.0187 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.4940, recall=0.5785 shows how much coverage you get with a larger queue.



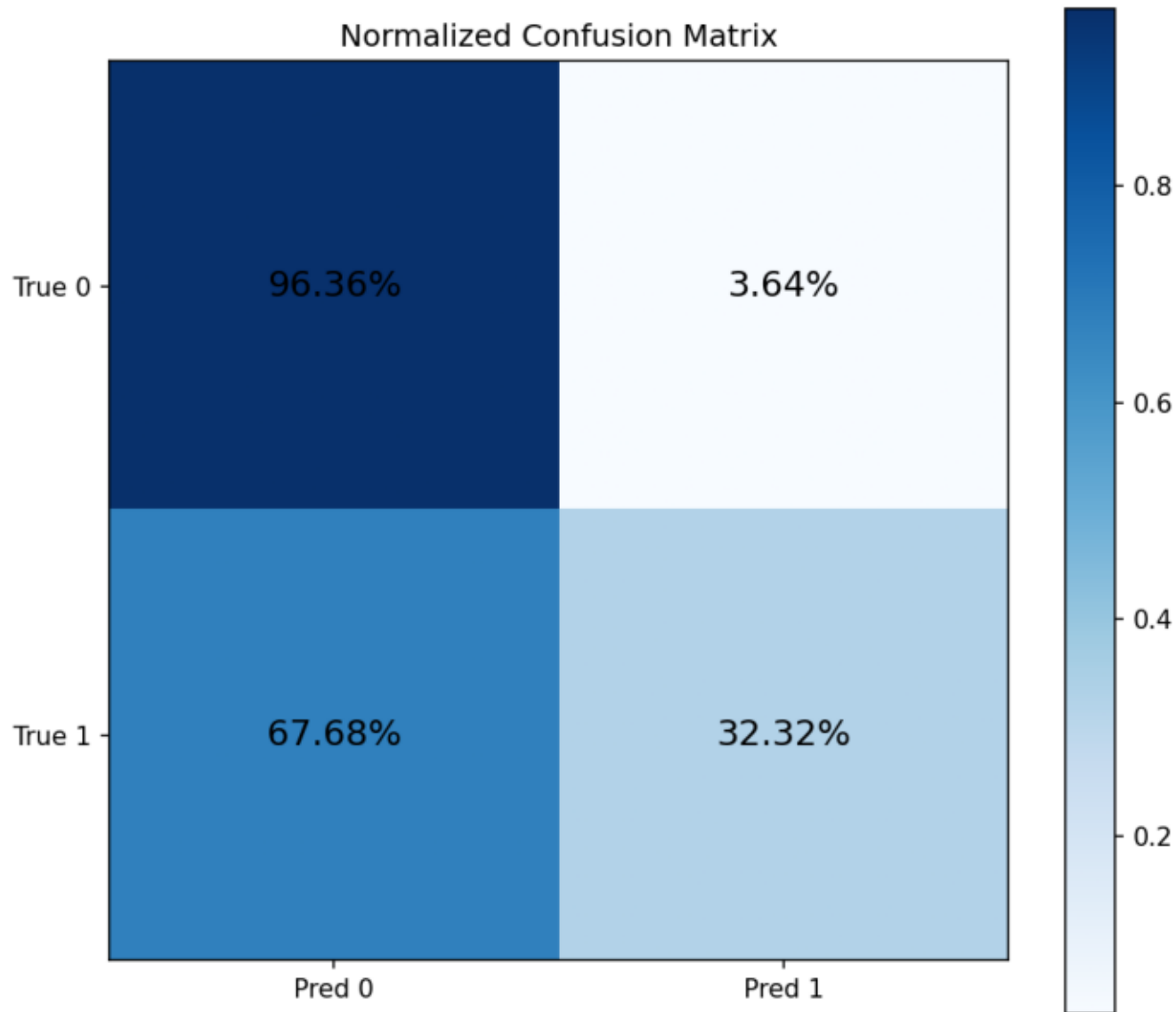
roc_curve: AUC-ROC=0.8432; curve above diagonal indicates strong separation.



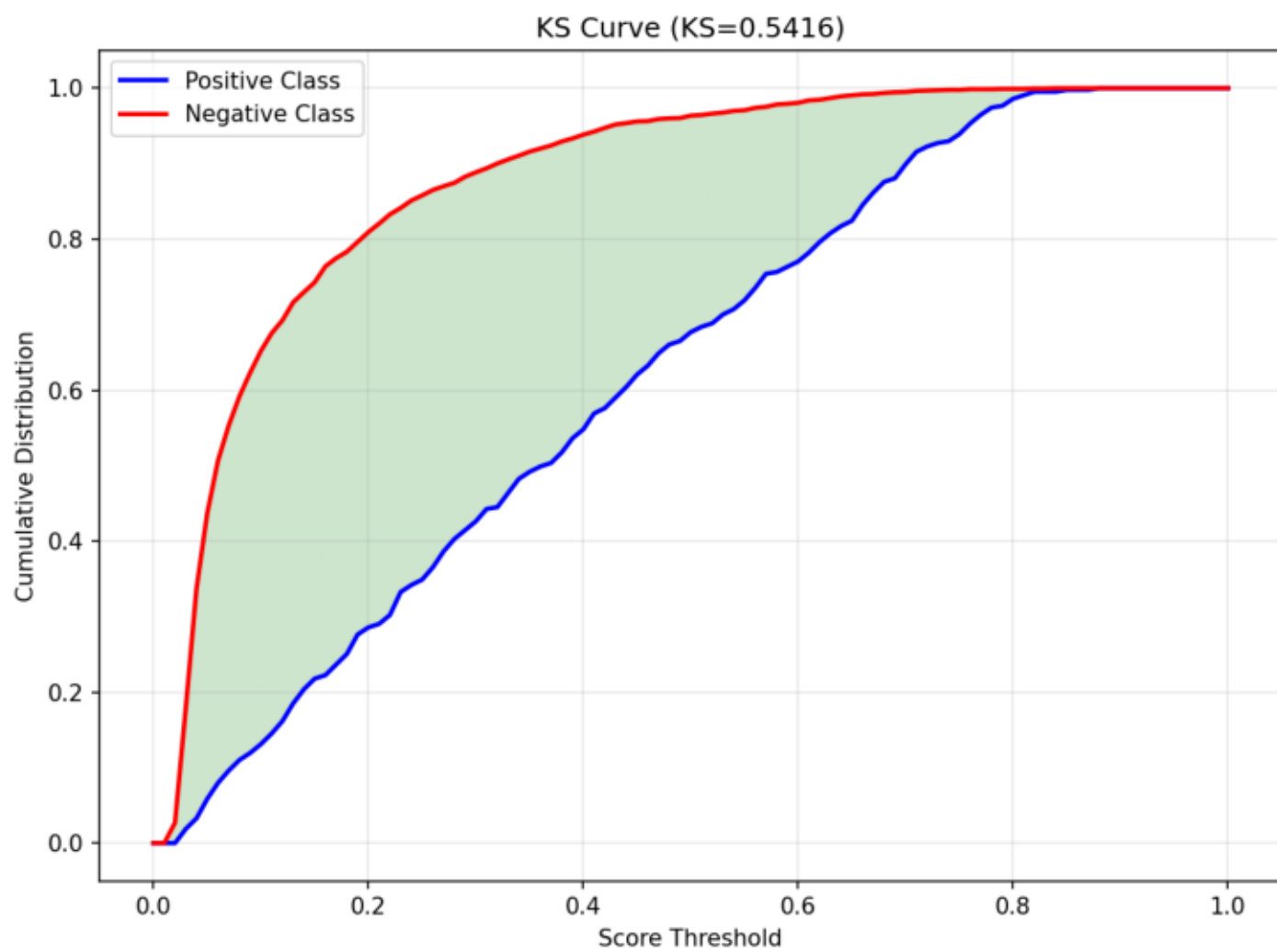
pr_curve: AUC-PR=0.5444 vs baseline 16.10%; higher curve implies better precision at recall.



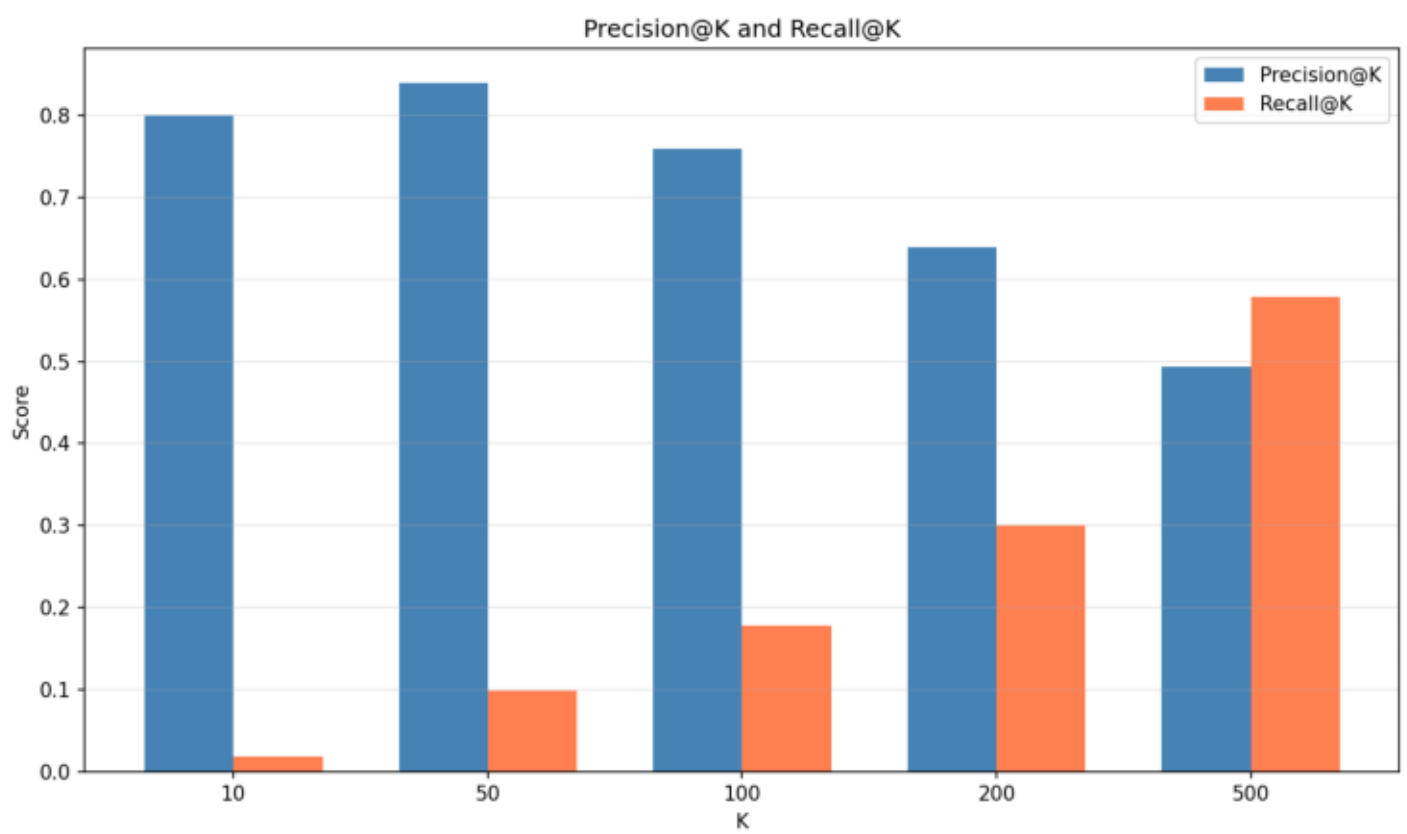
confusion_matrix: At threshold 0.50: TP=138, FP=81, FN=289, TN=2144.



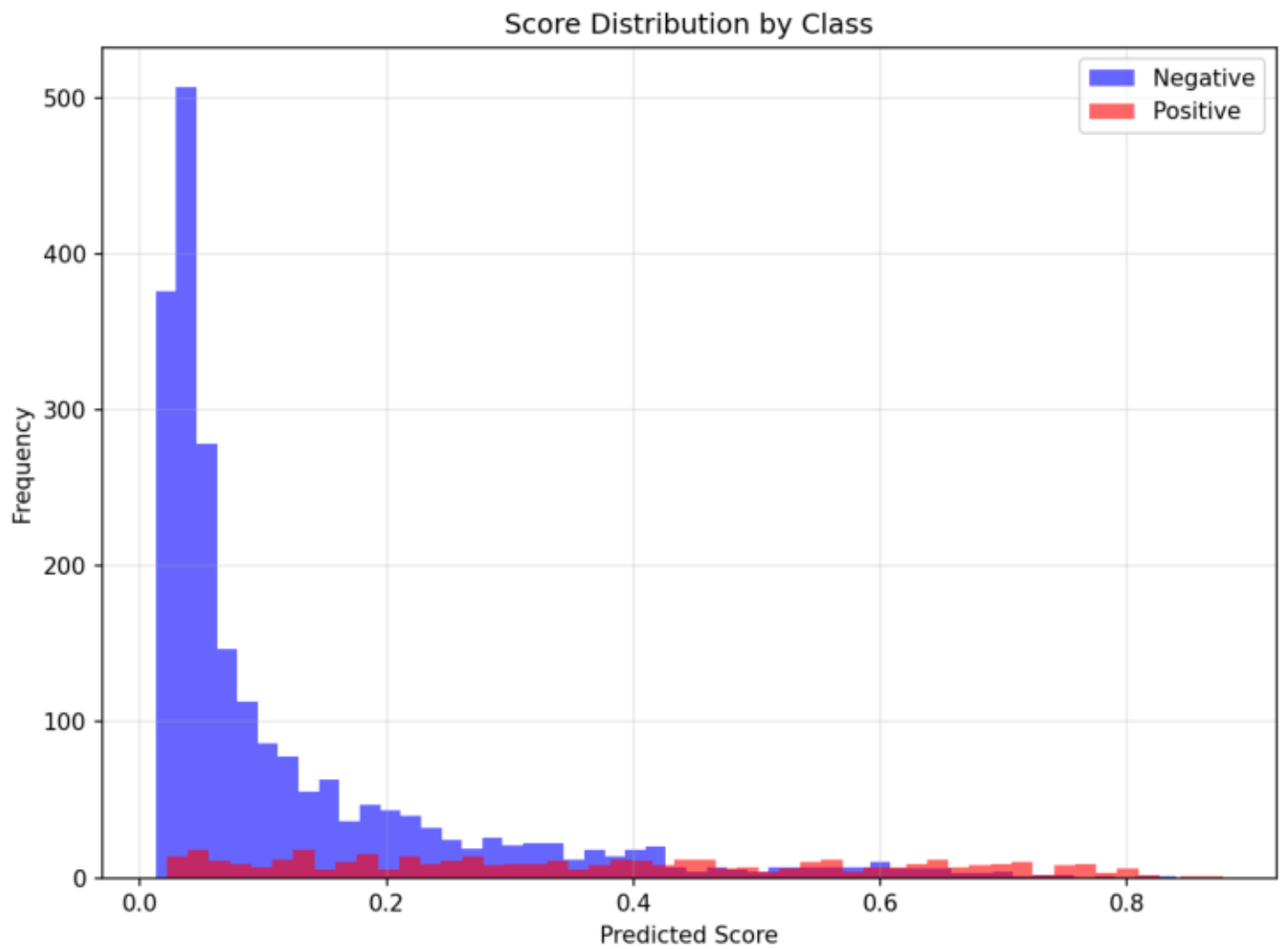
confusion_matrix_norm: Normalized rates: TPR=32.32%, TNR=96.36%.



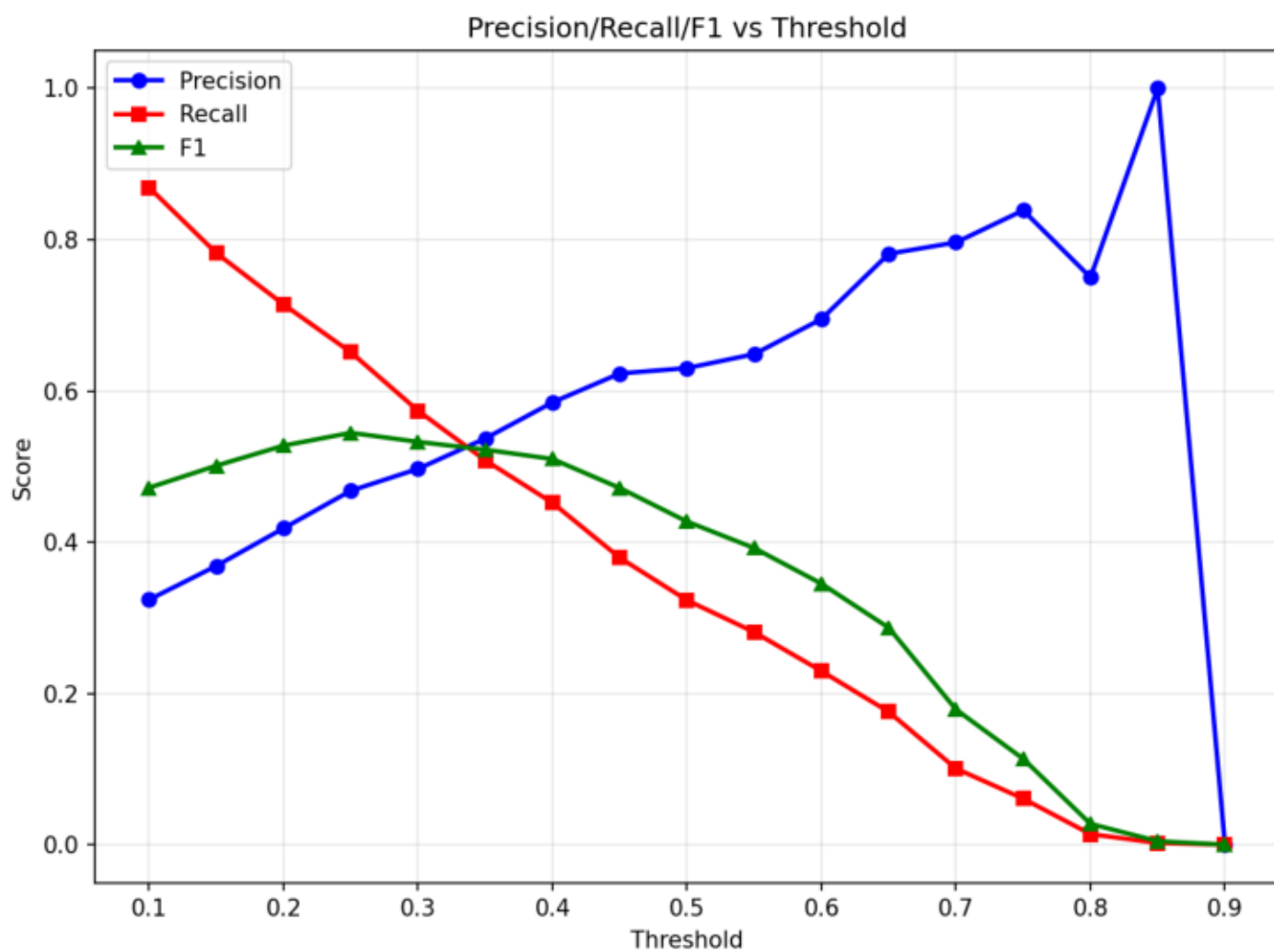
ks_curve: Maximum separation KS=0.5416 at threshold 0.16.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.378, negative=0.122 (larger gap is better).



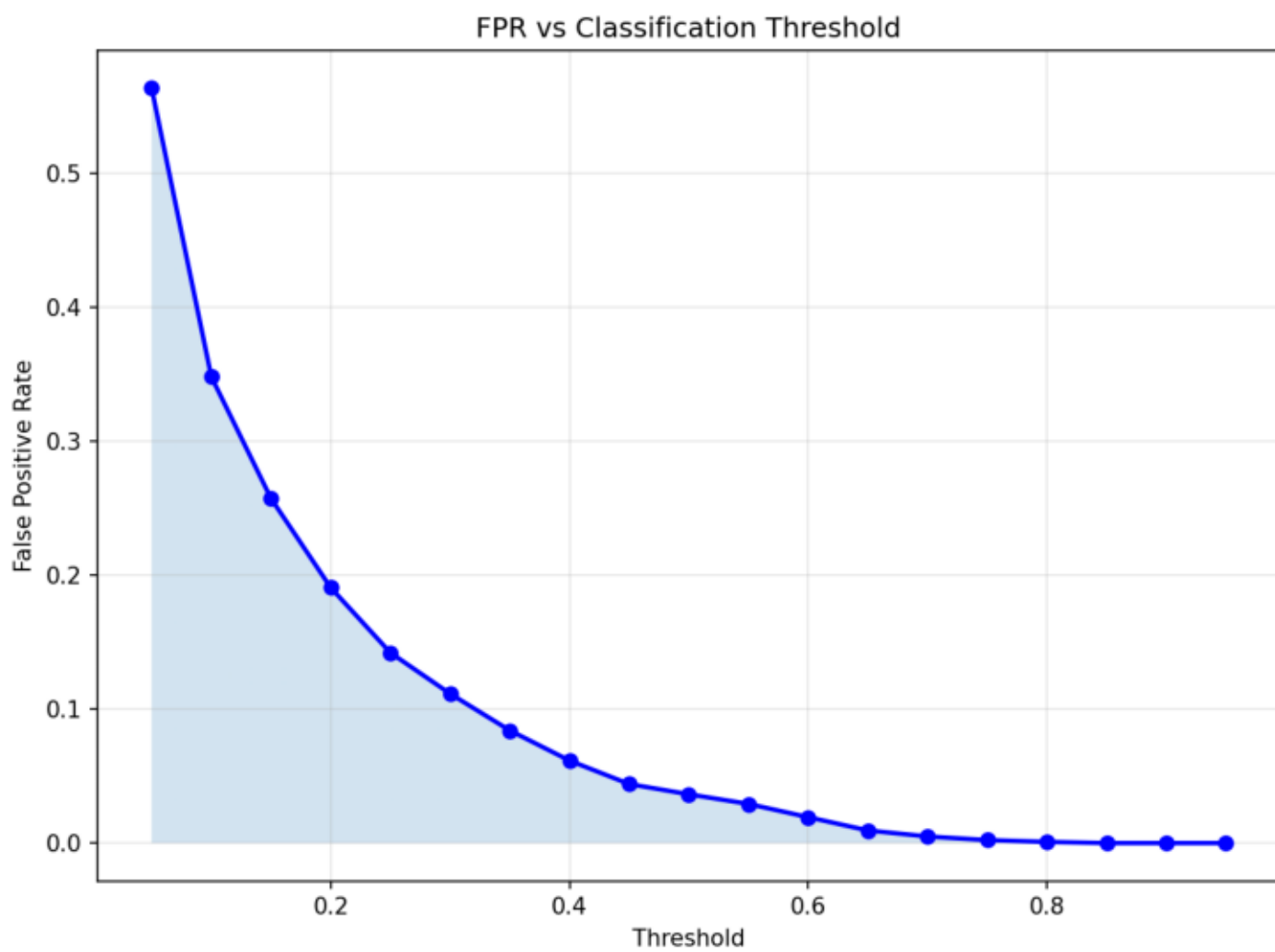
threshold_analysis: Best F1 in tested grid is 0.5446 at threshold 0.25.

value_date_quarter=2024Q3 - 2) Model Efficiency

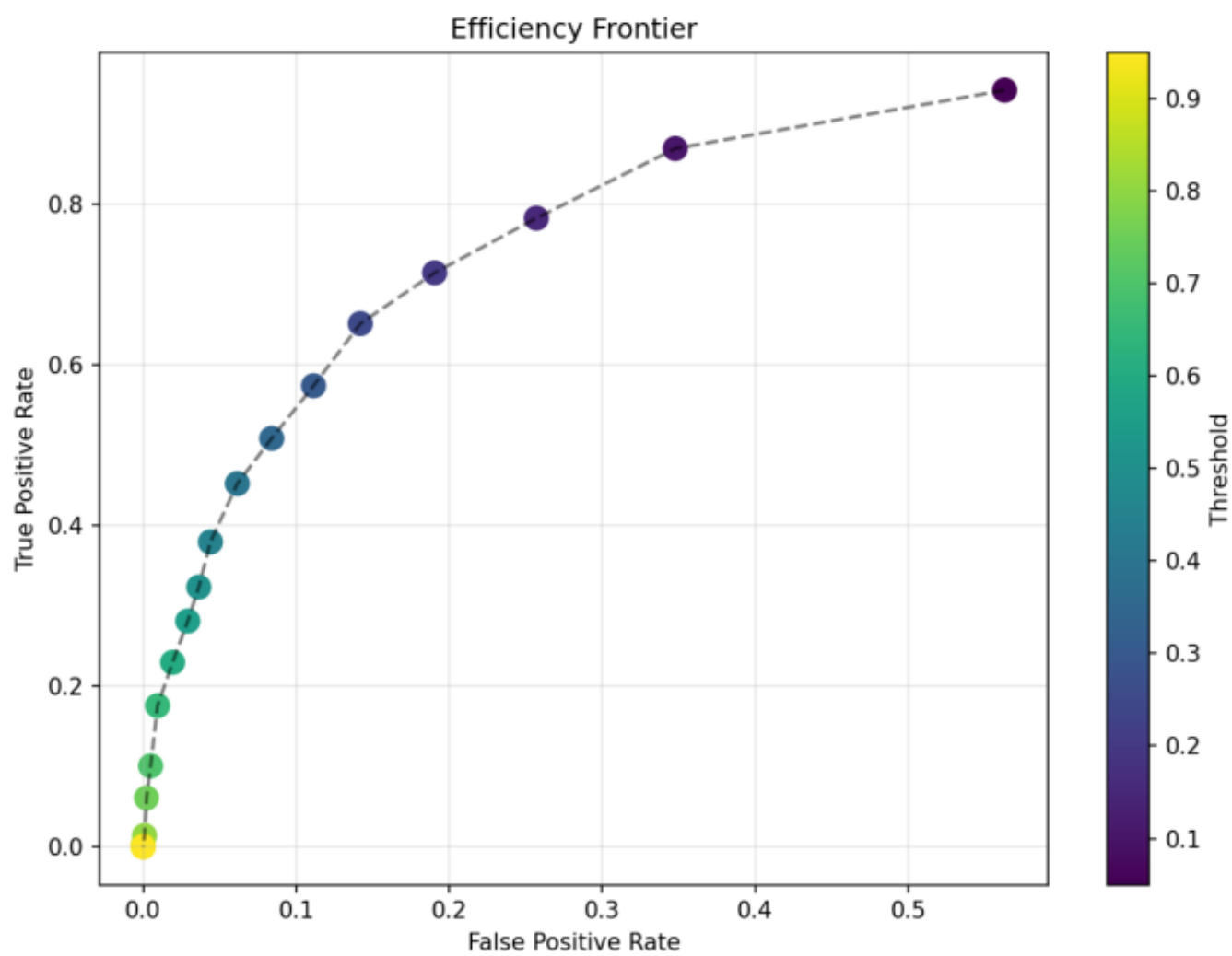
- fpr: 0.0364
- tn: 2144
- fp: 81
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.5631
- fpr_at_thresholds.t_0.10: 0.3479
- fpr_at_thresholds.t_0.15: 0.2571
- fpr_at_thresholds.t_0.20: 0.1906
- fpr_at_thresholds.t_0.25: 0.1420
- fpr_at_thresholds.t_0.30: 0.1115
- fpr_at_thresholds.t_0.35: 0.0840
- fpr_at_thresholds.t_0.40: 0.0616
- fpr_at_thresholds.t_0.45: 0.0440
- fpr_at_thresholds.t_0.50: 0.0364
- fpr_at_thresholds.t_0.55: 0.0292
- fpr_at_thresholds.t_0.60: 0.0193
- fpr_at_thresholds.t_0.65: 0.0094
- fpr_at_thresholds.t_0.70: 0.0049
- fpr_at_thresholds.t_0.75: 0.0022
- fpr_at_thresholds.t_0.80: 0.0009
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q3 - 2) Model Efficiency - Explanations

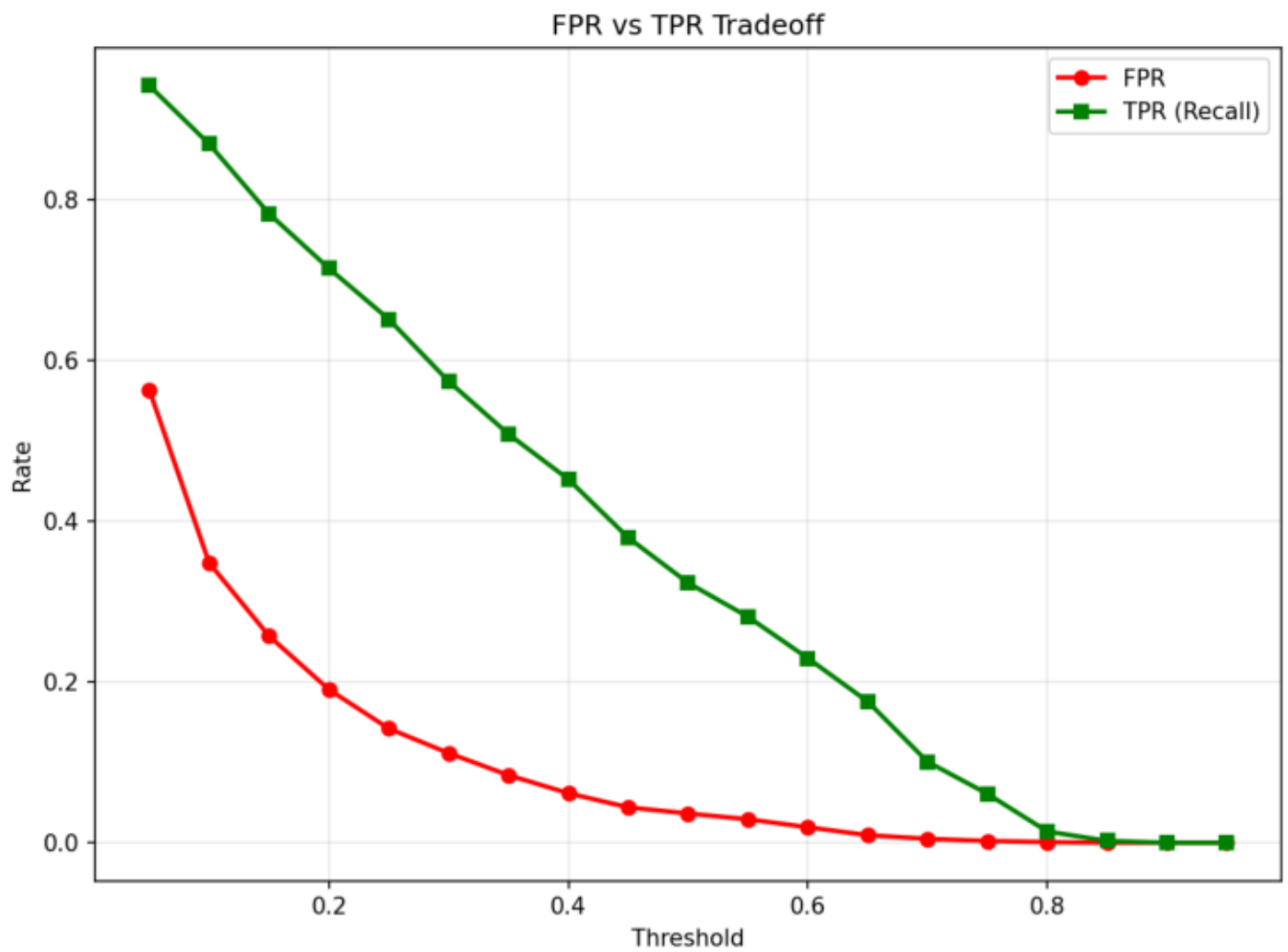
- At threshold 0.50, FPR=0.0364 (low); FP=81 out of 2225 negatives.
- At the same threshold, TPR=0.3232 with TP=138 and FN=289, showing the capture rate of positives.
- A threshold near 0.45 yields FPR $\approx$ 0.0440 with TPR $\approx$ 0.3794 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

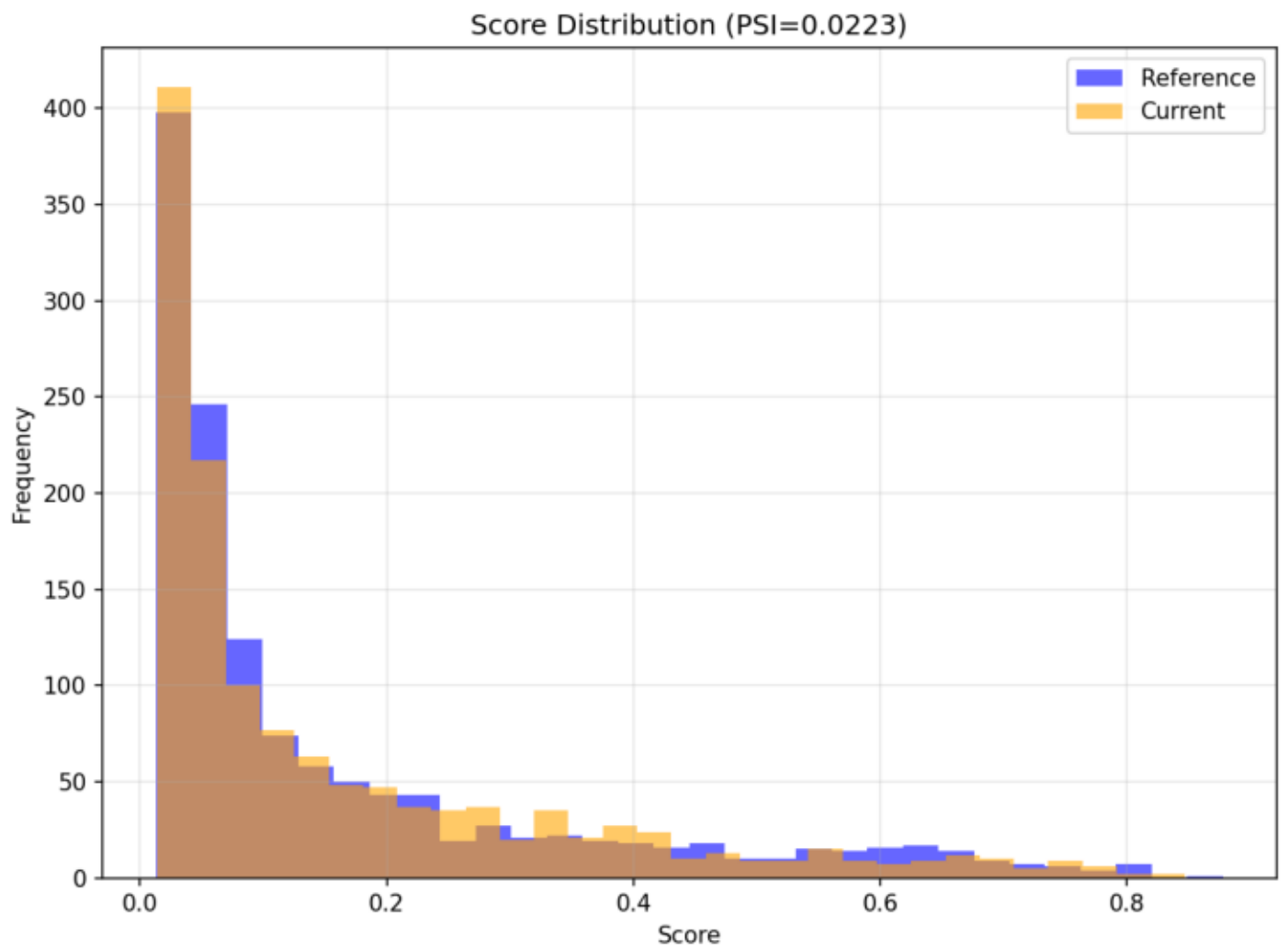


value_date_quarter=2024Q3 - 3) Model Stability

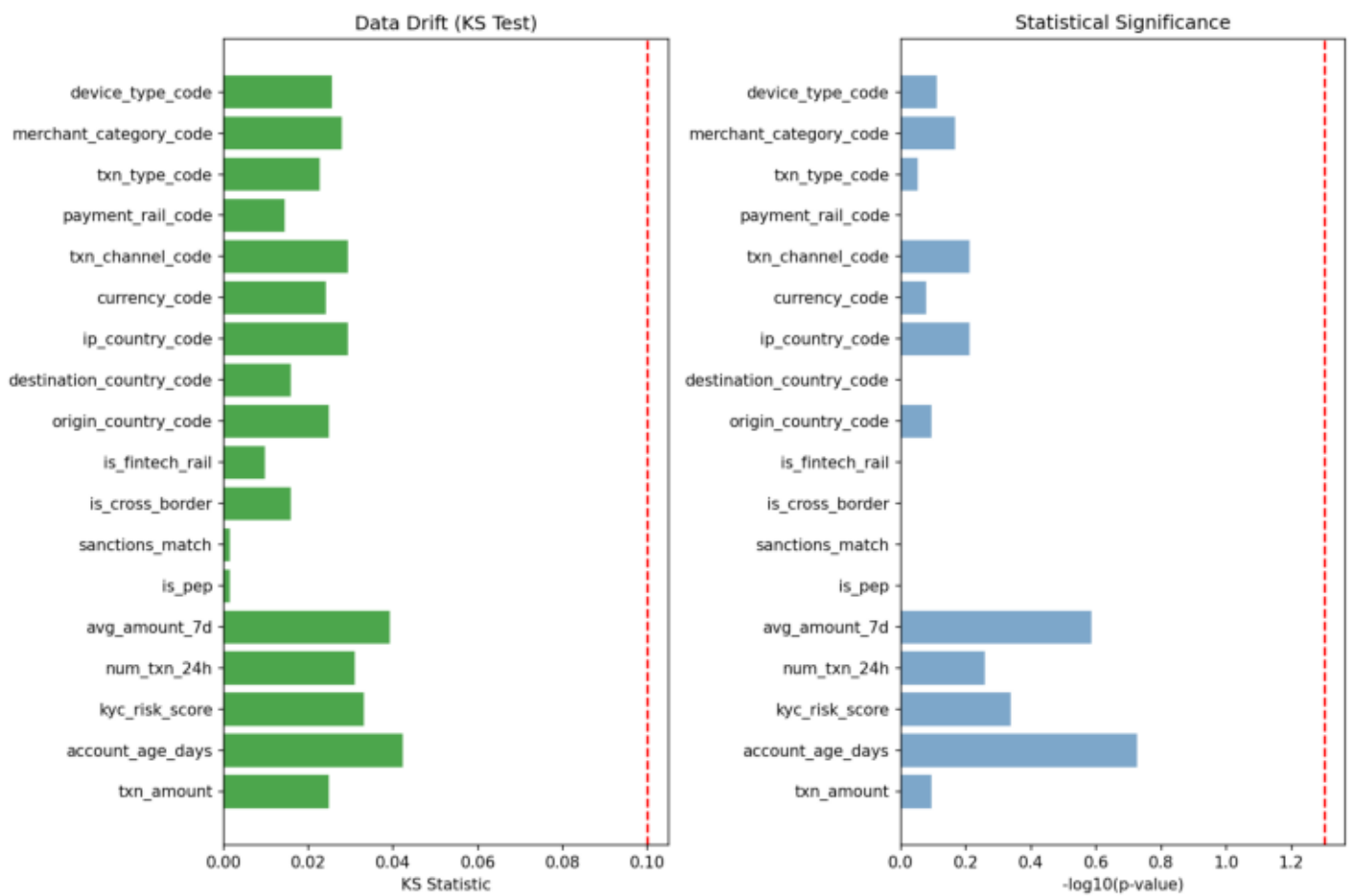
- psi: 0.0223
- cv_auc_roc_mean: 0.8434
- cv_auc_roc_std: 0.0139
- cv_auc_pr_mean: 0.5532
- cv_auc_pr_std: 0.0536
- bootstrap_auc_roc_mean: 0.8466
- bootstrap_auc_roc_ci_lower: 0.8205
- bootstrap_auc_roc_ci_upper: 0.8711
- concept_drift_detected: False
- concept_drift_score: 0.0112

value_date_quarter=2024Q3 - 3) Model Stability - Explanations

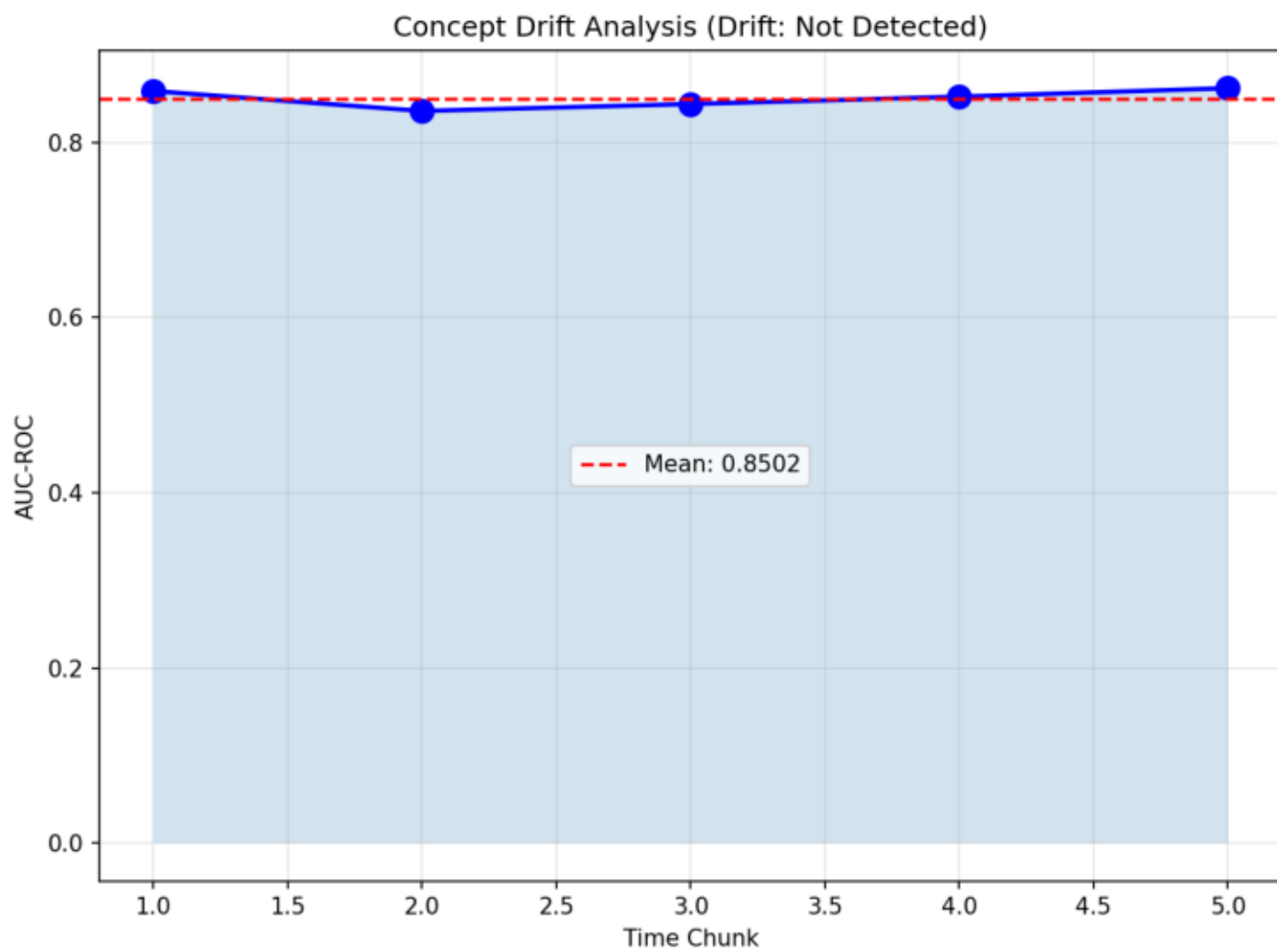
- PSI is 0.0223, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8434 ± 0.0139 (stable across folds).
- CV AUC-PR is 0.5532 ± 0.0536 (variable across folds).
- Bootstrap AUC-ROC mean is 0.8466 with 95% CI [0.8205, 0.8711] (width 0.0506).
- No concept drift detected (score 0.0112), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['account_age_days', 'avg_amount_7d', 'kyc_risk_score'].



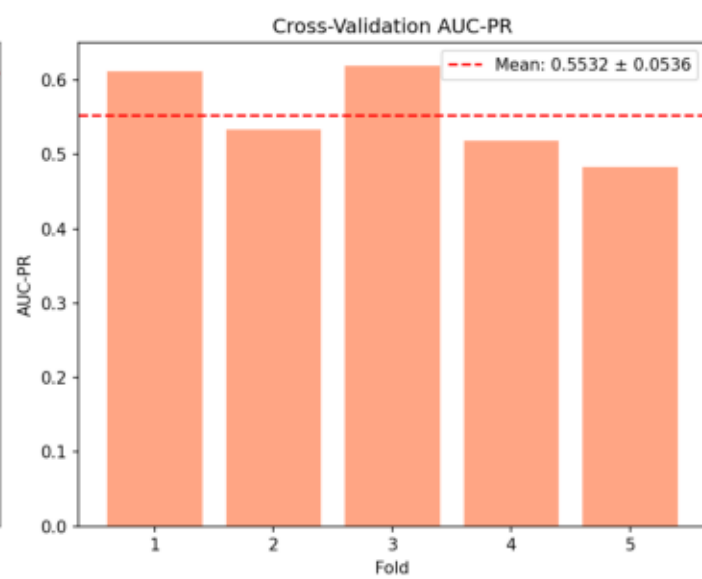
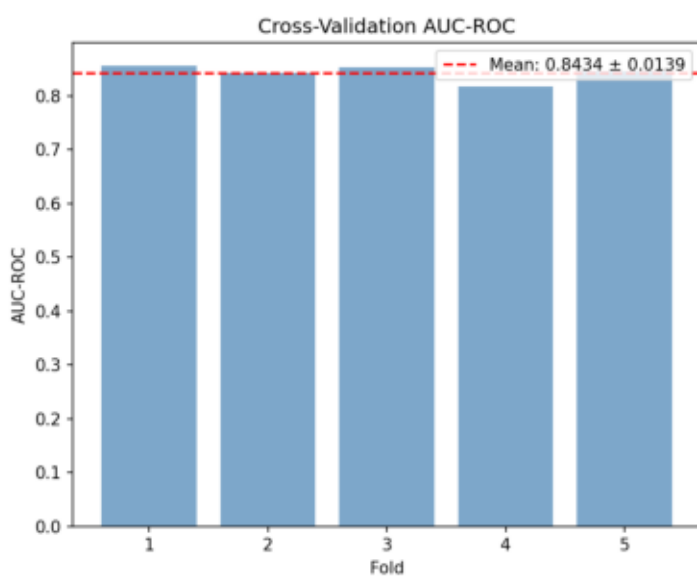
psi_distribution: Score distributions overlap consistent with PSI=0.0223.



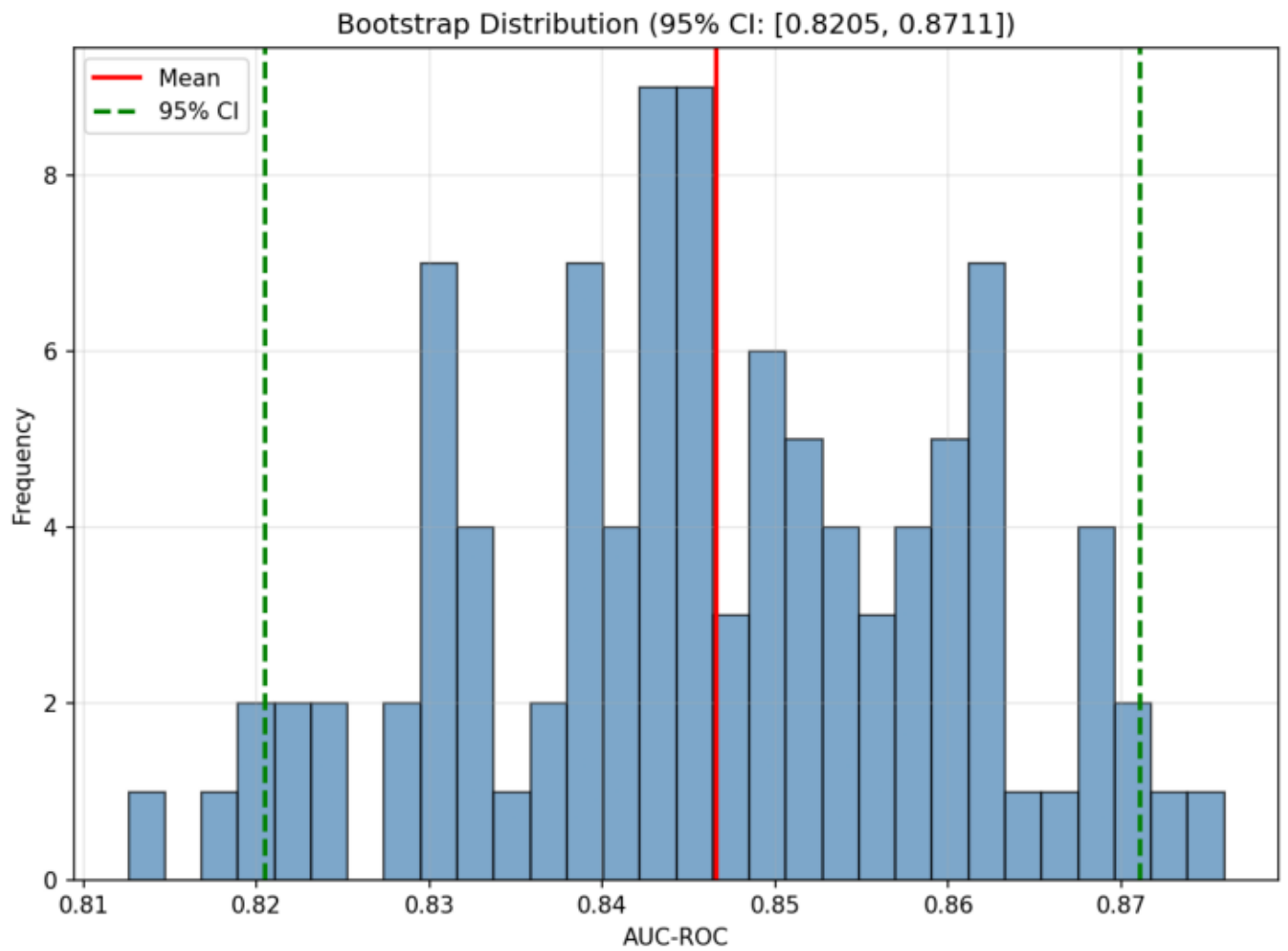
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



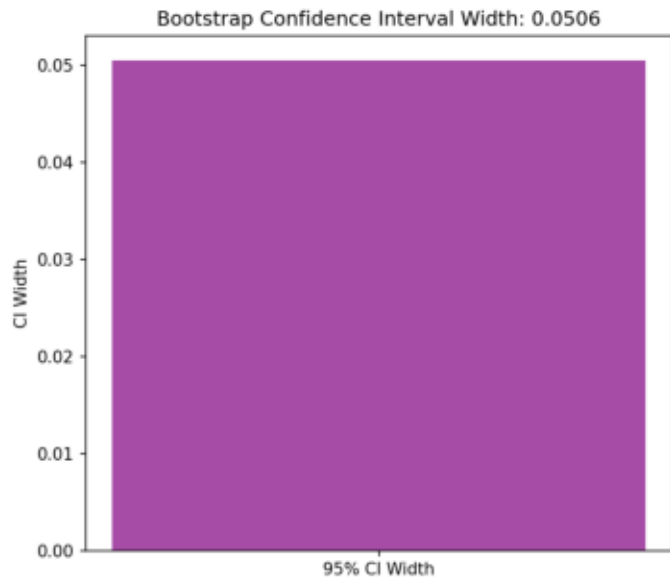
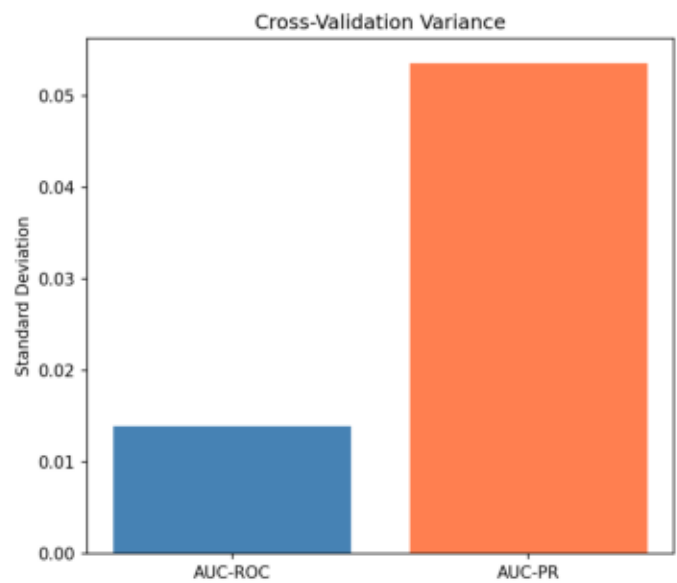
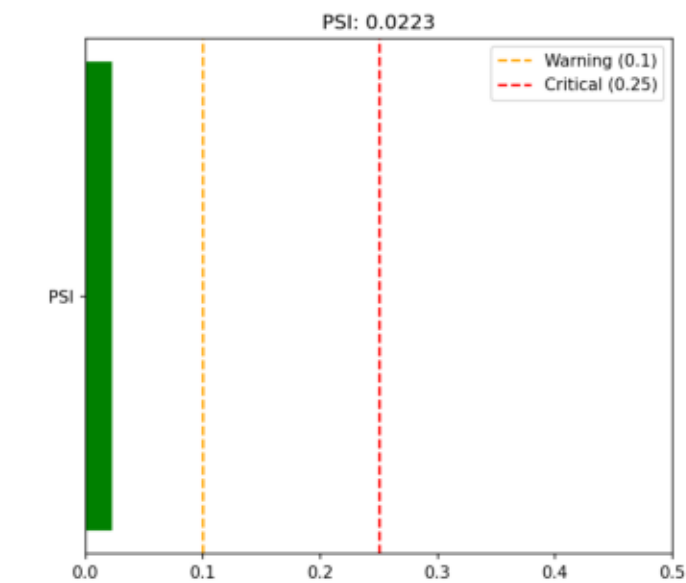
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8434 and PR 0.5532.



bootstrap_distribution: Bootstrap CI [0.8205, 0.8711] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0223 (OK)

CV AUC-ROC: 0.8434 ± 0.0139

CV AUC-PR: 0.5532 ± 0.0536

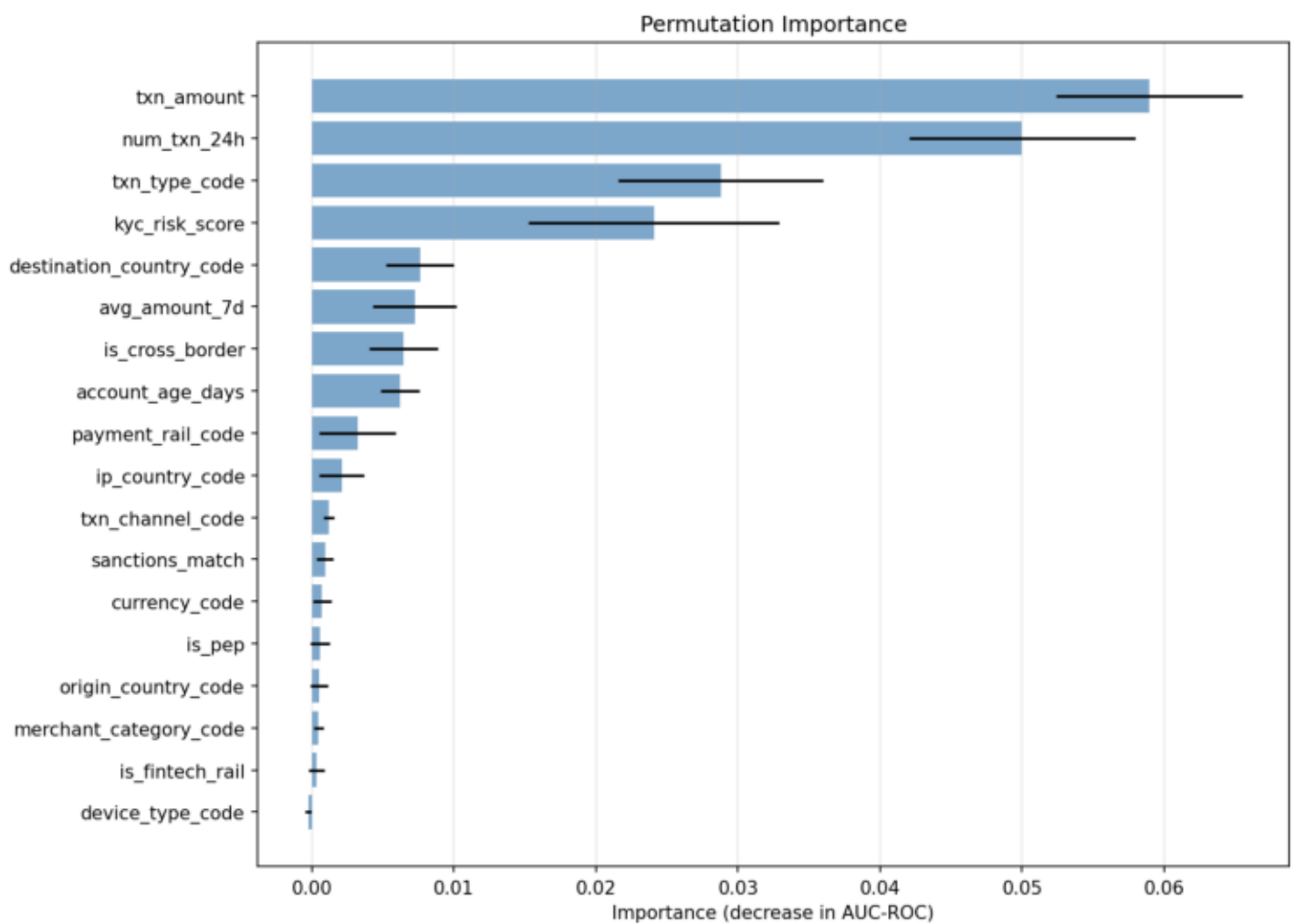
Bootstrap 95% CI: [0.8205, 0.8711]

value_date_quarter=2024Q3 - 4) Model Interpretability

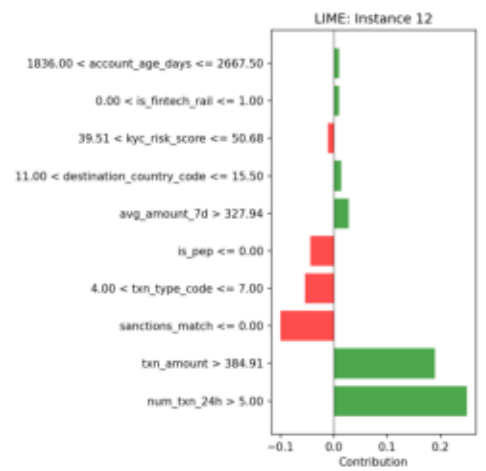
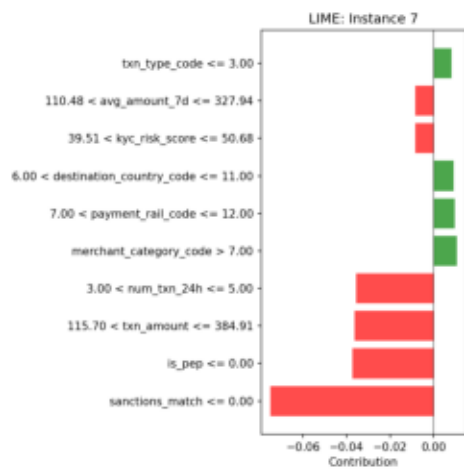
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'lime']
- shap_status: unavailable
- shap_reason: No module named 'shap'
- perm_top_features: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score', 'destination_country_code']...
- lime_instances: 3
- pdp_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q3 - 4) Model Interpretability - Explanations

- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'txn_type_code', 'kyc_risk_score', 'destination_country_code'].
- SHAP plots not generated because SHAP is unavailable (No module named 'shap').
- LIME generated local explanations for 3 instances; top contributors for instance_7: ['sanctions_match <= 0.00', 'is_pep <= 0.00', '115.70 < txn_amount <= 384.91', '3.00 < num_txn_24h <= 5.00', 'merchant_category_code > 7.00'].
- PDP plots show average effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'ip_country_code'].
- ICE plots show individual-level effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score'].

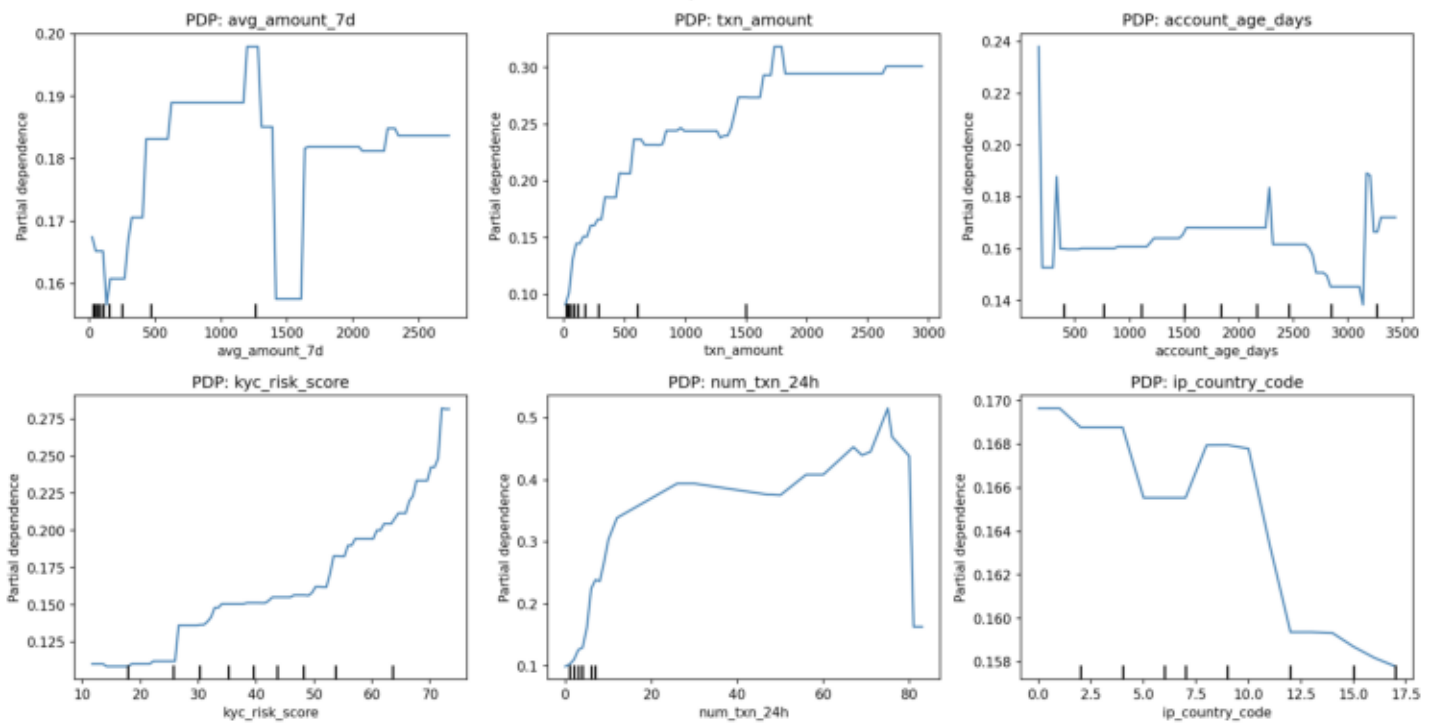


permutation_importance: Bars show how much AUC drops when each feature is permuted.



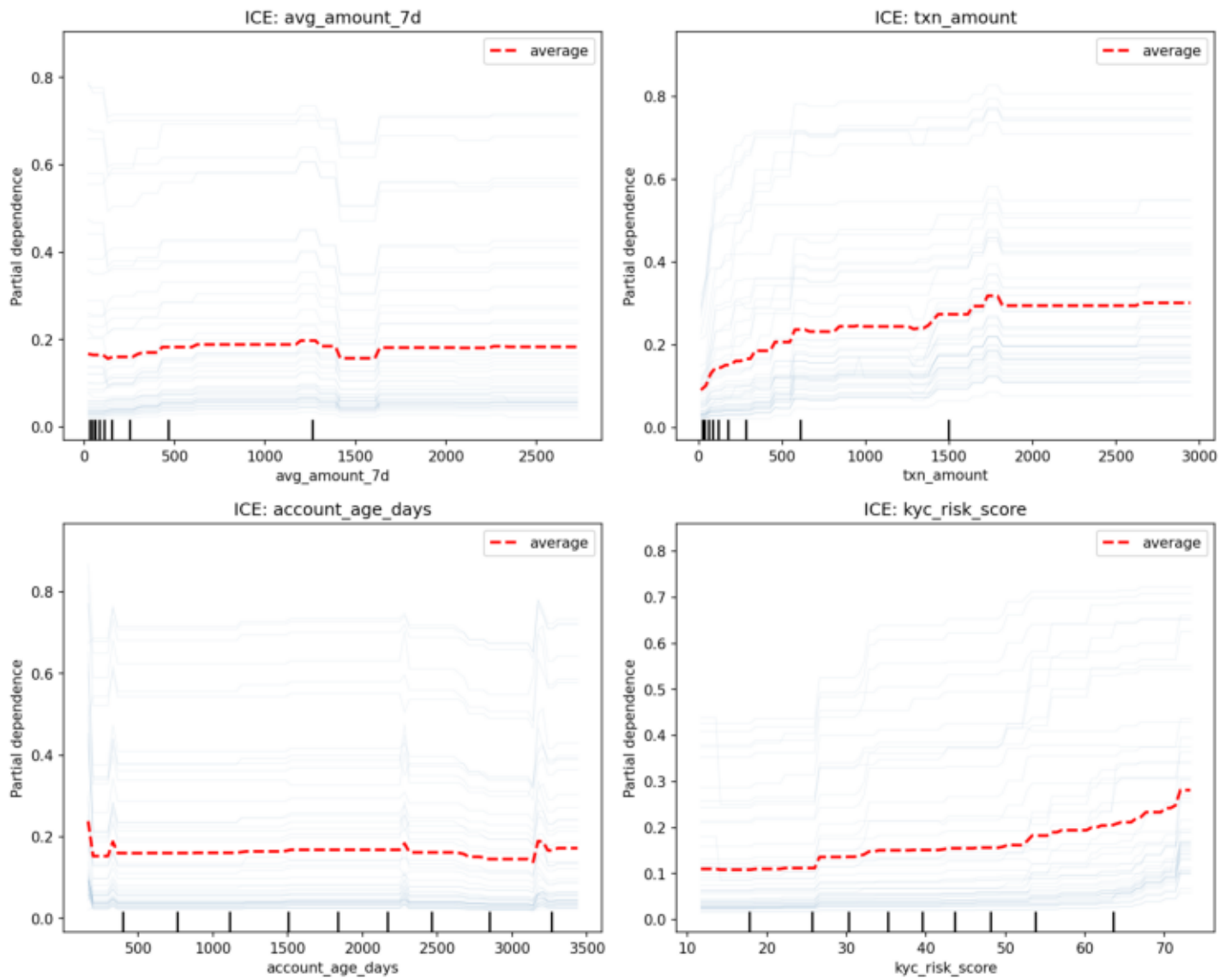
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



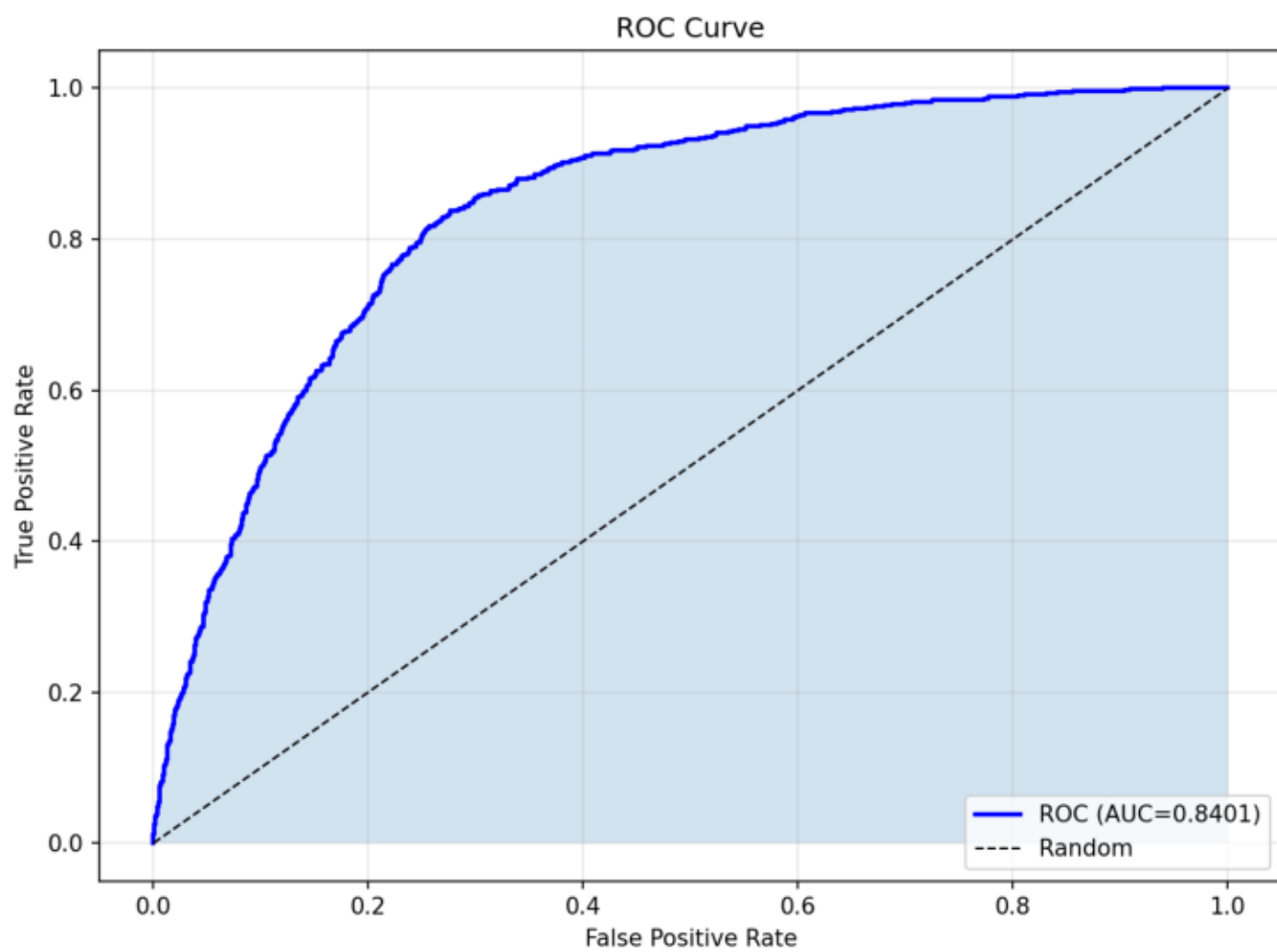
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q4 - 1) Model Effectiveness

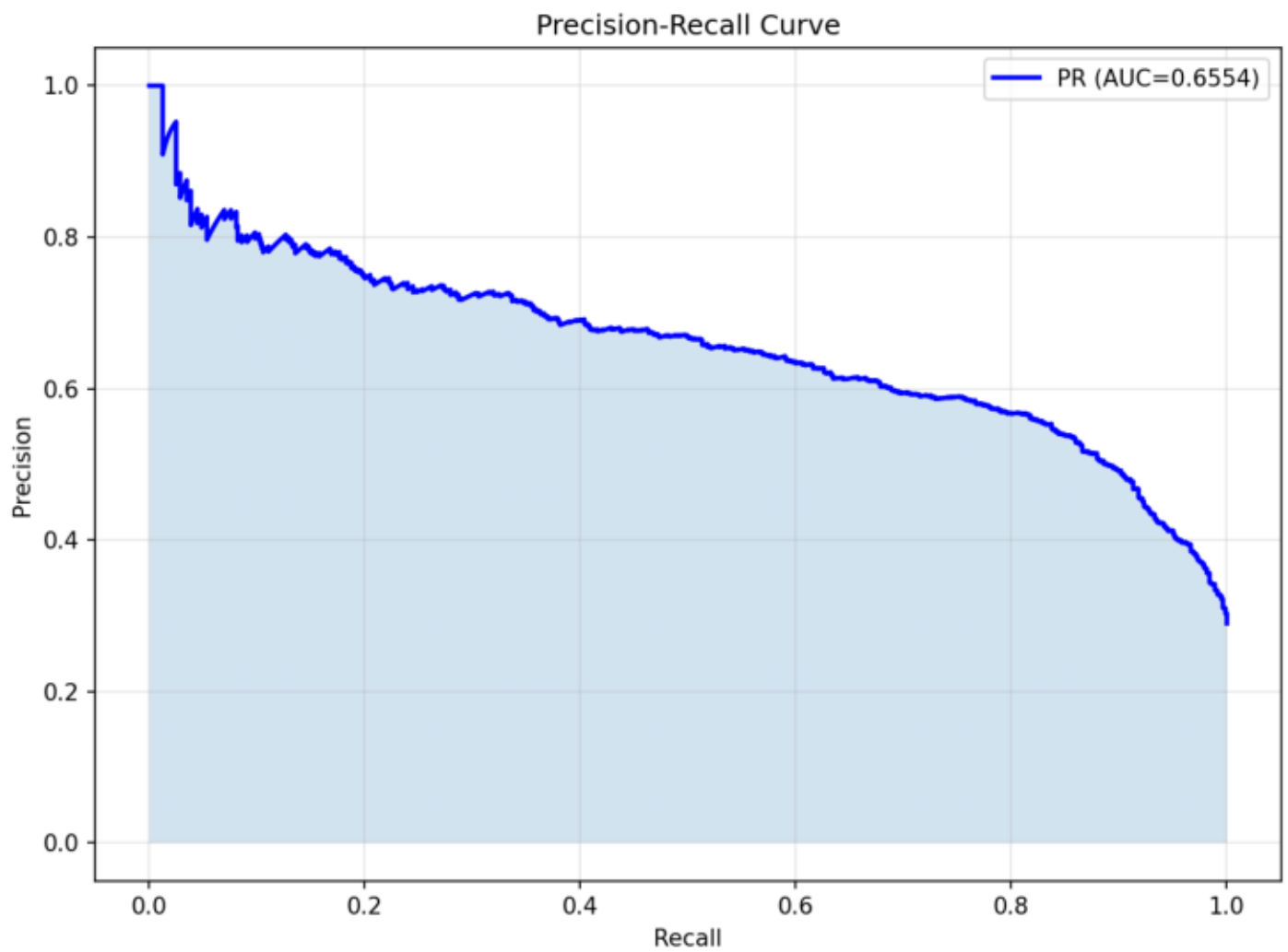
- auc_roc: 0.8401
- auc_pr: 0.6554
- precision: 0.6536
- recall: 0.5366
- f1: 0.5894
- ks_statistic: 0.5595
- ks_threshold: 0.2400
- confusion_matrix.TN: 1744
- confusion_matrix.FP: 229
- confusion_matrix.FN: 373
- confusion_matrix.TP: 432
- precision_at_k.10: 1.0000
- precision_at_k.50: 0.8200
- precision_at_k.100: 0.8000
- precision_at_k.200: 0.7600
- precision_at_k.500: 0.6780
- recall_at_k.10: 0.0124
- recall_at_k.50: 0.0509
- recall_at_k.100: 0.0994
- recall_at_k.200: 0.1888
- recall_at_k.500: 0.4211

value_date_quarter=2024Q4 - 1) Model Effectiveness - Explanations

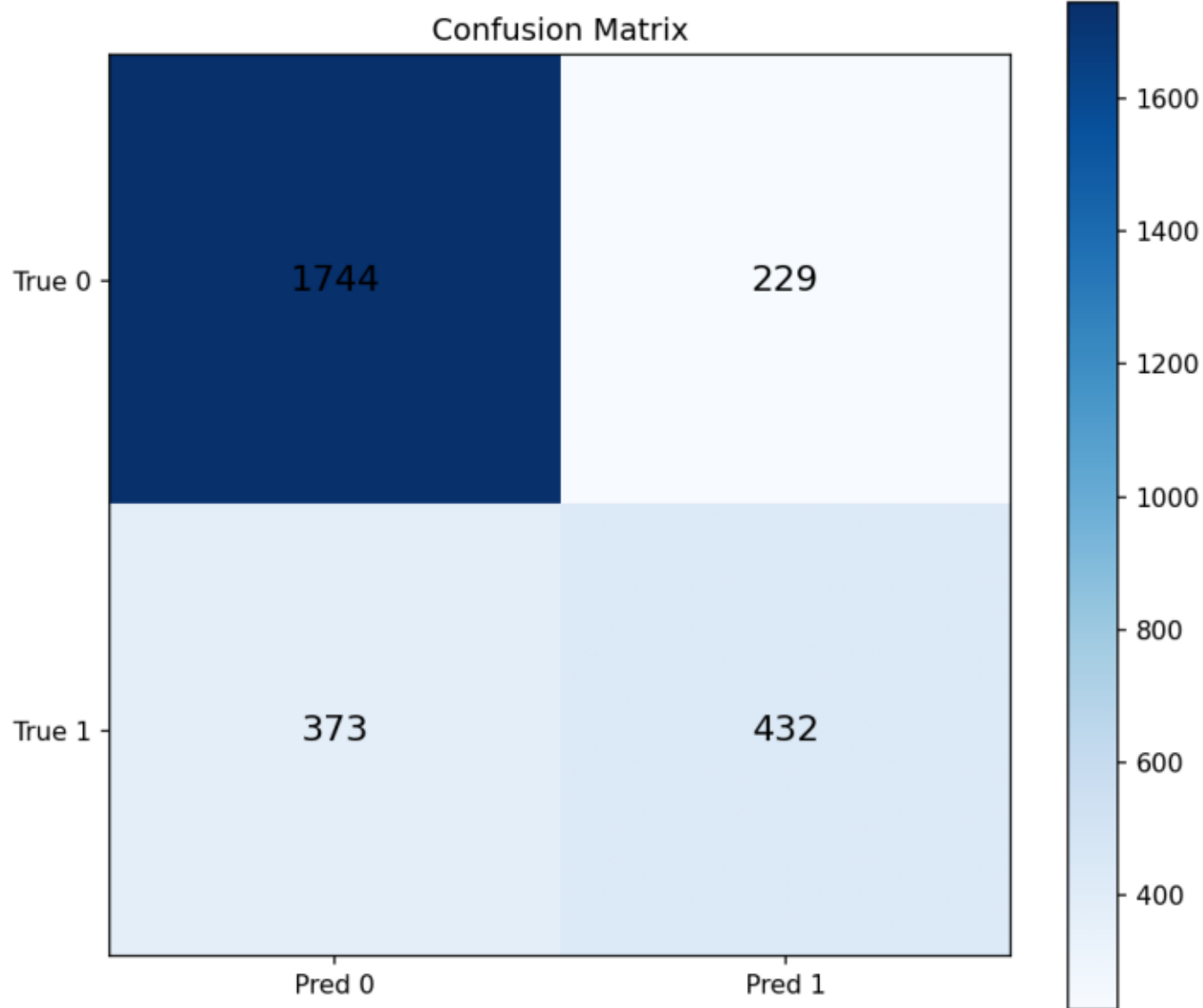
- Positive rate is 28.98%; AUC-ROC is 0.8401 (strong ranking).
- AUC-PR is 0.6554, above the base rate 28.98%, indicating meaningful lift.
- At threshold 0.50, precision=0.6536, recall=0.5366, F1=0.5894 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=432, FP=229, FN=373, TN=1744 (FPR=11.61%, FNR=46.34%).
- KS is 0.5595 at threshold 0.24, indicating very strong separation.
- Top-10: precision=1.0000, recall=0.0124 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.6780, recall=0.4211 shows how much coverage you get with a larger queue.



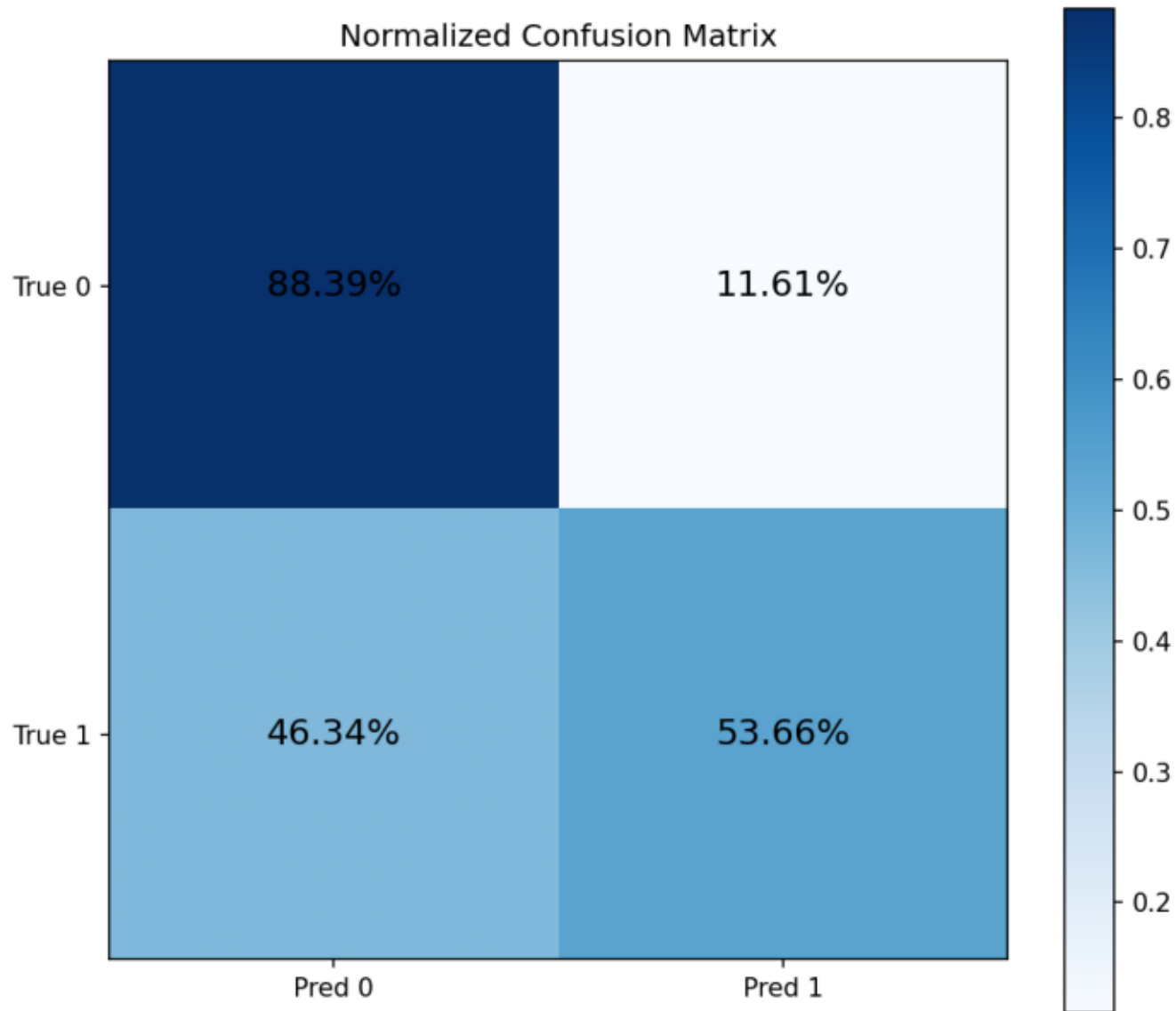
roc_curve: AUC-ROC=0.8401; curve above diagonal indicates strong separation.



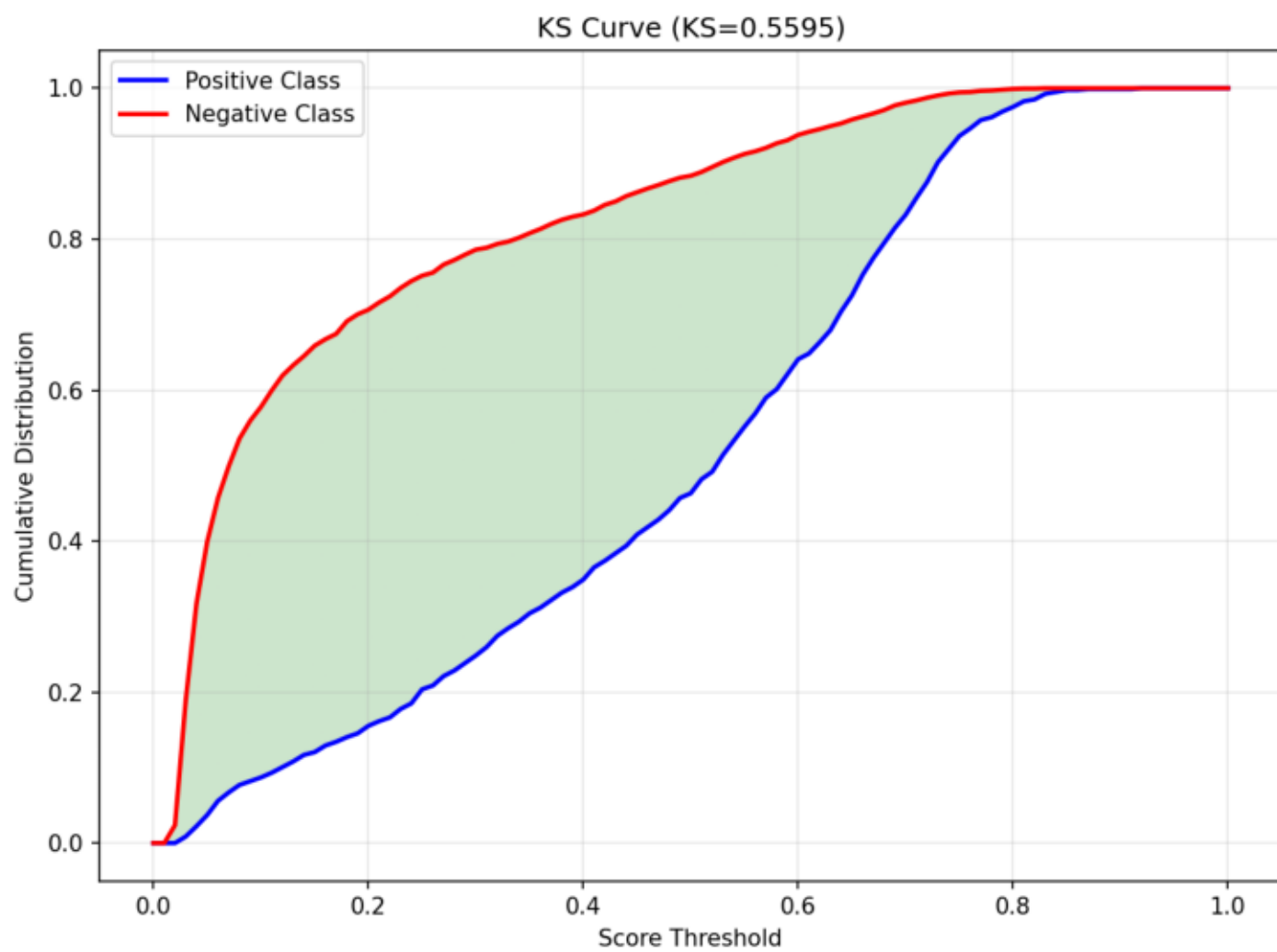
pr_curve: AUC-PR=0.6554 vs baseline 28.98%; higher curve implies better precision at recall.



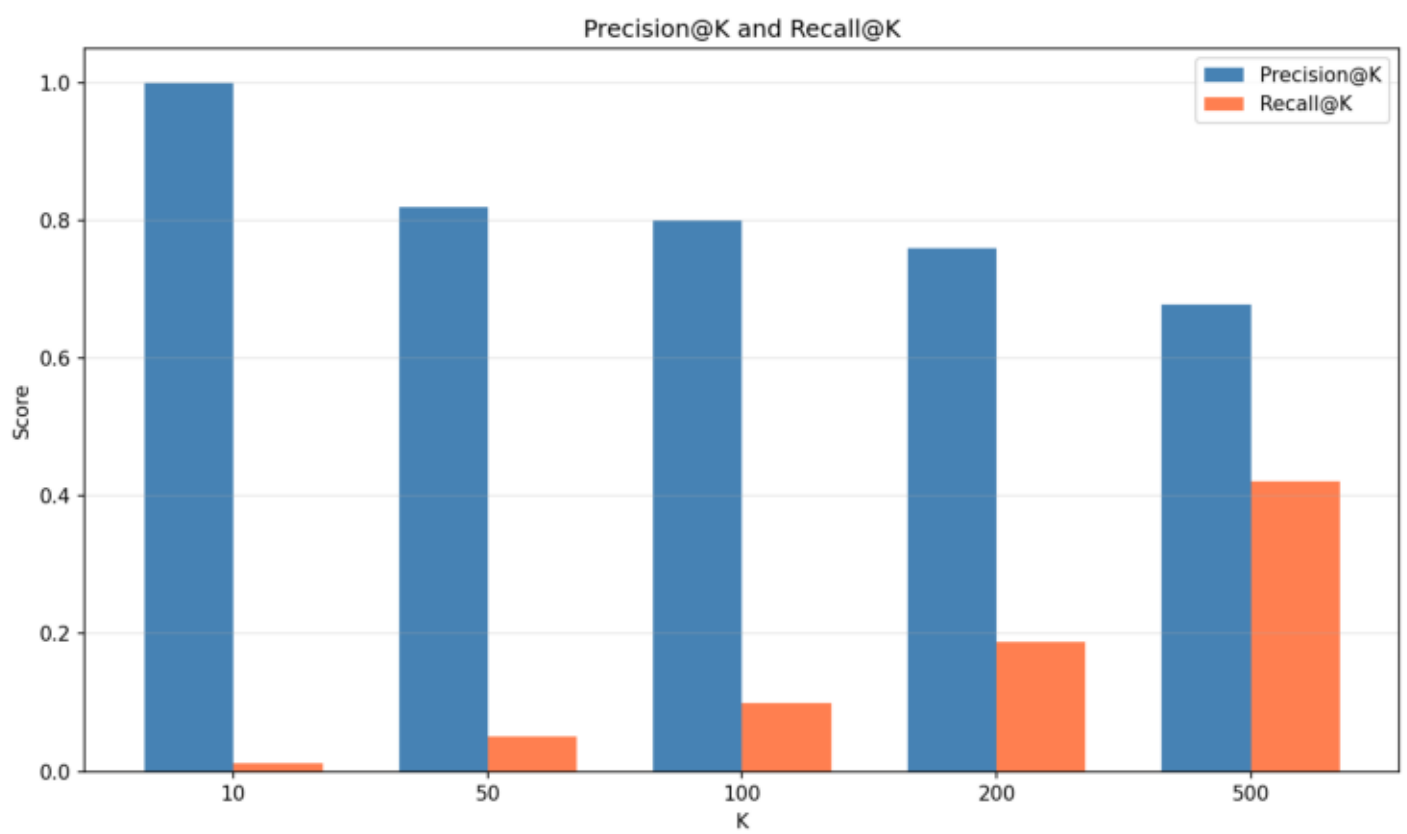
confusion_matrix: At threshold 0.50: TP=432, FP=229, FN=373, TN=1744.



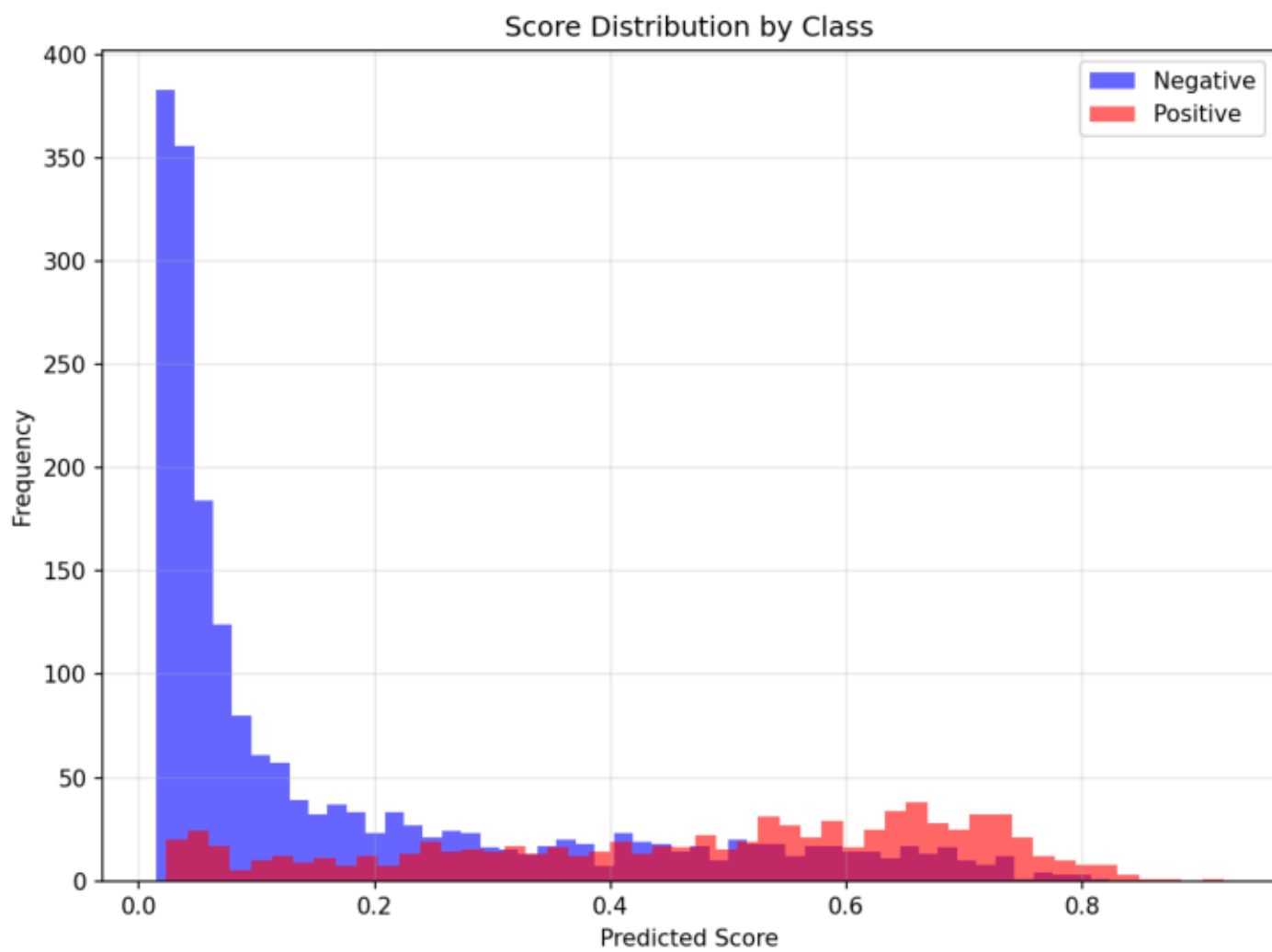
confusion_matrix_norm: Normalized rates: TPR=53.66%, TNR=88.39%.



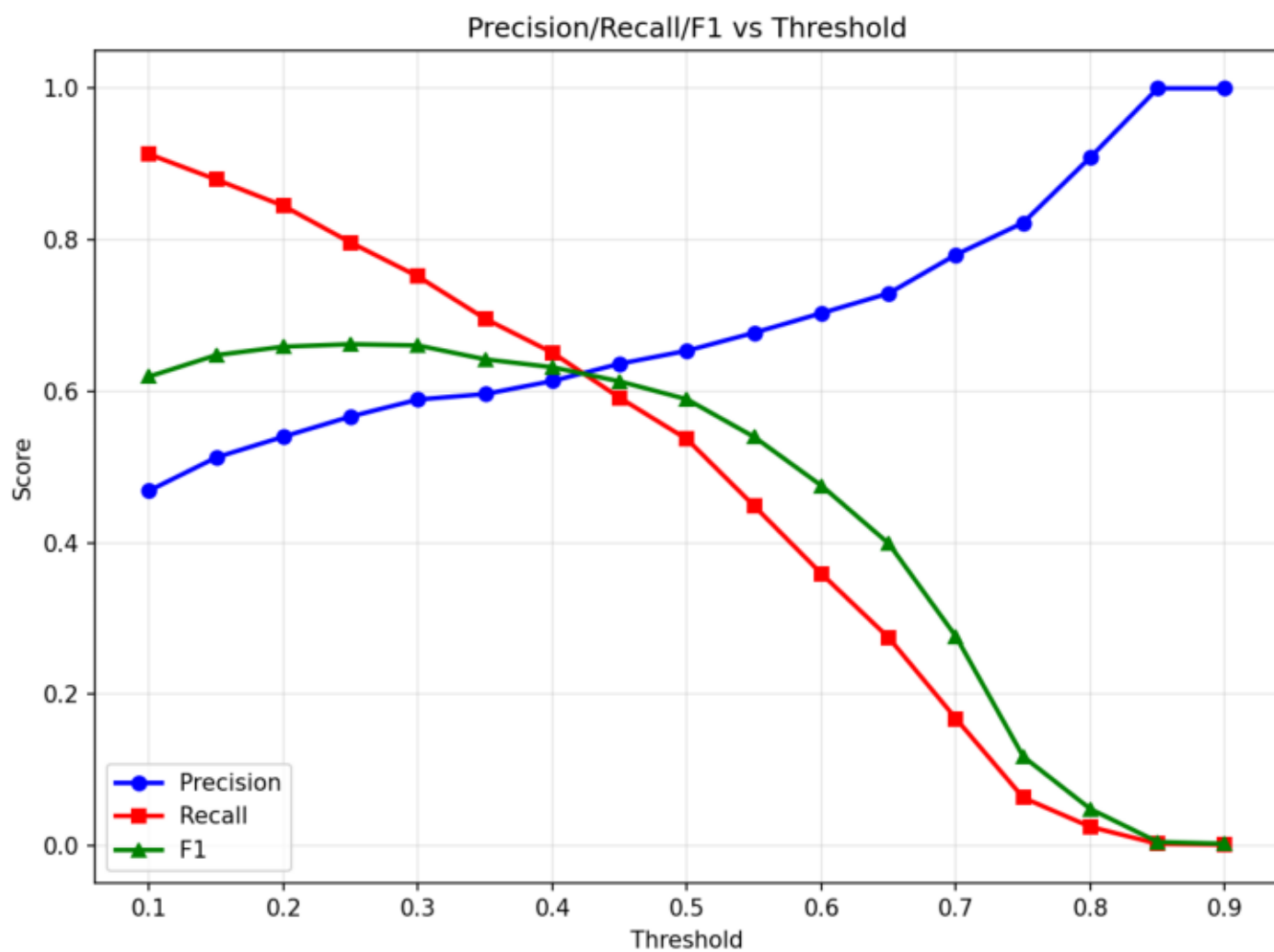
ks_curve: Maximum separation KS=0.5595 at threshold 0.24.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.474, negative=0.174 (larger gap is better).



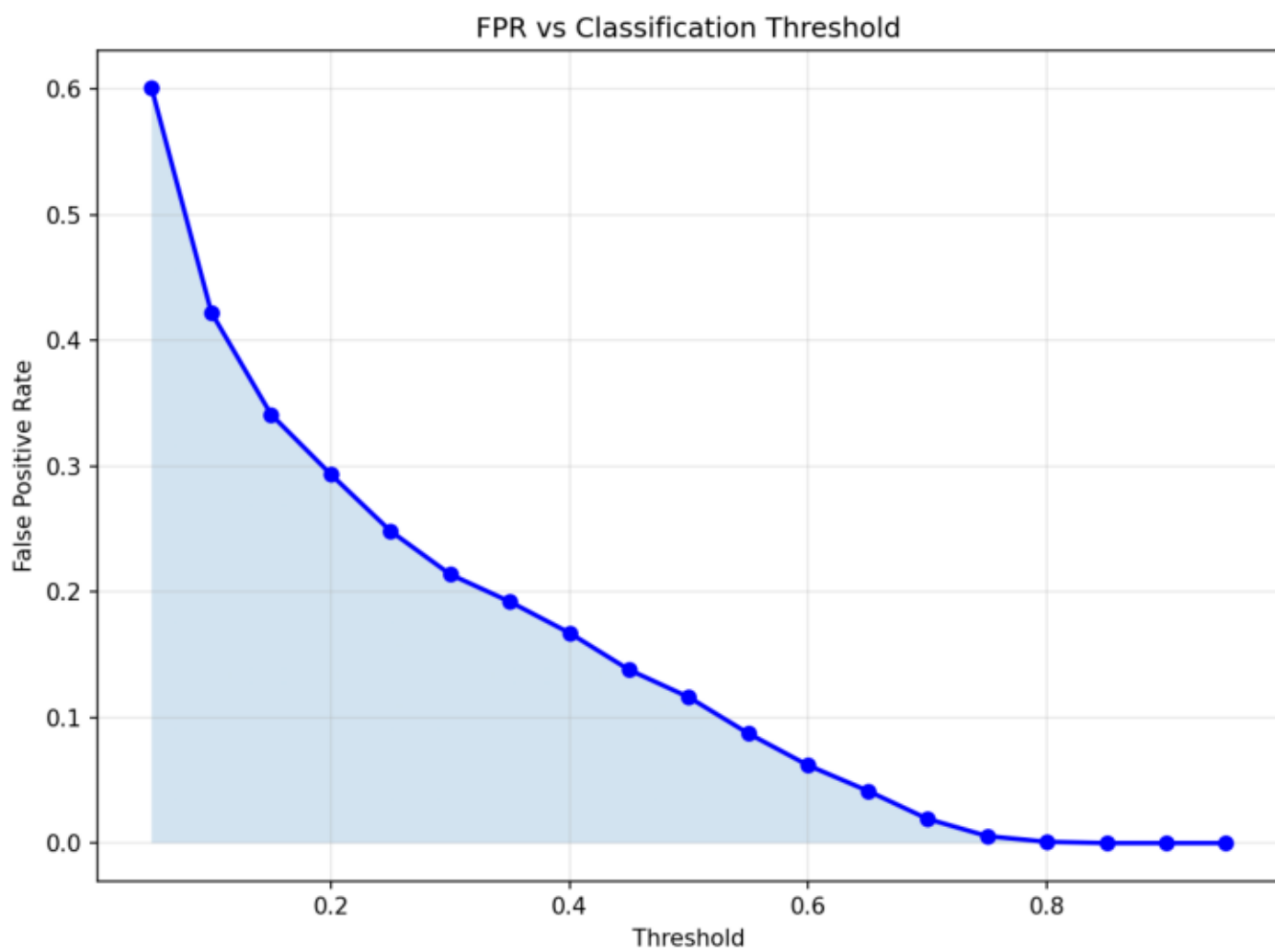
threshold_analysis: Best F1 in tested grid is 0.6622 at threshold 0.25.

value_date_quarter=2024Q4 - 2) Model Efficiency

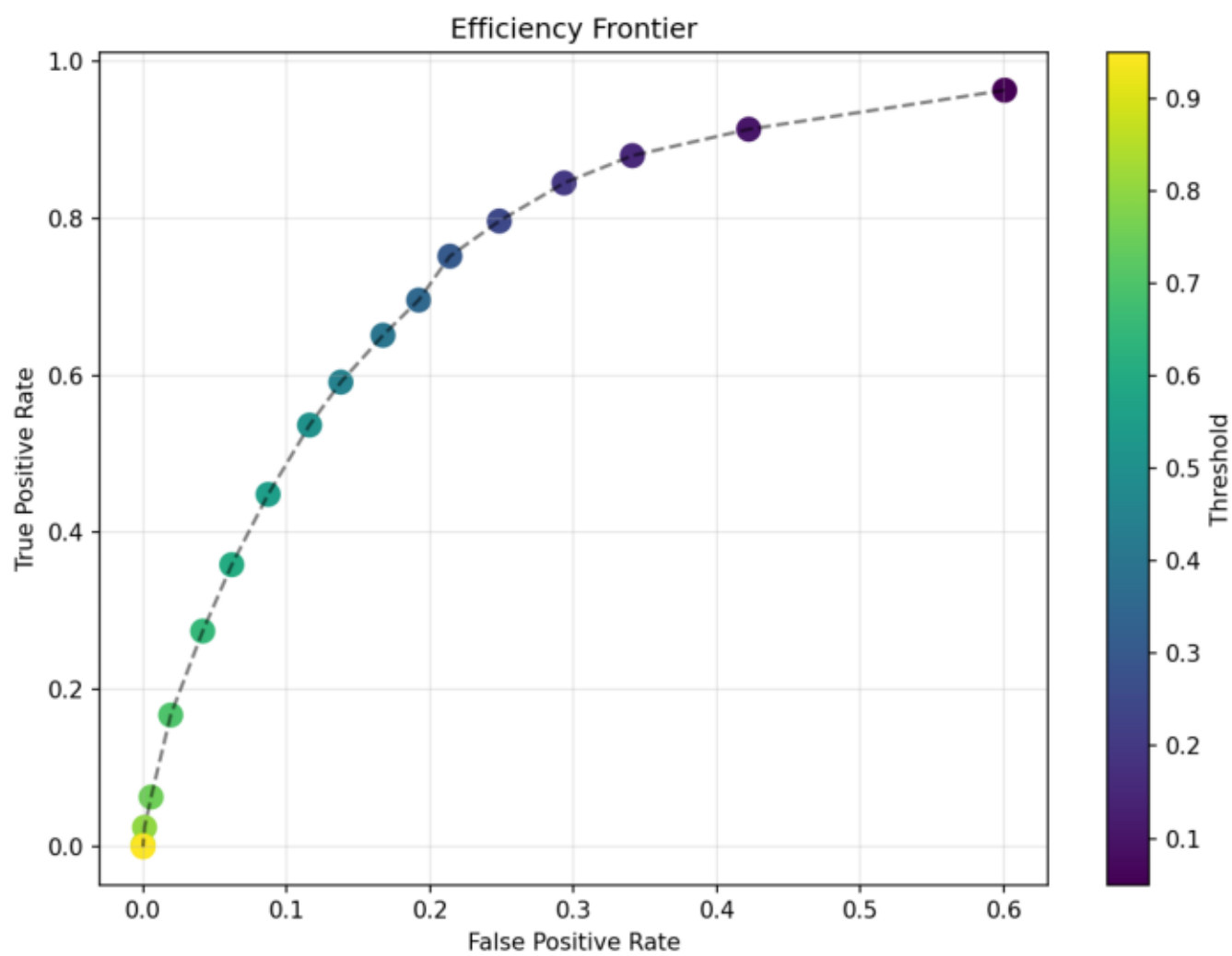
- fpr: 0.1161
- tn: 1744
- fp: 229
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.6006
- fpr_at_thresholds.t_0.10: 0.4222
- fpr_at_thresholds.t_0.15: 0.3411
- fpr_at_thresholds.t_0.20: 0.2935
- fpr_at_thresholds.t_0.25: 0.2484
- fpr_at_thresholds.t_0.30: 0.2139
- fpr_at_thresholds.t_0.35: 0.1921
- fpr_at_thresholds.t_0.40: 0.1673
- fpr_at_thresholds.t_0.45: 0.1379
- fpr_at_thresholds.t_0.50: 0.1161
- fpr_at_thresholds.t_0.55: 0.0872
- fpr_at_thresholds.t_0.60: 0.0618
- fpr_at_thresholds.t_0.65: 0.0416
- fpr_at_thresholds.t_0.70: 0.0193
- fpr_at_thresholds.t_0.75: 0.0056
- fpr_at_thresholds.t_0.80: 0.0010
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q4 - 2) Model Efficiency - Explanations

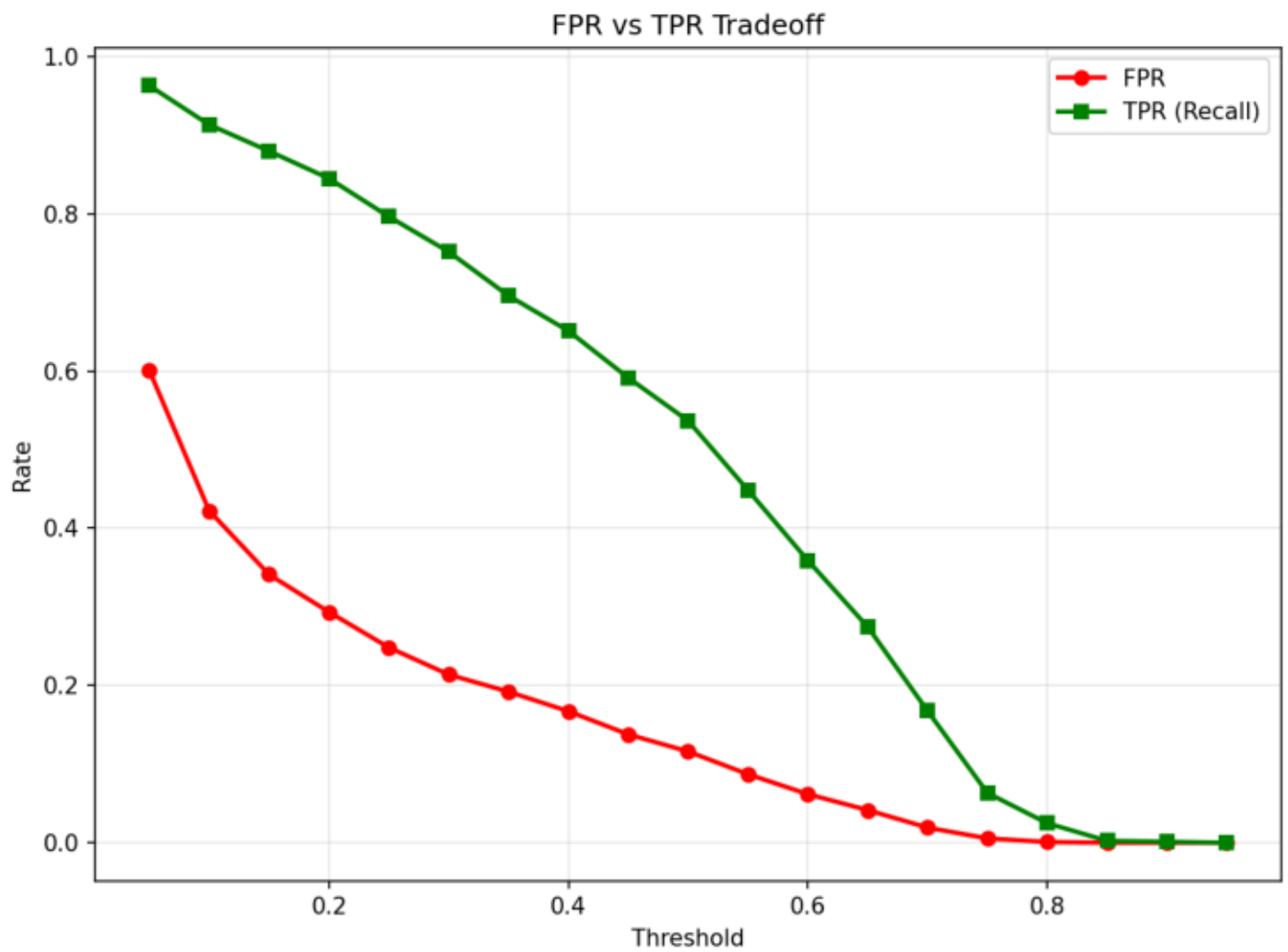
- At threshold 0.50, FPR=0.1161 (high); FP=229 out of 1973 negatives.
- At the same threshold, TPR=0.5366 with TP=432 and FN=373, showing the capture rate of positives.
- A threshold near 0.65 yields FPR $\approx$ 0.0416 with TPR $\approx$ 0.2745 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

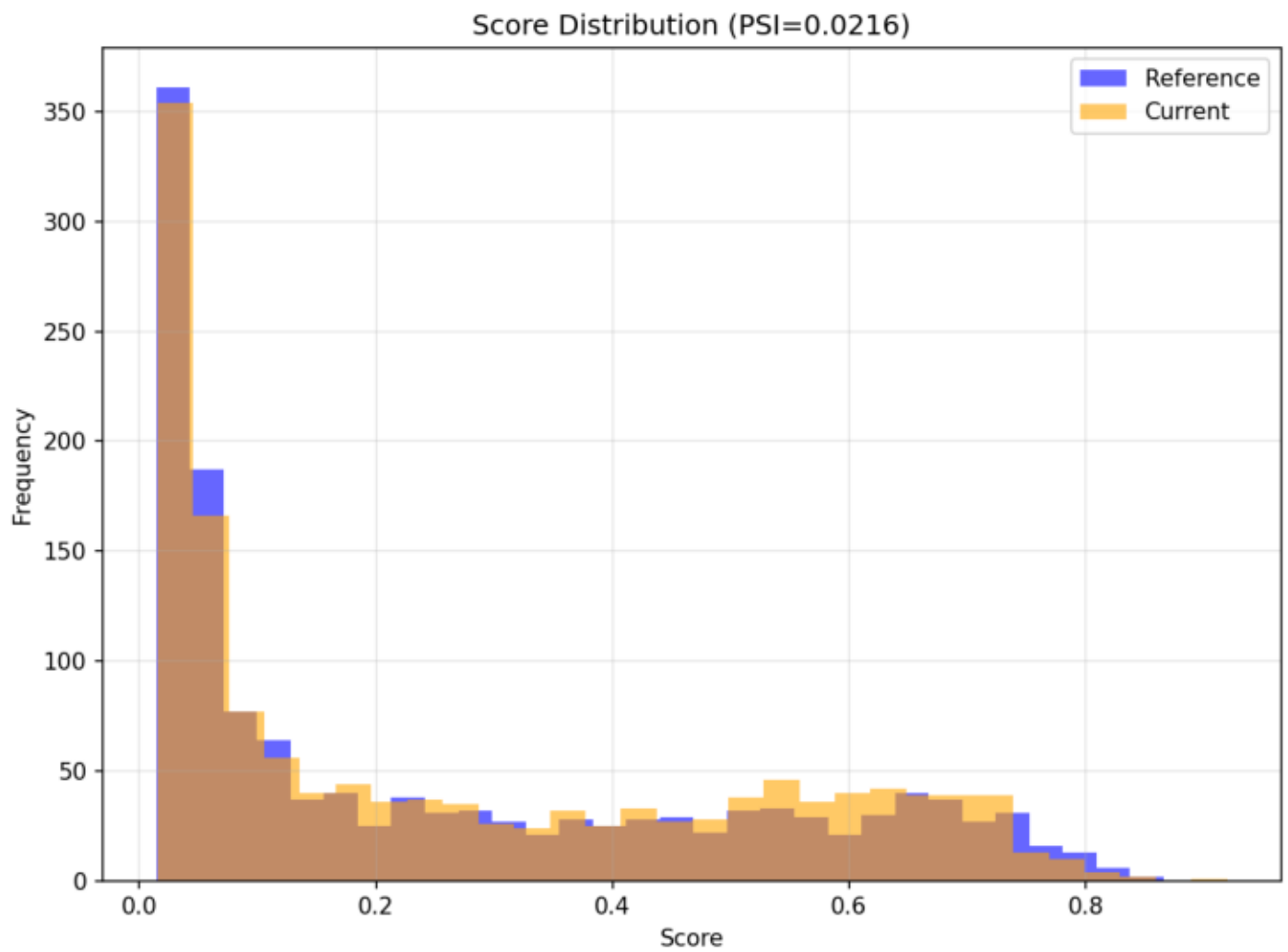


value_date_quarter=2024Q4 - 3) Model Stability

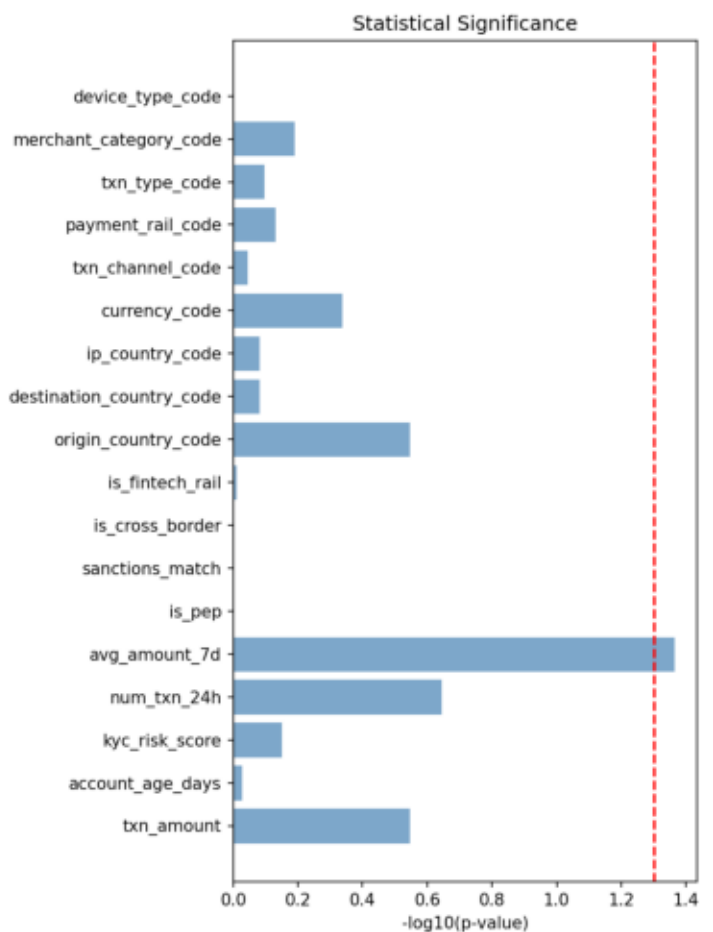
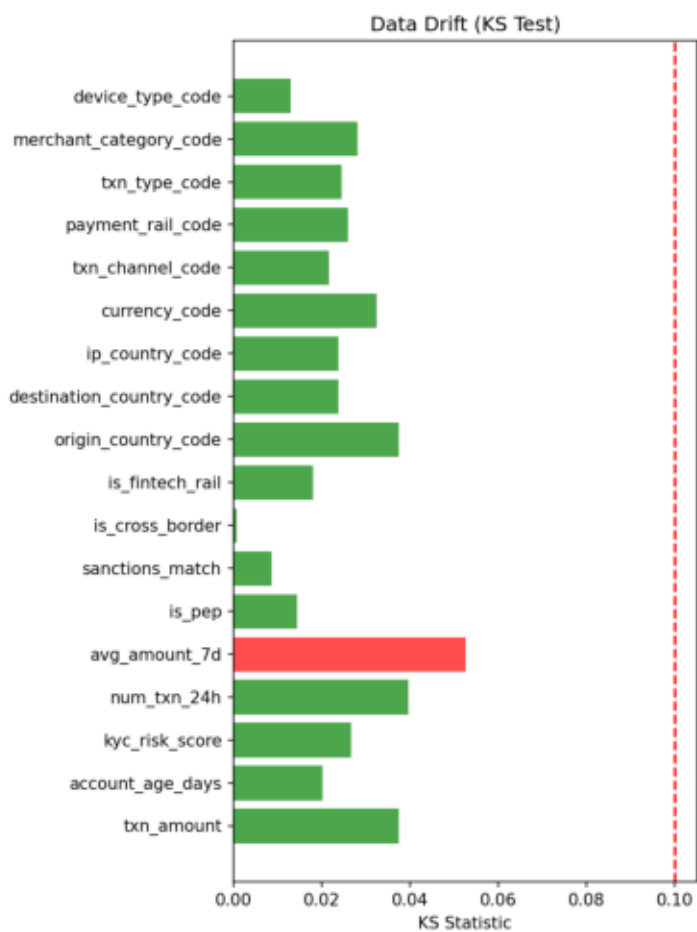
- psi: 0.0216
- cv_auc_roc_mean: 0.8398
- cv_auc_roc_std: 0.0188
- cv_auc_pr_mean: 0.6578
- cv_auc_pr_std: 0.0323
- bootstrap_auc_roc_mean: 0.8129
- bootstrap_auc_roc_ci_lower: 0.7923
- bootstrap_auc_roc_ci_upper: 0.8354
- concept_drift_detected: False
- concept_drift_score: 0.0261

value_date_quarter=2024Q4 - 3) Model Stability - Explanations

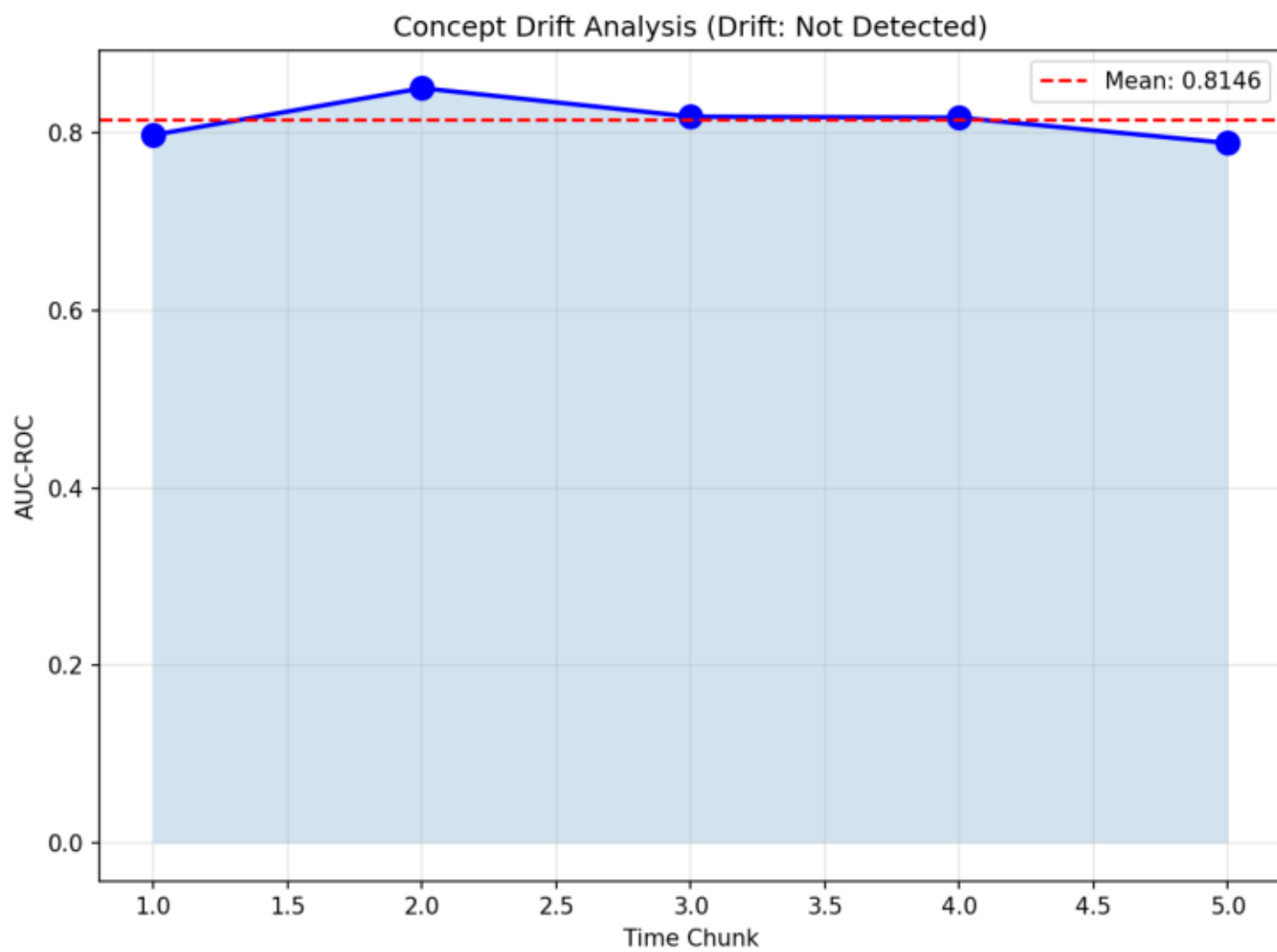
- PSI is 0.0216, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8398 ± 0.0188 (moderate across folds).
- CV AUC-PR is 0.6578 ± 0.0323 (moderate across folds).
- Bootstrap AUC-ROC mean is 0.8129 with 95% CI [0.7923, 0.8354] (width 0.0431).
- No concept drift detected (score 0.0261), suggesting stable performance over time.
- Data drift flagged 1/18 numeric features; top drift signals: ['avg_amount_7d', 'num_txn_24h', 'txn_amount'].



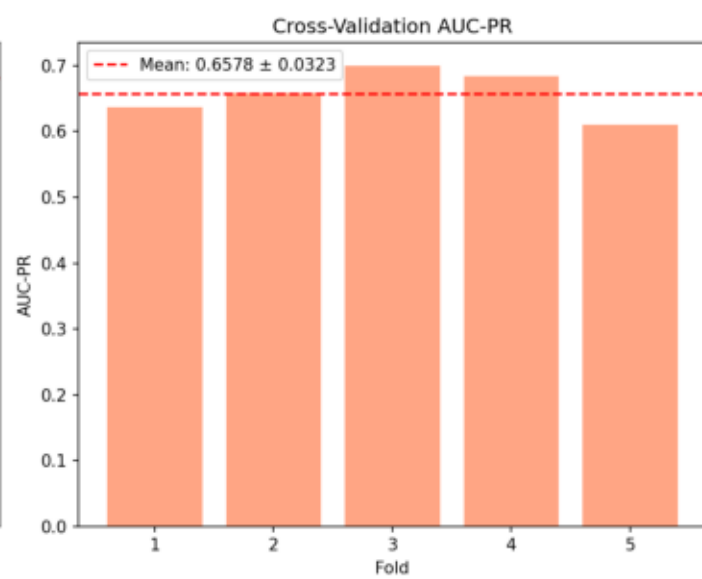
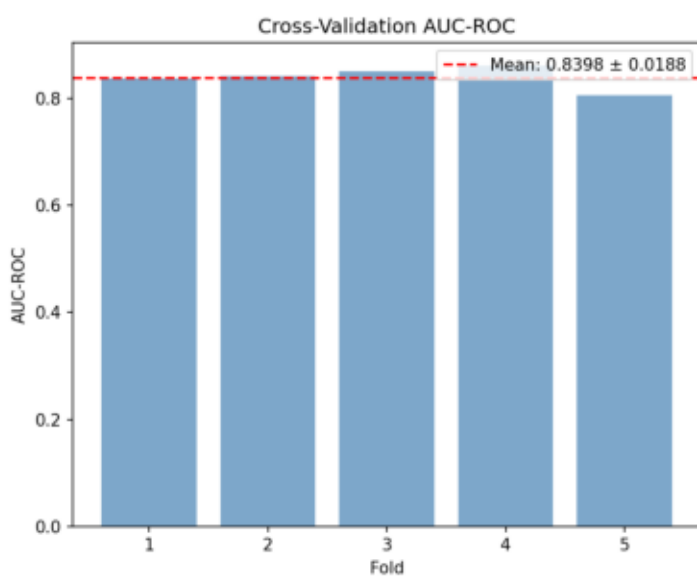
psi_distribution: Score distributions overlap consistent with PSI=0.0216.



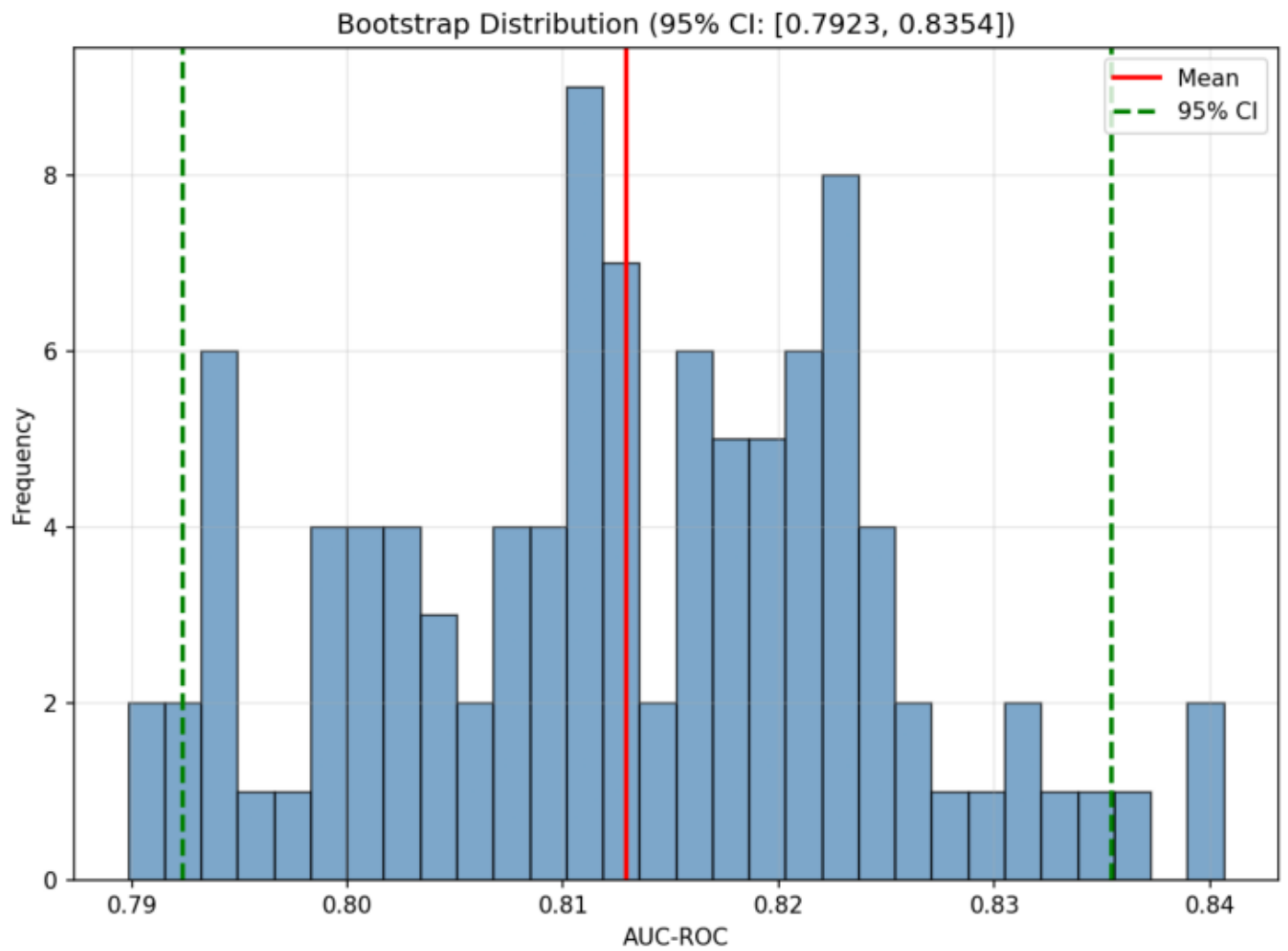
data_drift_heatmap: 1/18 features show drift; red bars highlight flagged features.



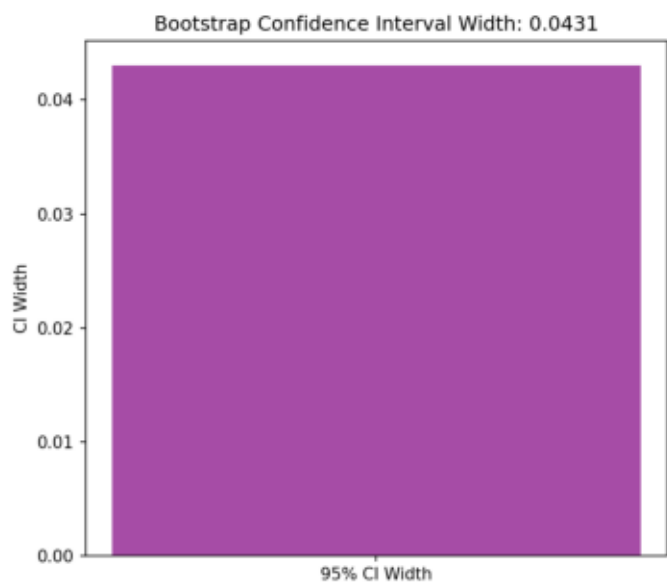
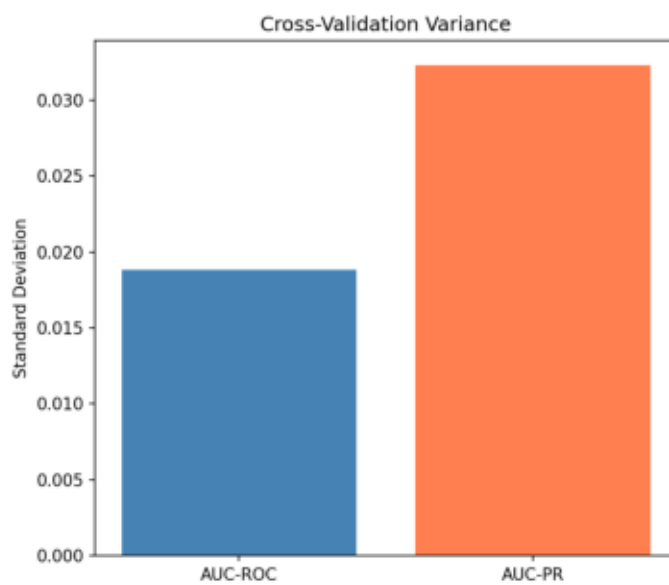
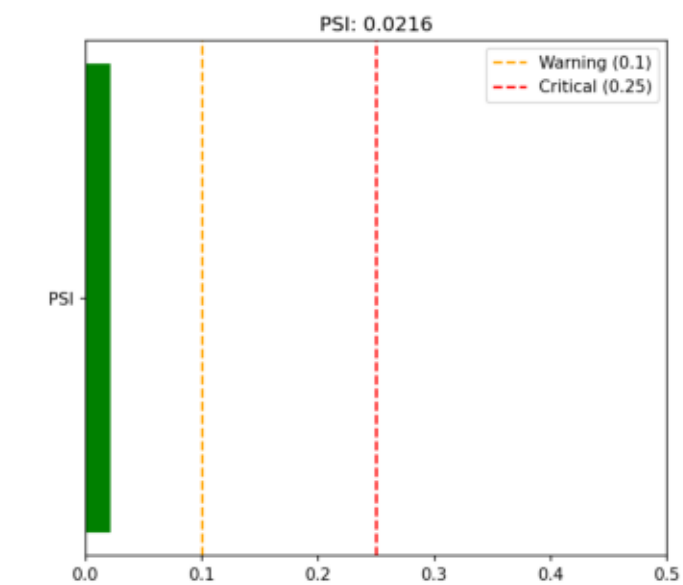
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8398 and PR 0.6578.



bootstrap_distribution: Bootstrap CI [0.7923, 0.8354] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0216 (OK)

CV AUC-ROC: 0.8398 $\pm$ 0.0188

CV AUC-PR: 0.6578 $\pm$ 0.0323

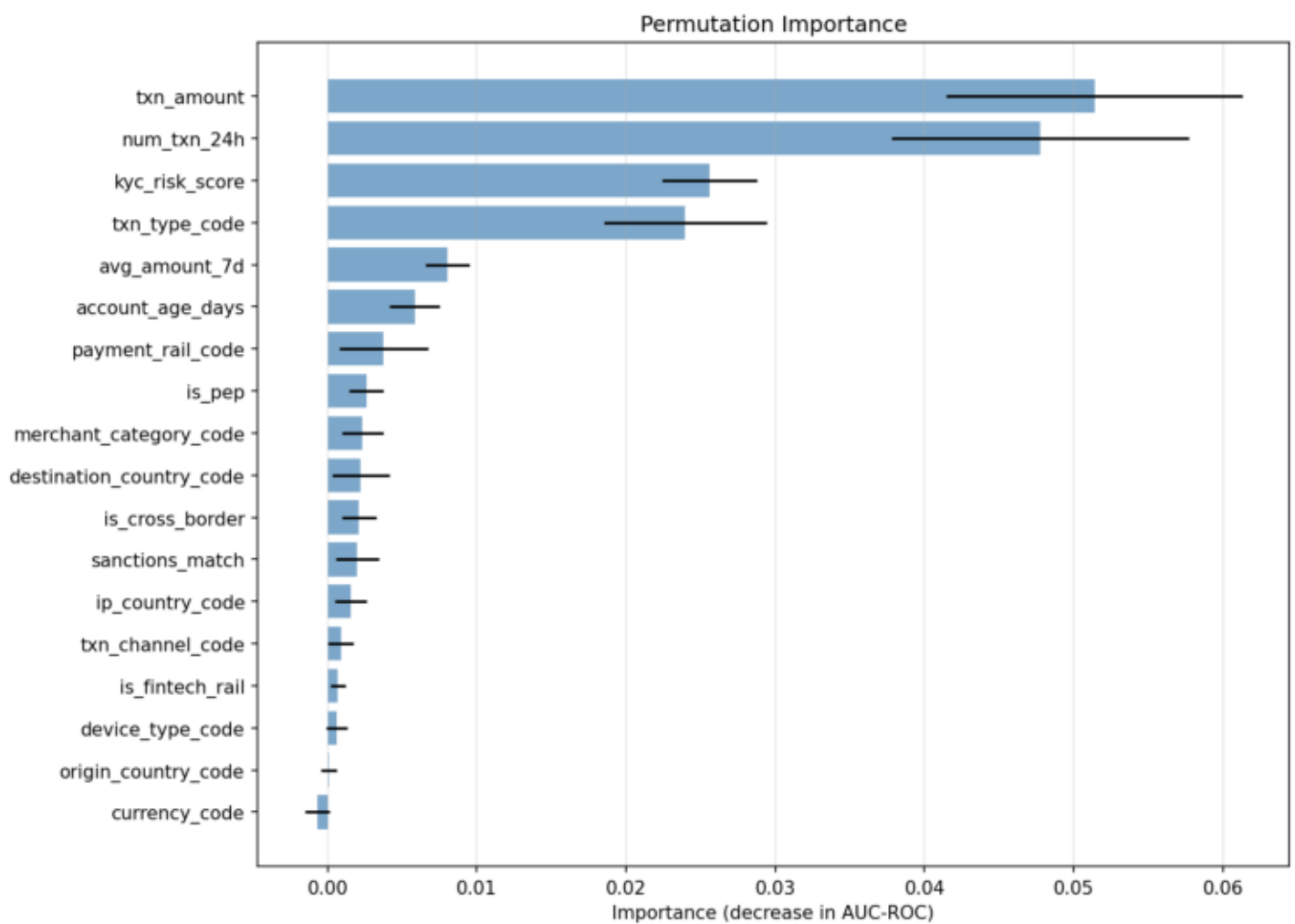
Bootstrap 95% CI: [0.7923, 0.8354]

value_date_quarter=2024Q4 - 4) Model Interpretability

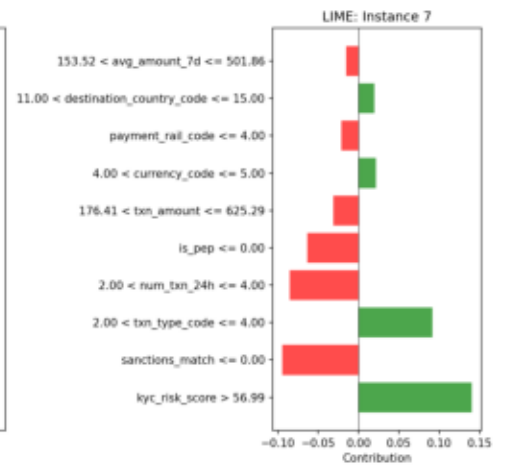
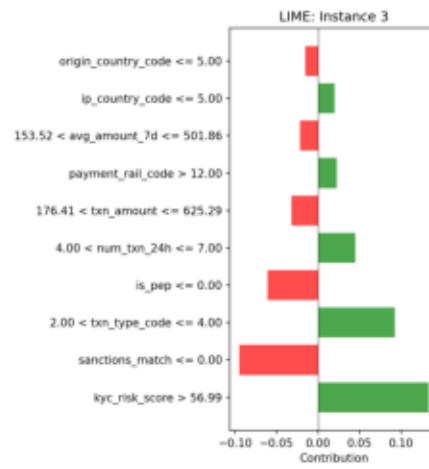
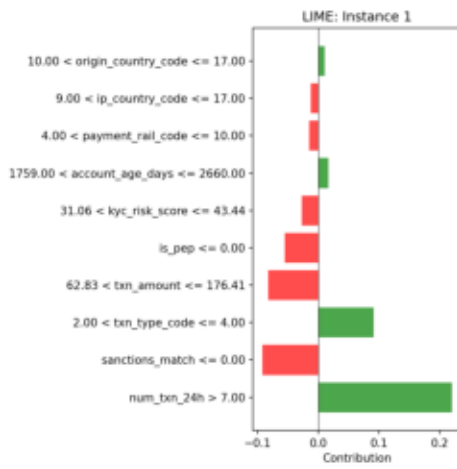
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'lime']
- shap_status: unavailable
- shap_reason: No module named 'shap'
- perm_top_features: ['txn_amount', 'num_txn_24h', 'kyc_risk_score', 'txn_type_code', 'avg_amount_7d']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q4 - 4) Model Interpretability - Explanations

- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'kyc_risk_score', 'txn_type_code',
- 'avg_amount_7d'].
- SHAP plots not generated because SHAP is unavailable (No module named 'shap').
- LIME generated local explanations for 3 instances; top contributors for instance_1: ['num_txn_24h > 7.00',
- 'sanctions_match <= 0.00', '2.00 < txn_type_code <= 4.00', '62.83 < txn_amount <= 176.41', 'is_pep <= 0.00'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score',
- 'num_txn_24h', 'ip_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days',
- 'kyc_risk_score'].

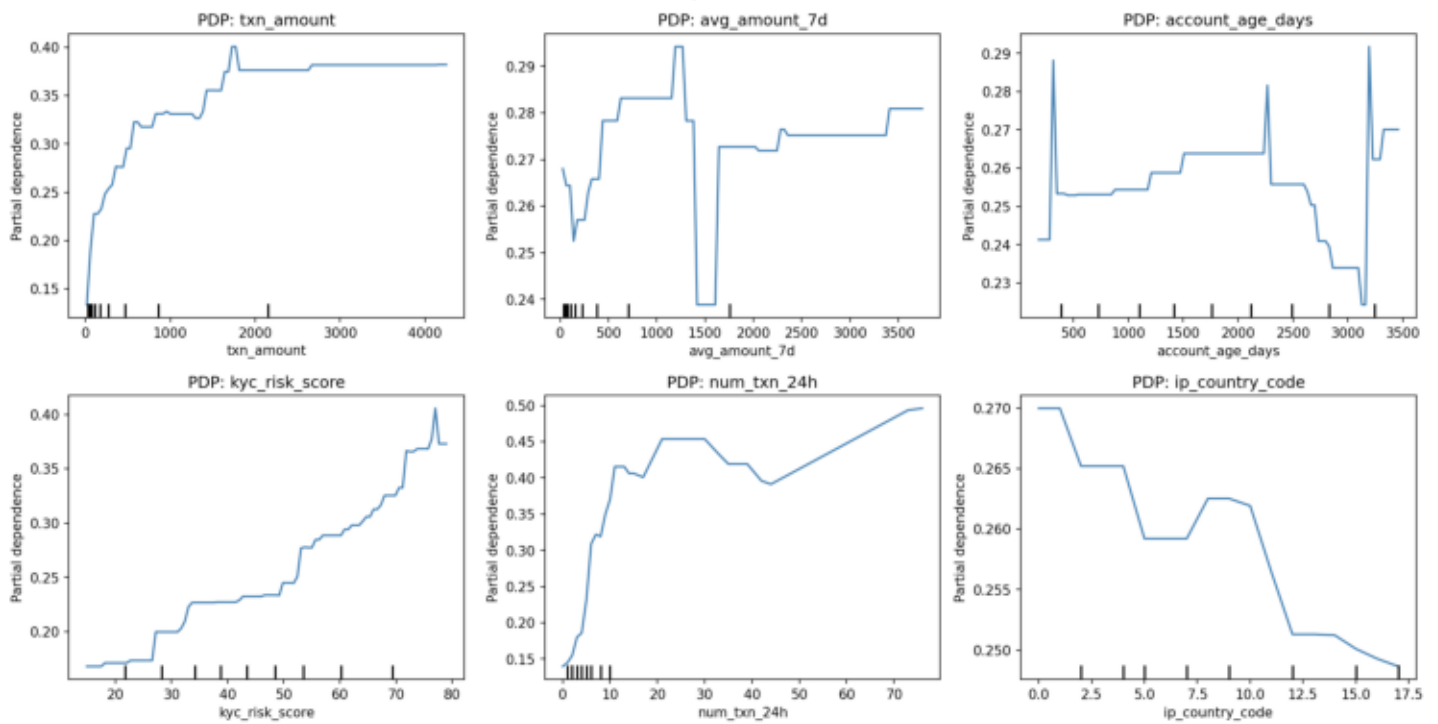


permutation_importance: Bars show how much AUC drops when each feature is permuted.



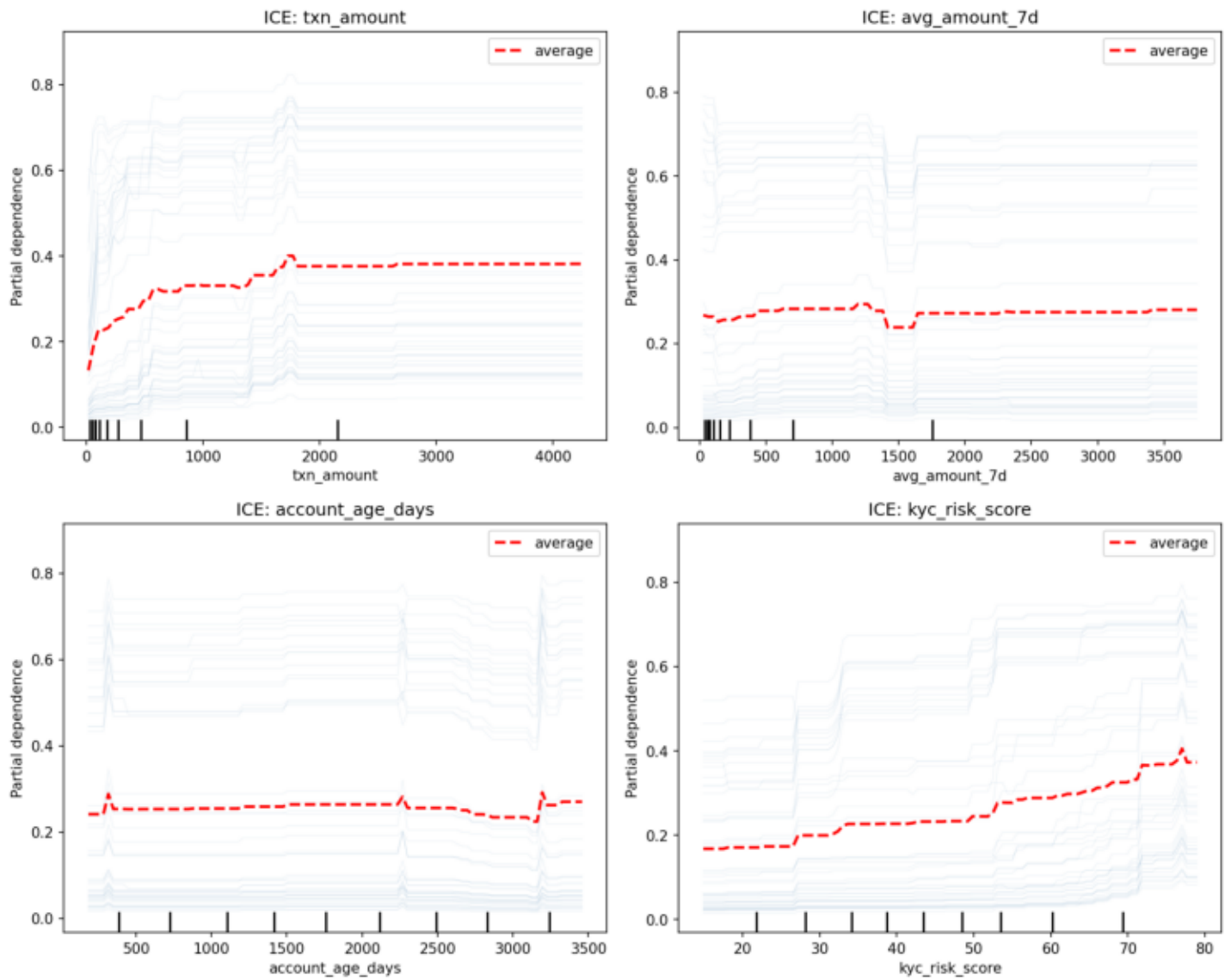
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.